- Thermodynamic and Fisher-Geometric Teacher-Side Antidistillation Under Black-Box Querying

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Abstract

Problem. Anthropic's recent public allegation of industrial-scale distillation attacks against Claude highlights a growing security risk for frontier language model APIs: black-box querying can be used to harvest outputs at scale and train student models that inherit valuable reasoning, coding, and instruction-following capabilities. Existing defenses against model extraction are either largely reactive, relying on telemetry or platform-level monitoring, or they perturb outputs heuristically without a principled account of why certain traces are more distillable than others. This exposes a fundamental gap: ***how can a teacher preserve API utility while making its outputs intrinsically difficult to convert into downstream student capability?*** ***Method.*** We introduce ***FROST***, a thermodynamic and Fisher-geometric framework for ***teacher-side antidistillation under black-box querying***. FROST models defended generation as a constrained free-energy minimization problem, constructing a Gibbs-reweighted output distribution that penalizes candidate responses according to an anti-copying energy measuring their learnability for a family of attacker students. Our central instantiation defines this energy through ***Fisher-geometric learning progress***, assigning higher cost to outputs that induce large natural-gradient updates in proxy students, while preserving proximity to the nominal teacher through a KL trust region. This yields a stateless, prompt-local defense that requires no account tracking, traffic metadata, or user-identity signals. We further provide an information-theoretic interpretation of defended decoding as reducing the per-query leakage of latent capability-bearing structure, leading to query-complexity lower bounds for successful extraction. ***Results and Implications.*** To evaluate teacher-side antidistillation, we design a benchmark protocol spanning multi-budget black-box harvesting, cross-architecture student distillation, and utility-preserving teacher evaluation on reasoning, knowledge, and code-generation tasks. Across multiple teachers, students, and transcript regimes, FROST substantially reduces capability transfer efficiency relative to nominal decoding and prior teacher-side defenses while maintaining the teacher's downstream utility and response quality. These results suggest that antidistillation should be understood not as heuristic output perturbation, but as a ***thermodynamic and information-geometric control problem over the learning value of emitted traces***. More broadly, our framework reframes API defense as the design of ***non-transferable yet still useful outputs***, opening a new direction for safeguarding frontier model capabilities under deployment.

1 Weaponizing Distillation as Capability and Data Stealing

From compression to attack surface. ***Distillation*** is traditionally framed as a benign compression primitive: a smaller student learns from a stronger teacher to reduce inference cost while preserving performance. ***However***, once a frontier model is exposed via a black-box API, the same teacher–student pipeline can be ***weaponized***: attackers can harvest outputs at scale and repurpose them as high-value supervision. In particular, Anthropic stated on February 23, 2026 that it had identified "industrial-scale distillation attacks" against Claude, alleging that fraudulent black-box access was used to harvest outputs for downstream model improvement; Reuters reported the same allegation and highlighted that the targeted capabilities included ***reasoning, coding, and tool use***. Importantly, these are *public allegations*, not adjudicated findings; nevertheless, they sharpen a broader security reality: ***deployment converts model outputs into an asset that can be systematically harvested and reused as training signal*** (Anthropic, 2026; Babu, 2026).

Why black-box distillation is plausible (and historically expected). This concern is not isolated to one incident. The model-extraction literature has long established that ***black-box query access*** can suffice to recover high-fidelity surrogates of deployed models. Tramèr et al. demonstrated efficient

extraction attacks against prediction APIs, and Orekondy et al. showed that competitive "knockoff" models can be trained from queried input–output pairs even when the surrogate architecture and query distribution differ from the victim's (Tramèr et al., 2016; Orekondy et al., 2019). In the LLM setting, the threat surface is amplified because the API output channel is far richer than labels or probabilities: it may expose ***long-form explanations, decompositions, code, rubrics, and reasoning traces***. Recent surveys therefore treat API-based functionality extraction, training-data extraction, and prompt-targeted attacks as a unified threat landscape for deployed language models (Zhao et al., 2025).

What exactly can be stolen? A three-channel leakage view. In the LLM setting, an attacker may extract far more than raw benchmark behavior. The output channel can leak at least three distinct forms of value:

- ***Capability***: reusable input–output behavior (e.g., reasoning patterns, coding competence, tool-use conventions, evaluation heuristics) that supports downstream student training (Tramèr et al., 2016; Orekondy et al., 2019; Zhao et al., 2025).

- ***Data***: memorized training content recoverable through repeated querying. Carlini et al. showed that large language models can emit verbatim training examples; more recently, Nasr et al. studied ***scalable extraction*** and report that adversaries can extract large quantities of training data from both open and production language models, including aligned systems, via targeted querying strategies (Carlini et al., 2021; Nasr et al., 2023).

- ***Control logic***: hidden prompts and behavioral scaffolds that shape the model's behavior. Zhang et al. (EMNLP 2024) propose output2prompt, a black-box method that reconstructs prompts from normal outputs without access to logits or jailbreaking; complementary work demonstrates systematic prompt extraction and benchmarks prompt-theft susceptibility in LLM-integrated applications (Zhang et al., 2024a,b; Wang et al., 2024).

Why structured traces are especially dangerous (high-density supervision). Not all outputs are equally distillable. Long-form explanations, step-by-step derivations, explicit decompositions, and reusable response templates provide substantially denser training signal than terse answers. This intuition is now reflected in the technical literature: *Antidistillation Sampling* is motivated by the observation that extended reasoning traces can facilitate downstream distillation, and proposes modifying a teacher's sampling procedure so that emitted traces remain useful to end users while becoming less effective training material for students (Savani et al., 2025). Likewise, prompt-theft benchmarks explicitly catalog multi-step extraction strategies and show broad susceptibility in the absence of defenses (Wang et al., 2024). Collectively, these results indicate that ***capability transfer is increasingly mediated by the structure and pedagogy of the teacher's outputs***, not only by the underlying model weights.

Distillation as capability theft vs. data theft: two attacker objectives. Weaponized distillation admits two closely related objectives:

1. ***Capability stealing*** (*functionality extraction*): train a surrogate that matches the teacher on target tasks while bypassing access controls, cost, and deployment governance (Tramèr et al., 2016; Orekondy et al., 2019; Zhao et al., 2025).

2. ***Data stealing*** (*extractable memorization / inversion*): recover private training artifacts or proprietary control prompts from outputs, either directly via memorization or indirectly via inversion/reconstruction (Carlini et al., 2021; Nasr et al., 2023; Zhang et al., 2024a,b).

Under this view, unauthorized distillation is not merely an IP dispute: it is a joint problem of ***capability theft, data leakage, and control-logic exfiltration*** (Zhao et al., 2025).

Provenance as a partial countermeasure (and why it is insufficient). A natural response to output-mediated theft is to embed provenance signals (e.g., watermarks) into generated text. Prior work proposes watermarking schemes for LLMs that bias token sampling so that outputs become statistically detectable without access to the underlying model (Kirchenbauer et al., 2023). More recently, SynthID-Text provides a production-focused watermarking scheme and reports large-scale and controlled human evaluations indicating ***quality-neutral*** watermarking under their settings (Dathathri et al., 2024). However, provenance primarily supports *attribution* rather than *prevention*: even when outputs are detectable as AI-generated, they may still carry valuable supervision for distillation. This motivates defenses that directly control the ***learnability*** of emitted traces, rather than only their detectability.

Implication for this paper. This perspective motivates a stronger defensive objective than post-hoc monitoring or identity-based controls. If outputs can be repurposed as high-value supervision for downstream surrogates, then securing a frontier API requires controlling not only *what* the teacher can do for legitimate users, but also ***how learnable*** each emitted trace is for an adversary. This is precisely the goal of ***FROST***: to treat teacher-side generation as a ***thermodynamic control problem over the learning value of outputs***, preserving utility for normal use while making capability transfer under black-box querying substantially harder.

2 The FROST Framework

Overview and central challenge. ***The central challenge addressed by FROST is deceptively simple:*** a frontier model exposed through a black-box API must remain ***useful*** to legitimate users, yet its outputs should not become ***cheap supervision*** for downstream student models. This tension is what makes teacher-side antidistillation fundamentally difficult. If the teacher emits rich, explicit, and pedagogically structured responses, it maximizes utility for users but also increases the transferability of capability to an attacker harvesting outputs at scale. If, on the other hand, the teacher aggressively suppresses, truncates, or randomizes its generations, it may reduce distillability only by collapsing the very API value it is meant to provide. Existing black-box extraction work establishes that queried outputs can suffice to recover high-fidelity surrogates (Tramèr et al., 2016; Orekondy et al., 2019; Zhao et al., 2025), while recent teacher-side defenses such as *Antidistillation Sampling* show that output shaping can indeed reduce downstream student performance (Savani et al., 2025). ***FROST is built around the stronger principle that defended generation should be formulated as a utility-constrained control problem over the learnability of emitted traces.*** Concretely, FROST treats each candidate response not merely as text to be returned, but as a potential training example whose downstream learning value must be regulated through ***thermodynamic reweighting*** and ***Fisher-geometric anti-copying***.

2.1 Problem Setup and Threat Model

Motivation. ***Knowledge distillation*** is classically introduced as a ***compression*** mechanism: a smaller student model is trained to imitate a stronger teacher so as to retain predictive utility at lower inference cost (Hinton et al., 2015). However, once a frontier model is deployed behind a query interface, the same teacher–student pipeline acquires a security interpretation. A long line of work on ***model extraction*** shows that black-box query access can suffice to recover high-fidelity surrogates of deployed models (Tramèr et al., 2016; Orekondy et al., 2019). Recent surveys argue that the LLM setting only broadens this attack surface, because outputs now include not just labels or confidence values, but also natural-language explanations, code, structured traces, and reusable reasoning patterns (Zhao et al., 2025). More recently, *Antidistillation Sampling* explicitly studies the teacher-side problem of making reasoning traces ***less useful for student training while preserving teacher utility*** (Savani et al., 2025). FROST is designed for precisely this deployment regime.

Teacher as a deployed stochastic kernel. We model the deployed teacher not merely as a fixed set of weights, but as a ***conditional output channel***. Let

$$x \in \mathcal{X} \qquad \text{and} \qquad y \in \mathcal{Y}$$

denote, respectively, a user prompt and a generated response. Let

$$P_0(y \mid x)$$

be the ***nominal teacher distribution***: the response distribution induced by the deployed frontier model under its standard decoding stack. This formulation is intentionally general. It absorbs the base language model, safety layers, refusal policies, prompt formatting, and standard decoding settings into a single conditional distribution over outputs. FROST will construct a defended distribution

$$Q(y \mid x),$$

which is the only object exposed to the attacker at inference time.

Black-box adaptive attacker. We consider a black-box adversary that can query the teacher repeatedly, but does ***not*** observe logits, gradients, parameters, optimizer state, hidden activations, or any internal telemetry. The attacker is allowed to choose prompts *adaptively*. Formally, for a total query budget B, the attacker collects the transcript

$$\mathcal{T}_B = \{(x_t, y_t)\}_{t=1}^{B}, \qquad y_t \sim Q(\cdot \mid x_t),$$

where x_t may depend arbitrarily on the past:

$$x_t \sim \pi_t(\cdot \mid x_{<t}, y_{<t}).$$

This captures the operational reality emphasized in the model-extraction literature: attacks need not use a fixed prompt set, and can actively refine future queries using previous outputs (Tramèr et al., 2016; Orekondy et al., 2019; Zhao et al., 2025).

Student learning from harvested traces. After collecting $\mathcal{T}_B$, the attacker trains a student S_θ by imitation. At a minimum, this can be modeled as standard supervised fine-tuning on teacher outputs:

$$\hat{\mathcal{L}}_{\text{imit}}(\theta; \mathcal{T}_B) = \frac{1}{B} \sum_{t=1}^{B} \ell_\theta(x_t, y_t), \qquad \ell_\theta(x_t, y_t) := -\log p_\theta(y_t \mid x_t).$$

This formulation subsumes several practical attack instantiations:

- ***answer-only distillation***, where the student imitates final responses;
- ***trace-rich distillation***, where the student is trained on longer explanations, decompositions, or code;
- ***cross-architecture transfer***, where the student architecture differs from the teacher;
- ***parameter-efficient attacks***, such as LoRA-style fine-tuning on harvested transcripts.

These attack families are consistent with both the broader extraction literature and the recent antidistillation literature, which explicitly treats model outputs as valuable downstream supervision (Orekondy et al., 2019; Zhao et al., 2025; Savani et al., 2025).

What the defender can and cannot use. A central design decision in FROST is that the defense is ***identity-free*** and ***prompt-local***. That is, FROST does *not* assume access to user accounts, IP addresses, cross-session telemetry, or metadata-level coordination signals. Instead, it acts only through the ***output distribution conditioned on the current prompt***. This makes the defense conceptually stronger and operationally cleaner: even if the attacker rotates accounts, proxies requests, or otherwise hides telemetry, the output channel itself remains protected.

Core threat-model assumptions. For clarity, we summarize the main assumptions:

1. The attacker has ***black-box query access*** to the teacher.
2. The attacker may choose queries ***adaptively*** based on previous responses.
3. The attacker trains a downstream student using harvested teacher outputs as supervision.
4. The defender operates purely at the level of ***teacher-side output shaping***, with no dependence on identity or telemetry.

Design goal. Under this threat model, the goal of FROST is ***not*** to make extraction information-theoretically impossible in an absolute sense. Rather, the goal is to make each emitted response ***intrinsically less useful as training signal*** for downstream students, while preserving the teacher's utility for legitimate users. In other words, FROST treats the output channel itself as the object of protection.

2.2 Utility-Preserving Antidistillation Objective

Why utility must remain explicit. A trivial antidistillation strategy would be to degrade outputs arbitrarily: shorten them, randomize them, or return nonsensical responses. Such strategies are useless in practice because they collapse the API's value for normal users. Accordingly, FROST is built around a stricter requirement: ***teacher-side defense must preserve utility while reducing learnability***. This principle is consistent with recent teacher-side defenses such as *Antidistillation Sampling*, which explicitly targets lower distillability at comparable teacher performance, and with inference-time watermarking methods such as SynthID-Text and the watermark of Kirchenbauer et al., both of which modify sampling while aiming to preserve output quality (Savani et al., 2025; Kirchenbauer et al., 2023; Dathathri et al., 2024).

Utility as a constrained resource. Let

$$u(x, y) \in \mathbb{R}$$

denote a ***utility functional*** measuring the quality of response y to prompt x. This quantity is intentionally broad: depending on the evaluation slice, it may encode task accuracy, helpfulness, code correctness, reasoning quality, formatting fidelity, or calibrated refusal behavior. For a prompt distribution $\mathcal{D}$, the expected teacher utility under defended decoding Q is

$$U(Q; \mathcal{D}) = \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{E}_{y\sim Q(\cdot|x)}[u(x, y)]\right].$$

FROST does not maximize $U(Q; \mathcal{D})$ from scratch; rather, it seeks to remain close to the utility of the nominal teacher distribution P_0. This distinction is important: the problem is ***utility-preserving protection***, not generic response optimization.

Learnability as the defended quantity. To formalize teacher-side antidistillation, we introduce an ***anti-copying energy***

$$E_x(y) \geq 0,$$

which quantifies how useful output y would be as supervision for a downstream student if emitted in response to prompt x. Intuitively:

- ***low-energy outputs*** are weak training signal for attackers;
- ***high-energy outputs*** are rich, reusable, student-helpful traces.

At this stage, $E_x(y)$ is kept abstract. In the next subsection, FROST will instantiate it using ***Fisher-geometric learning progress***. For now, it serves as a generic measure of downstream learnability.

The constrained antidistillation program. FROST seeks a defended teacher distribution Q that makes emitted traces difficult to distill while preserving utility. The most transparent formulation is the following constrained program:

$$\min_{Q} \quad \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{E}_{y\sim Q(\cdot|x)}[E_x(y)]\right]$$

subject to

$$U(Q; \mathcal{D}) \geq U_{\min}$$

and

$$\mathbb{E}_{x\sim\mathcal{D}}[D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x))] \leq \delta.$$

These two constraints play distinct roles:

1. ***Utility floor*** $U_{\min}$: the defended teacher must remain useful to legitimate users.
2. ***Trust-region constraint*** δ: the defended distribution must remain close to the nominal teacher, preventing arbitrary or degenerate perturbations.

Why the KL trust region matters. The KL term is not only mathematically convenient; it formalizes a key deployment desideratum. FROST should be understood as a ***minimal intervention*** on the original teacher. By constraining

$$D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)),$$

we force the defense to remain close to standard behavior unless doing so would expose disproportionately high-value training signal. This gives a clean interpretation: FROST is not a separate assistant layered on top of the teacher, but a ***defended perturbation*** of the teacher's own response distribution.

Lagrangian form. Introducing Lagrange multipliers yields the equivalent unconstrained objective

$$\min_{Q} \quad \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{E}_{y\sim Q(\cdot|x)}[E_x(y)] + \lambda_u\,\Phi_u(Q;x) + \lambda_{\mathrm{KL}}\,D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x))\right],$$

where $\Phi_u(Q;x)$ is a penalty for falling below the desired utility level on prompt x. In practice, this formulation makes explicit the ***three-way tradeoff*** at the heart of teacher-side antidistillation:

lower learnability vs. ***higher utility*** vs. ***smaller deviation from nominal behavior***.

Interpretation. This objective already distinguishes FROST from simpler output-space defenses:

- it is ***not*** mere randomization, because perturbations are scored against a learnability objective;
- it is ***not*** mere provenance marking, because the primary target is transfer efficiency, not only detectability (Kirchenbauer et al., 2023; Dathathri et al., 2024);
- it is ***not*** post-hoc monitoring, because the intervention is built directly into generation itself (Savani et al., 2025).

The role of the next subsection is therefore to answer the central remaining question: ***how should one measure the learnability of an emitted trace?*** FROST answers this by moving from a generic anti-copying energy $E_x(y)$ to a ***Fisher-geometric*** notion of student learning progress.

2.3 Free-Energy Formulation of Defended Decoding

Why a free-energy formulation? The constrained objective in Section 3.2 makes clear that teacher-side antidistillation is not a standard decoding problem. The defender does not merely wish to maximize utility, nor merely to minimize deviation from the nominal teacher, nor merely to suppress structured traces. Rather, the defender must solve a ***three-way control problem***:

low downstream learnability ***high teacher utility*** ***small deviation from nominal behavior***.

Objectives of the general form "risk plus KL divergence to a reference distribution" arise naturally in several adjacent literatures, including ***Gibbs/PAC-Bayesian posteriors*** and ***KL-regularized control***. In PAC-Bayesian settings, the posterior is chosen by trading off empirical performance against KL divergence from a prior (Seldin and Tishby, 2009; Sahu and Hemachandra, 2019; Dziugaite and Roy, 2018). In modern language-model control, KL-regularized reinforcement learning is used to keep generations close to a base model while steering them toward desired reward signals, and *Controlled Decoding* shows that tokenwise controlled generation can be interpreted as sampling from the solution of such a regularized objective (Mudgal et al., 2024; Leibfried and Grau-Moya, 2020). FROST adopts this variational perspective, but with a different control signal: ***the learnability of an emitted trace for downstream students***. :contentReference[oaicite:1]index=1

Prompt-local defended decoding. Fix a prompt $x \in \mathcal{X}$. Let

$$P_0(\cdot \mid x)$$

denote the nominal teacher distribution, and let

$$Q(\cdot \mid x)$$

denote a candidate defended decoding distribution. Let

$$E_x(y) \geq 0$$

be the anti-copying energy introduced in Section 3.2, interpreted as the ***downstream learning value*** of response y for an adversarial student. At this stage, we do not yet assume any specific form for $E_x(y)$; the Fisher-geometric instantiation is deferred to Section 3.4. The core idea of FROST is to choose $Q(\cdot \mid x)$ by minimizing a prompt-local ***free-energy functional***:

$$\mathcal{F}_\beta(Q; x) = \mathbb{E}_{y \sim Q(\cdot|x)}[E_x(y)] + \frac{1}{\beta} D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)),$$

where $\beta > 0$ is an inverse-temperature parameter controlling the strength of antidistillation.

This objective has a direct operational meaning:

- the first term penalizes ***distillable outputs***;
- the second term penalizes ***deviation from the nominal teacher***;
- β controls how aggressively the defender trades utility-preserving fidelity for anti-copying robustness.

Thus, FROST does not search over arbitrary responses. It computes the ***least disruptive reweighting*** of the original teacher distribution that suppresses high-learnability traces.

Variational definition of FROST. For each prompt x, the defended decoding rule is defined as

$$Q^\star(\cdot \mid x) = \arg\min_{Q(\cdot|x)} \mathcal{F}_\beta(Q; x).$$

This choice yields an explicit Gibbs form.

Proposition 1 (Variational optimality of FROST). For any fixed prompt x, the unique minimizer of

$$\mathcal{F}_\beta(Q; x) = \mathbb{E}_Q[E_x(y)] + \frac{1}{\beta} D_{\mathrm{KL}}(Q \| P_0)$$

over all distributions $Q(\cdot \mid x)$ absolutely continuous with respect to $P_0(\cdot \mid x)$ is

$$Q^\star(y \mid x) = \frac{1}{Z_\beta(x)} P_0(y \mid x) \exp(-\beta E_x(y)),$$

where

$$Z_\beta(x) = \sum_{y \in \mathcal{Y}} P_0(y \mid x) \exp(-\beta E_x(y))$$

is the partition function.

Proof sketch. Define

$$Q^\star_\beta(y \mid x) := \frac{1}{Z_\beta(x)} P_0(y \mid x) e^{-\beta E_x(y)}.$$

Consider the KL divergence between any candidate Q and $Q^\star_\beta$:

$$D_{\mathrm{KL}}(Q \| Q^\star_\beta) = \sum_y Q(y \mid x) \log \frac{Q(y \mid x)}{Q^\star_\beta(y \mid x)}.$$

Substituting the definition of $Q^\star_\beta$,

$$\log \frac{Q(y \mid x)}{Q^\star_\beta(y \mid x)} = \log \frac{Q(y \mid x)}{P_0(y \mid x)} + \beta E_x(y) + \log Z_\beta(x).$$

Taking expectation under Q gives

$$D_{\mathrm{KL}}(Q \| Q^\star_\beta) = D_{\mathrm{KL}}(Q \| P_0) + \beta \mathbb{E}_Q[E_x(y)] + \log Z_\beta(x).$$

Rearranging,

$$\mathcal{F}_\beta(Q; x) = \frac{1}{\beta} D_{\mathrm{KL}}(Q \| Q^\star_\beta) - \frac{1}{\beta} \log Z_\beta(x).$$

Since KL divergence is nonnegative and equals zero if and only if $Q = Q^\star_\beta$, the minimizer is uniquely given by the Gibbs-reweighted distribution above. □

Interpretation of the Gibbs form. Proposition 1 reveals the exact structure of the defense. FROST does ***not*** replace the nominal teacher with a new model, nor does it enforce crude post-hoc truncation. Instead, it performs a ***Boltzmann-style reweighting*** of the original output law:

$$P_0(y \mid x) \;\longrightarrow\; Q^\star(y \mid x) \propto P_0(y \mid x) e^{-\beta E_x(y)}.$$

Outputs with large anti-copying energy are exponentially downweighted; outputs with small anti-copying energy are preserved. The partition function $Z_\beta(x)$ normalizes the defended distribution and encapsulates the prompt-specific cost of protection.

This interpretation is important for two reasons. First, it makes FROST a ***minimal-intervention*** defense: every protected output is still sampled from the support of the teacher's own response law. Second, it aligns FROST with the same variational family that appears in Gibbs posteriors and KL-regularized control, while changing the semantics of the "energy" term from *task loss* or *reward* to *distillability*. :contentReference[oaicite:2]index=2

Inverse temperature as a defense knob. The inverse temperature β has a clean operational role:

- as $\beta \to 0$, the KL term dominates and

  $$Q^\star(\cdot \mid x) \to P_0(\cdot \mid x),$$

  so the defense vanishes;

- as β increases, the decoding distribution becomes increasingly concentrated on low-energy, less-distillable traces;

- for excessively large β, the defense may over-regularize generation, potentially harming utility.

Thus, β induces a ***security–utility frontier***: low β yields high fidelity but weak protection, while high β yields stronger anti-copying at the risk of reduced helpfulness.

Thermodynamic diagnostics. The partition function

$$Z_\beta(x) = \sum_y P_0(y \mid x) e^{-\beta E_x(y)}$$

also gives FROST a useful thermodynamic interpretation. Define the prompt-level free energy

$$\Psi_\beta(x) := -\frac{1}{\beta} \log Z_\beta(x).$$

Then standard differentiation identities yield

$$-\frac{\partial}{\partial \beta} \log Z_\beta(x) = \mathbb{E}_{y \sim Q^\star(\cdot \mid x)}[E_x(y)],$$

and

$$\frac{\partial^2}{\partial \beta^2} \log Z_\beta(x) = \mathrm{Var}_{y \sim Q^\star(\cdot \mid x)}(E_x(y)).$$

These quantities have natural interpretations:

- the first derivative gives the ***expected defended distillability*** of generated responses;

- the second derivative gives the ***susceptibility*** of the defense, i.e., how sensitive defended decoding is to changes in β.

In other words, FROST is not merely inspired by thermodynamics metaphorically; it inherits an actual ***free-energy geometry over output distributions***.

Connection to controlled decoding. It is useful to contrast FROST with nearby inference-time control methods. Controlled Decoding formulates language-model generation as a tokenwise KL-regularized RL problem and uses a prefix scorer so that inference-time sampling follows the optimizer of a regularized control objective (Mudgal et al., 2024). Antidistillation Sampling, in turn, modifies next-token probabilities to make emitted traces less useful for downstream distillation while preserving practical model utility (Savani et al., 2025). FROST is conceptually aligned with both directions, but differs in the ***semantics of control***:

- in KL-regularized alignment, the control signal is usually ***reward***;
- in watermarking, the control signal is ***detectability or provenance***;
- in FROST, the control signal is ***downstream student learnability***.

This shift is crucial. FROST does not ask which outputs are most preferred, nor which are easiest to attribute, but rather which are ***least transferable as training signal*** under a utility-preserving trust region. :contentReference[oaicite:3]index=3

From constrained utility to free energy. The free-energy objective above can also be viewed as the Lagrangian relaxation of the constrained antidistillation program from Section 3.2. If the original problem is

$$\min_{Q} \ \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim Q(\cdot|x)}[E_x(y)] \quad \text{s.t.} \quad U(Q;\mathcal{D}) \geq U_{\min}, \quad \mathbb{E}_x D_{\mathrm{KL}}(Q\|P_0) \leq \delta,$$

then for fixed Lagrange multipliers, the prompt-local subproblem reduces exactly to the minimization of

$$\mathbb{E}_Q[E_x(y)] + \frac{1}{\beta} D_{\mathrm{KL}}(Q\|P_0),$$

possibly with an additional utility-penalty term absorbed into the energy. This observation matters because it makes the defended distribution ***derivable***, not heuristic: the Gibbs form is the optimizer of a well-defined variational program.

What remains unspecified. At this point, the formal structure of defended decoding is complete, but one critical ingredient remains open: ***how should the anti-copying energy*** $E_x(y)$ ***be defined?*** The entire effectiveness of FROST depends on whether this energy actually captures the downstream learnability of an emitted trace. The next subsection answers this question by introducing a ***Fisher-geometric*** notion of student learning progress, yielding a principled and attacker-aware instantiation of $E_x(y)$.

2.4 Fisher-Geometric Learnability as Anti-Copying Energy

From abstract energy to intrinsic learnability. Section 3.3 established that FROST protects the teacher by Gibbs-reweighting the nominal output distribution according to an anti-copying energy $E_x(y)$. The remaining question is therefore the most important one: ***what makes an emitted trace easy or hard for a student to learn from?*** A naive answer would score outputs using Euclidean gradient norms under a proxy student. However, this is unsatisfactory because the parameter space of a statistical model is not naturally Euclidean. As emphasized in information geometry, the same predictive distribution may admit many parametrizations, and ordinary gradients can therefore be highly coordinate-dependent. Amari's natural-gradient framework resolves this by endowing the statistical model with the ***Fisher information metric***, under which steepest descent is measured intrinsically rather than in an arbitrary parameter coordinate system (Amari, 1998, 2016). FROST adopts precisely this viewpoint: ***an output is dangerous when it induces large intrinsic learning progress in plausible attacker students.***

Student family and imitation loss. Let $\mathcal{Q}_S$ denote a distribution over plausible attacker students. This family may vary in architecture, scale, parameterization, fine-tuning recipe, and optimization strategy, but all members are assumed to consume harvested pairs (x, y) as supervision. For a given student S with parameters θ, define the token- or sequence-level imitation loss

$$\ell_S(\theta; x, y) := -\log p_\theta(y \mid x).$$

Its gradient

$$g_S(\theta; x, y) := \nabla_\theta \ell_S(\theta; x, y)$$

describes the immediate update pressure induced by seeing (x, y) as a training example. Intuitively, if g_S is large in an intrinsic sense, then the teacher output contains a strong training signal for the student; if it is small, the response is comparatively uninformative as supervision.

Fisher geometry of the student manifold. For each student S, let

$$F_S(\theta) = \mathbb{E}_{(x,y)\sim p_\theta}\left[\nabla_\theta \log p_\theta(y \mid x)\, \nabla_\theta \log p_\theta(y \mid x)^\top\right]$$

denote the Fisher information matrix. The Fisher matrix plays a distinguished role in information geometry: it defines the canonical Riemannian metric on the manifold of probability distributions induced by the model family (Amari, 1998, 2016). Under this metric, the natural gradient direction is

$$\widetilde{\nabla}_\theta \ell = F_S(\theta)^{-1} \nabla_\theta \ell,$$

which is the steepest descent direction in distribution space rather than parameter space. This distinction matters because ordinary gradients are sensitive to reparameterization, while the Fisher metric captures the local geometry of the predictive family itself (Amari, 1998).

Natural-gradient learning progress. FROST defines the learnability of a candidate output using the quadratic form induced by the Fisher metric:

$$\mathcal{G}_S(\theta; x, y) := g_S(\theta; x, y)^\top F_S(\theta)^{-1} g_S(\theta; x, y).$$

This quantity has a direct geometric meaning. It is the squared norm of the ordinary gradient measured in the inverse-Fisher metric, or equivalently the inner product between the Euclidean gradient and the natural-gradient direction. In other words, $\mathcal{G}_S(\theta; x, y)$ measures how much the sample (x, y) can move the student *intrinsically* along the statistical manifold.

Why this quantity captures learnability. The justification follows from a first-order natural-gradient update. Consider the update

$$\Delta\theta = -\eta F_S(\theta)^{-1} g_S(\theta; x, y),$$

where $\eta > 0$ is a small stepsize. A first-order Taylor expansion of the loss around θ yields

$$\ell_S(\theta + \Delta\theta; x, y) \approx \ell_S(\theta; x, y) + \nabla_\theta \ell_S(\theta; x, y)^\top \Delta\theta + O(\|\Delta\theta\|^2).$$

Substituting the natural-gradient step,

$$\ell_S(\theta + \Delta\theta; x, y) \approx \ell_S(\theta; x, y) - \eta\, g_S(\theta; x, y)^\top F_S(\theta)^{-1} g_S(\theta; x, y) + O(\eta^2).$$

Thus, to first order, the loss reduction induced by the sample is exactly controlled by $\mathcal{G}_S(\theta; x, y)$. This gives a precise interpretation:

- if $\mathcal{G}_S(\theta; x, y)$ is ***large***, then (x, y) is a ***high-leverage training example*** for the student;
- if $\mathcal{G}_S(\theta; x, y)$ is ***small***, then (x, y) induces comparatively weak first-order learning progress.

Accordingly, FROST regards large-$\mathcal{G}_S$ responses as dangerous and seeks to downweight them during decoding.

Proposition 2 (Fisher-geometric progress of a harvested trace). Let S be a proxy student with parameters θ, loss $\ell_S(\theta; x, y)$, and Fisher matrix $F_S(\theta)$. Under the natural-gradient update

$$\Delta\theta = -\eta F_S(\theta)^{-1} g_S(\theta; x, y),$$

the first-order decrease in imitation loss induced by the sample (x, y) is

$$\ell_S(\theta + \Delta\theta; x, y) - \ell_S(\theta; x, y) = -\eta\, g_S(\theta; x, y)^\top F_S(\theta)^{-1} g_S(\theta; x, y) + O(\eta^2).$$

Hence the Fisher-quadratic form $g_S^\top F_S^{-1} g_S$ is a first-order measure of how learnable the emitted trace is for student S.

Implication for antidistillation. Proposition 2 provides exactly the quantity needed by FROST. If the goal is to make teacher outputs ***intrinsically less useful as training signal***, then the defense should penalize outputs that maximize first-order learning progress across plausible attacker students. This motivates the following Fisher-geometric anti-copying energy:

$$E_x^{\mathrm{geom}}(y) = \mathbb{E}_{S\sim\mathcal{Q}_S}\left[g_S(\theta_S;x,y)^\top F_S(\theta_S)^{-1} g_S(\theta_S;x,y)\right].$$

Here θ_S denotes the current parameter state of the proxy student S. The outer expectation over $\mathcal{Q}_S$ makes the defense ***multi-student robust***: FROST does not tailor itself to a single surrogate architecture, but instead suppresses responses that are broadly learnable across a family of likely attacker models.

Expectation vs. worst-case robustness. There are two natural robustifications:

$$E_x^{\mathrm{avg}}(y) = \mathbb{E}_{S\sim\mathcal{Q}_S}\left[g_S^\top F_S^{-1} g_S\right], \qquad E_x^{\mathrm{wc}}(y) = \sup_{S\in\mathcal{S}}\left[g_S^\top F_S^{-1} g_S\right].$$

The ***expectation form*** is smoother and often easier to estimate in practice, while the ***worst-case form*** offers a minimax interpretation: the teacher downweights outputs that are maximally useful to the strongest plausible student. In this paper, we take the expectation form as the primary instantiation of FROST because it yields lower-variance scoring and better computational stability, while still capturing broad attacker uncertainty.

Why Euclidean gradient norms are insufficient. It is important that FROST uses

$$g^\top F^{-1} g$$

rather than simpler alternatives such as $\|g\|_2^2$. The Euclidean norm measures sensitivity in parameter coordinates, not in the space of predictive distributions. Two models that represent the same local distributional behavior can have very different Euclidean gradients under different parameterizations. By contrast, the Fisher metric is invariant to smooth reparameterization and therefore measures update magnitude at the level that actually matters for extraction: ***how much the student's predictive behavior changes as a result of the harvested trace*** (Amari, 1998, 2016). This is precisely why the Fisher-geometric quantity is a more faithful notion of *transferable learning value*.

Interpreting anti-copying as geodesic resistance. Information geometry offers an even deeper interpretation. The model family equipped with Fisher metric forms a statistical manifold, and natural-gradient learning follows steepest descent with respect to this manifold's intrinsic geometry (Amari, 2016). From this viewpoint, FROST treats an emitted response as dangerous when it creates a ***geodesically efficient update direction*** for the student. Downweighting such responses increases the effective geometric resistance of the output channel: the attacker still receives fluent responses, but the induced student updates become less aligned with fast intrinsic progress on the manifold. In this sense, FROST does not merely obscure information; it ***reshapes the geometry of what is easy to learn from***.

Composite energy and modularity. The Fisher-geometric term is the core of FROST, but it need not be the only term. In the most general form, the anti-copying energy can be written as

$$E_x(y) = \lambda_{\mathrm{geom}} E_x^{\mathrm{geom}}(y) + \lambda_{\mathrm{struct}} E_x^{\mathrm{struct}}(y) + \lambda_{\mathrm{prov}} E_x^{\mathrm{prov}}(y),$$

where:

- $E_x^{\mathrm{geom}}(y)$ captures ***Fisher-geometric student learnability***;
- $E_x^{\mathrm{struct}}(y)$ can penalize highly reusable scaffolds such as explicit decompositions or trace-heavy outputs;
- $E_x^{\mathrm{prov}}(y)$ can encourage provenance-supporting properties such as watermark retention.

This modularity is deliberate. It allows FROST to unify geometry-aware antidistillation with structured-trace hardening and provenance-aware generation inside a single Gibbs-reweighted decoding rule. Nevertheless, the geometric term remains the conceptual heart of the method, because it is the component that directly measures ***how much the student can learn from a single emitted trace***.

From geometry to defended decoding. Substituting the Fisher-geometric energy into the variational program of Section 3.3 yields the defended output distribution

$$Q^\star(y \mid x) \propto P_0(y \mid x) \exp\Big(-\beta\, \mathbb{E}_{S\sim\mathcal{Q}_S}[g_S(\theta_S; x, y)^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, y)]\Big).$$

This expression is the mathematical core of FROST. It states that defended generation is a ***Gibbs posterior over candidate outputs***, where the energy of a response is not task loss, reward, or toxicity, but rather the ***intrinsic natural-gradient progress it affords to downstream students***. The role of the next subsection is to complement this geometric view with an information-theoretic one, showing that reducing Fisher-geometric learnability also reduces the effective extraction rate of latent capability-bearing structure.

2.5 Multi-Student Robustness

Why single-student defenses are insufficient. A teacher-side antidistillation defense is only as strong as its robustness to the ***attacker model family*** it implicitly assumes. If the defense is optimized against a single proxy student, then its protection may be brittle: an attacker can simply change architecture, scale, fine-tuning regime, or transcript formatting strategy, and recover capability through a different student that the defense did not anticipate. This concern is not hypothetical. Classical functionality-extraction work already shows that black-box stealing can transfer across architectures and even across mismatched query distributions (Orekondy et al., 2019). Recent surveys of LLM extraction attacks likewise emphasize that the attack surface is inherently heterogeneous, spanning API-based distillation, prompt-targeted extraction, training-data extraction, and varied downstream student pipelines (Zhao et al., 2025). Moreover, *Antidistillation Sampling* itself is evaluated with distinct teacher, proxy, and student models, underscoring that practical antidistillation must generalize beyond a single surrogate (Savani et al., 2025). For this reason, FROST is designed not as a defense against *one* student, but as a defense against a ***family of plausible students***.

Student family as an uncertainty set. Let $\mathcal{S}$ denote the set of plausible attacker students. An element $S \in \mathcal{S}$ may vary along several axes:

- ***architecture***: e.g., LLaMA-family, Qwen-family, Mistral-family, or other dense transformer variants;
- ***scale***: small, medium, or relatively strong student models;
- ***fine-tuning regime***: full supervised fine-tuning, LoRA-style PEFT, adapter-based training, or other lightweight adaptation;
- ***transcript format***: answer-only supervision, rationale-rich supervision, or structured traces such as decompositions or code;
- ***optimization state***: different initial checkpoints, optimization hyperparameters, or local parameter geometries.

Rather than assuming that the defender knows the exact student chosen by the adversary, FROST models this uncertainty explicitly through either:

(i) a distribution $\mathcal{Q}_S$ over students, or (ii) a worst-case uncertainty set $\mathcal{S}$.

This mirrors the logic of ***distributionally robust optimization***: if the deployment environment contains uncertainty about future adversaries, one should optimize against a neighborhood or family of plausible threats rather than a single nominal one (Sinha et al., 2018).

Expected Fisher-geometric anti-copying energy. Recall from Section 3.4 that for a fixed student S with parameters θ_S, the Fisher-geometric learnability of a harvested trace (x, y) is measured by

$$\mathcal{G}_S(x, y) := g_S(\theta_S; x, y)^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, y),$$

where g_S is the imitation-loss gradient and F_S is the Fisher information matrix. The most natural multi-student extension is to average this quantity over the student family:

$$E_x^{\text{avg}}(y) = \mathbb{E}_{S\sim\mathcal{Q}_S}[\mathcal{G}_S(x, y)] = \mathbb{E}_{S\sim\mathcal{Q}_S}\left[g_S(\theta_S; x, y)^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, y)\right].$$

This defines the ***expected anti-copying energy*** of output y under prompt x: a response is penalized when it is broadly learnable across many plausible student instantiations, not merely when it is informative to one specific proxy.

Worst-case Fisher-geometric anti-copying energy. A stronger, minimax-style alternative is to define

$$E_x^{\mathrm{wc}}(y) = \sup_{S \in \mathcal{S}} \mathcal{G}_S(x, y) = \sup_{S \in \mathcal{S}} \left[g_S(\theta_S; x, y)^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, y) \right].$$

Under this formulation, the defender treats an output as dangerous if *any* plausible student can learn rapidly from it. The resulting decoding rule prioritizes responses that are hard to distill even for the strongest or most compatible attacker in the uncertainty set. Conceptually, this is the most conservative form of teacher-side protection.

Average-case vs. worst-case robustness. The two formulations reflect different operational philosophies:

- ***Average-case robustness*** is appropriate when the defender has a meaningful prior over likely attacker pipelines and wishes to optimize expected resistance.
- ***Worst-case robustness*** is appropriate when the defender wants guarantees against the strongest plausible student in a threat set.

In practice, these choices induce different tradeoffs. The average-case objective is smoother, lower-variance, and easier to estimate using a finite proxy ensemble. The worst-case objective is more conservative, but can be numerically unstable and overly pessimistic if $\mathcal{S}$ is very broad. In this paper, we adopt the expectation form as the primary instantiation of FROST and treat the worst-case form as a conceptual upper envelope.

Robust FROST objective. Substituting the expected multi-student energy into the free-energy formulation of Section 3.3 yields the robust defended distribution

$$Q^\star(y \mid x) \propto P_0(y \mid x) \exp\Big(- \beta E_x^{\mathrm{avg}}(y) \Big) = P_0(y \mid x) \exp\Big(- \beta \, \mathbb{E}_{S \sim \mathcal{Q}_S}[\mathcal{G}_S(x, y)] \Big).$$

Equivalently, the prompt-local variational problem becomes

$$Q^\star(\cdot \mid x) = \arg\min_{Q(\cdot|x)} \left\{ \mathbb{E}_{y \sim Q} \left[\mathbb{E}_{S \sim \mathcal{Q}_S}[\mathcal{G}_S(x, y)] \right] + \frac{1}{\beta} D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)) \right\}.$$

This is the robust decoding rule implemented by FROST. It says that a response should be downweighted whenever it offers ***high intrinsic learning progress across the attacker family as a whole***.

A minimax interpretation. The worst-case version admits a more explicitly adversarial form:

$$Q_{\mathrm{wc}}^\star(\cdot \mid x) = \arg\min_{Q(\cdot|x)} \left\{ \mathbb{E}_{y \sim Q} \left[\sup_{S \in \mathcal{S}} \mathcal{G}_S(x, y) \right] + \frac{1}{\beta} D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)) \right\}.$$

This objective can be read as a two-player game:

- the ***defender*** chooses a response distribution Q;
- the ***attacker*** chooses the student S in $\mathcal{S}$ that extracts the most learning value from the emitted trace.

The defender then minimizes the worst achievable student progress subject to remaining close to the nominal teacher. This places FROST in the broader conceptual family of robust and adversarial optimization, but at the ***output-channel level*** rather than the input or parameter level (Sinha et al., 2018).

Finite proxy ensembles. In practice, neither the expectation nor the supremum can be computed over an infinite student family. FROST therefore approximates $\mathcal{Q}_S$ using a finite ensemble of proxy students

$$\{S^{(1)}, S^{(2)}, \ldots, S^{(M)}\},$$

chosen to cover meaningful variation in architecture, scale, and fine-tuning style. The expected energy is then approximated by

$$\widehat{E}_x^{\text{avg}}(y) = \frac{1}{M} \sum_{m=1}^{M} g_{S^{(m)}}(\theta_m; x, y)^\top F_{S^{(m)}}(\theta_m)^{-1} g_{S^{(m)}}(\theta_m; x, y).$$

This approximation has two practical virtues. First, it reduces variance relative to single-proxy scoring. Second, it directly aligns the defense with the evaluation protocol: if distillation is later tested on held-out attacker students, the proxy ensemble can be viewed as the training-time analogue of a robust validation set.

Why robustness matters empirically. Without multi-student robustness, the defense risks learning ***proxy-specific anti-signals*** rather than true anti-distillation structure. For example, an output that is hard for one proxy student to fit may still be highly learnable for another student whose tokenizer, inductive bias, or fine-tuning geometry is better aligned with the teacher. Robust averaging mitigates this pathology by favoring outputs whose low learnability is ***stable across students*** rather than accidental under one architecture. This is especially important in black-box extraction, where the defender almost never knows which student model an adversary will actually deploy (Orekondy et al., 2019; Zhao et al., 2025).

Robustness as smooth anti-transfer. There is also a geometric advantage to the expectation form. Averaging Fisher-quadratic forms across a student family induces a smoother energy landscape over candidate outputs:

$$E_x^{\text{avg}}(y) = \mathbb{E}_{S \sim \mathcal{Q}_S}[g_S^\top F_S^{-1} g_S].$$

Smoother energies are easier to optimize in practice and less prone to brittle reweighting around artifacts of a single proxy model. In other words, multi-student robustness makes FROST less likely to overfit its defense to one local geometry, and more likely to suppress ***transferable learnability*** as such.

Relation to the broader FROST pipeline. This subsection upgrades FROST from a single-proxy defense to a ***robust teacher-side antidistillation framework***. The defense no longer asks: *"Is this output easy for one surrogate to learn from?"* Instead, it asks: ***"Is this output broadly learnable across the attacker family we care about?"*** That distinction is crucial for a deployment-time defense, because real attackers are free to choose whatever student best exploits the exposed output channel. The next subsection complements this robust geometric view with an information-theoretic one, showing how suppressing broad learnability also suppresses the effective leakage of latent capability-bearing structure.

2.6 Information-Theoretic Interpretation

Why an information-theoretic view is needed. The Fisher-geometric formulation in Section 3.4 explains ***how much first-order learning progress*** a harvested trace can induce in a downstream student. This is already a powerful notion of anti-copying. However, extraction attacks are not only about gradient progress; they are ultimately about ***recovering reusable capability-bearing structure*** from repeated black-box interaction. For this reason, it is useful to complement the geometric view with an information-theoretic one. The central question becomes:

How much information about the teacher's transferable capability is leaked by each emitted response?

This perspective is closely related to the broader ***information bottleneck*** principle, where one seeks representations that preserve task-relevant information while discarding unnecessary or dangerous detail (Tishby et al., 2000). It is also consistent with modern information-theoretic analyses of adaptive procedures, where mutual information quantifies how much can be learned from an interaction transcript (Russo and Zou, 2016; Steinke and Zakynthinou, 2020).

A latent capability variable. To make this concrete, introduce a latent random variable

$$W \in \mathcal{W},$$

which represents the ***capability-bearing structure*** the attacker wishes to recover from the teacher. The exact semantics of W may vary by setting. For example, W may encode:

- a reusable reasoning template,
- a latent task decomposition policy,
- a coding or tool-use scaffold,
- a family of evaluation heuristics, or
- a hidden structured behavior that distinguishes the teacher from weaker models.

The attacker never observes W directly. Instead, it interacts with the output channel and observes responses

$$Y_t \sim Q(\cdot \mid X_t),$$

where X_t is the current prompt. The information-theoretic security question is therefore not whether the attacker can read off W from a single response, but whether the full transcript reveals enough about W to support effective downstream imitation.

Per-query leakage. For a fixed prompt x, the most basic leakage quantity is

$$I(W; Y \mid X = x),$$

the mutual information between the latent capability variable and the emitted response, conditioned on the prompt. This quantity has a direct interpretation:

- if $I(W; Y \mid X = x)$ is ***large***, then the output reveals substantial information about the transferable structure the attacker cares about;
- if $I(W; Y \mid X = x)$ is ***small***, then the response is comparatively poor evidence about that structure.

From this perspective, a strong teacher-side defense should make the output channel ***informationally stingy*** with respect to W, while remaining behaviorally useful to legitimate users.

Transcript-level leakage under adaptive querying. The attacker, however, receives not one response but an adaptive transcript. Let

$$Z_B := (X_1, Y_1, \ldots, X_B, Y_B)$$

denote the full interaction transcript after B black-box queries. Since prompts may be chosen adaptively based on earlier responses, the relevant object is the mutual information

$$I(W; Z_B).$$

By the chain rule of mutual information,

$$I(W; Z_B) = \sum_{t=1}^{B} I(W; Y_t \mid X_t, Z_{t-1}),$$

where $Z_{t-1} = (X_1, Y_1, \ldots, X_{t-1}, Y_{t-1})$. This identity is exact and does not assume non-adaptivity: it simply decomposes total leakage into the sum of ***incremental per-query leakages*** (**?**Russo and Zou, 2016). Thus, even in an adaptive black-box attack, the transcript leaks information only query by query.

A leakage-budget interpretation. Suppose now that the defended teacher satisfies the prompt-local leakage bound

$$I(W; Y_t \mid X_t, Z_{t-1}) \leq \gamma \qquad \text{for all } t.$$

Then immediately,

$$I(W; Z_B) \leq B\gamma.$$

This inequality admits a simple interpretation: ***each query spends at most γ bits of capability leakage budget***. If FROST can reduce γ, then the attacker must spend proportionally more queries to acquire the same amount of information about W. The role of the defense is therefore not necessarily to drive leakage to zero, but to make each emitted trace ***low-bandwidth with respect to transferable capability***.

From leakage to query complexity via Fano. The operational consequence of low leakage is captured by ***Fano's inequality***, which relates mutual information to probability of identification error (**??**). Suppose the attacker's objective is to identify W among $|\mathcal{W}| = M$ possible hypotheses using the transcript Z_B, producing an estimate $\widehat{W}$. Then Fano's inequality implies

$$\Pr(\widehat{W} \neq W) \geq 1 - \frac{I(W; Z_B) + \log 2}{\log M}.$$

Combining this with the transcript leakage bound above yields

$$\Pr(\widehat{W} \neq W) \geq 1 - \frac{B\gamma + \log 2}{\log M}.$$

Equivalently, to drive the error probability below a fixed constant, the attacker must satisfy

$$B = \Omega\left(\frac{\log M}{\gamma}\right).$$

This gives the desired query-complexity interpretation: ***reducing per-query leakage increases the number of black-box interactions required for successful capability recovery.***

Proposition 3 (Query-complexity lower bound under bounded leakage). Let W be a latent capability-bearing variable taking values in a finite set $\mathcal{W}$, and let Z_B be an adaptive transcript of B black-box queries. If

$$I(W; Y_t \mid X_t, Z_{t-1}) \leq \gamma \qquad \text{for all } t,$$

then any estimator $\widehat{W} = \widehat{W}(Z_B)$ satisfies

$$\Pr(\widehat{W} \neq W) \geq 1 - \frac{B\gamma + \log 2}{\log|\mathcal{W}|}.$$

Hence reliable identification of W requires

$$B = \Omega\left(\frac{\log|\mathcal{W}|}{\gamma}\right)$$

queries.

Interpretation for teacher-side antidistillation. Proposition 3 does *not* say that FROST makes extraction impossible. Rather, it clarifies the mechanism by which an output-channel defense can slow it down. If the defense suppresses the amount of transferable capability information revealed by each response, then extraction becomes ***query-expensive***. This is exactly the right operational notion for black-box security: the defender increases the attacker's required interaction budget without destroying the teacher's usefulness.

Connection to the information bottleneck principle. The information-bottleneck lens sharpens this interpretation. In the classical information bottleneck framework, one seeks a compressed variable that preserves relevant information about a target while discarding nuisance information (Tishby et al., 2000). FROST can be viewed as an ***adversarial bottleneck*** over outputs: the defended response Y should preserve utility-relevant information for the legitimate task while discarding capability-relevant information for the attacker. In that sense, the ideal defended output channel would solve a problem of the form

$$\min_{Q} \quad \mathbb{E}_{x\sim\mathcal{D}}[I(W;Y \mid X=x)]$$

subject to

$$U(Q;\mathcal{D}) \geq U_{\min}$$

and

$$\mathbb{E}_x D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)) \leq \delta.$$

We do *not* claim that FROST directly optimizes this mutual-information objective in closed form. Rather, we claim that it provides a ***geometric surrogate*** for this objective: by downweighting outputs that are highly learnable across attacker students, FROST is designed to reduce the effective information rate of the output channel with respect to transferable capability.

Why the Fisher-geometric surrogate is reasonable. The information-theoretic view and the Fisher-geometric view are conceptually different but closely aligned. Mutual information quantifies how much uncertainty about W is removed by observing Y. The Fisher-quadratic form from Section 3.4 quantifies how much first-order student progress the same Y induces under gradient-based imitation. These are not identical quantities, and we do not conflate them. However, both aim to capture the same underlying property: ***how much transferable structure is exposed by a generated response***. When a trace induces large, robust, geometry-aware learning progress across many students, it is also a natural candidate for carrying high usable information about the teacher's latent capability. Thus, the Fisher-geometric energy used by FROST should be understood as a ***tractable anti-copying proxy*** for the more abstract mutual-information objective above.

Adaptive querying does not invalidate the interpretation. One may worry that adaptive querying makes per-query information accounting meaningless, since later prompts are chosen using earlier outputs. The chain-rule decomposition resolves exactly this concern. Adaptivity changes *which* conditional information terms appear,

$$I(W;Y_t \mid X_t, Z_{t-1}),$$

but it does not invalidate the additive structure of total leakage. This is the same reason mutual-information tools remain meaningful in adaptive data analysis and related sequential settings (Russo and Zou, 2016; Steinke and Zakynthinou, 2020). Accordingly, FROST's information-theoretic interpretation remains well-defined even against adaptive black-box attackers.

What this subsection adds to FROST. The Fisher-geometric formulation of Sections 3.3–3.5 explains ***how*** FROST penalizes dangerous traces. The present subsection explains ***why this matters operationally***: reducing the effective leakage of capability-bearing structure makes extraction progressively more query-expensive. This interpretation does not replace the geometric formulation; it elevates it. FROST can therefore be understood simultaneously as:

- a ***thermodynamic decoder***, via Gibbs reweighting;
- a ***Fisher-geometric defense***, via natural-gradient learnability suppression; and
- an ***information-theoretic bottleneck***, via reduced per-query leakage of transferable capability.

The next subsection returns to the algorithmic side and shows how these principles are instantiated in the composite FROST energy used in practice.

2.7 Composite FROST Energy

Why a composite energy is necessary. Sections 3.3–3.6 establish the conceptual foundation of FROST. Section 3.3 shows that defended decoding should be written as a Gibbs-reweighted variational problem. Section 3.4 defines the central anti-copying signal through ***Fisher-geometric student learnability***. Section 3.6 complements this with an information-theoretic interpretation in terms of ***per-query leakage of transferable capability***. However, a deployable teacher-side defense must ultimately operate in a richer response space than any single scalar criterion can capture. In practice, outputs are dangerous not only because they induce large first-order student progress, but also because they may expose ***highly reusable structural scaffolds*** such as explicit decompositions, long reasoning traces, code skeletons, and stepwise templates. Conversely, some applications also benefit from preserving ***provenance-supporting structure*** such as watermark compatibility, even if provenance alone is insufficient as a primary defense (Savani et al., 2025; Kirchenbauer et al., 2023; Dathathri et al., 2024). For this reason, FROST is instantiated through a ***composite energy*** that separates the *core anti-copying geometry* from *auxiliary structural and provenance controls*.

General form of the energy. For a prompt x and candidate response y, we define the defended energy as

$$E_x(y) = \lambda_{\text{geom}}\, E_x^{\text{geom}}(y) + \lambda_{\text{struct}}\, E_x^{\text{struct}}(y) + \lambda_{\text{prov}}\, E_x^{\text{prov}}(y),$$

where:

- $E_x^{\text{geom}}(y)$ is the ***Fisher-geometric anti-copying term***;
- $E_x^{\text{struct}}(y)$ is the ***structural trace-risk term***;
- $E_x^{\text{prov}}(y)$ is the ***provenance-aware shaping term***;
- $\lambda_{\text{geom}}, \lambda_{\text{struct}}, \lambda_{\text{prov}} \geq 0$ are tunable weights.

This decomposition is deliberate. It lets FROST express a clear priority ordering:

intrinsic learnability suppression $>$ ***structural hardening*** $>$ ***provenance support***.

Accordingly, in our formulation λ_{geom} is the principal coefficient, while the remaining terms play secondary, regularizing roles.

Core term: Fisher-geometric anti-copying energy. The principal term is inherited directly from Section 3.5:

$$E_x^{\text{geom}}(y) = \mathbb{E}_{S\sim\mathcal{Q}_S}\left[g_S(\theta_S; x, y)^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, y)\right].$$

This term measures the expected first-order natural-gradient progress induced by (x, y) across a family of plausible attacker students. It is the mathematical heart of FROST. A response receives a high geometric energy if it is broadly useful as supervision across attacker students; it receives a low geometric energy if it is comparatively inert as training signal. All other terms in the composite energy should therefore be understood as refinements around this core notion, not substitutes for it.

Why geometry alone is not always enough. Although the Fisher-geometric term is the most principled quantity in the framework, it is not always the whole story. In black-box extraction, attackers benefit disproportionately from outputs that are ***pedagogically explicit***: responses that expose reusable decomposition patterns, chain-like templates, code scaffolds, or highly regular reasoning layouts. Antidistillation Sampling is motivated by a closely related observation, namely that structured reasoning traces can be especially valuable for downstream distillation (Savani et al., 2025). This means that some traces may be dangerous not only because they induce large intrinsic gradient progress, but also because they expose ***portable structural templates*** that survive across students and fine-tuning regimes. To capture this phenomenon, FROST adds a separate structural term.

Structural trace-risk energy. We define

$$E_x^{\text{struct}}(y)$$

as a penalty on ***structurally reusable supervision***. In its most general form, this term is left abstract and implemented through measurable features of the generated response. Examples include:

- ***trace length***: long-form derivations, extended chain-like explanations, or verbose decompositions;
- ***template explicitness***: numbered procedures, algorithmic stages, recipe-style instructions, or systematic subgoal lists;
- ***code scaffold exposure***: reusable function skeletons, boilerplate, tool-call templates, or partially executable structures;
- ***structural regularity***: highly canonical response formats that compress task structure into a transferable template.

A generic parameterization is

$$E_x^{\text{struct}}(y) = \sum_{j=1}^{J} \alpha_j \, \phi_j(x, y),$$

where each $\phi_j(x, y)$ is a structural feature and $\alpha_j \geq 0$ is a learned or tuned coefficient. This formulation keeps FROST flexible: the structural term can be adapted to the domain without changing the geometric core.

Interpretation of the structural term. The structural term does ***not*** replace the Fisher-geometric term; rather, it complements it by targeting a different failure mode. The geometric term asks:

How much intrinsic learning progress does this trace induce?

The structural term asks:

How easily can this trace be reused as a portable supervision template?

These are related but distinct questions. A response can have moderate local gradient impact yet still reveal a high-value structured scaffold. Conversely, a response may induce strong local gradients without exposing much reusable formatting. The composite energy is designed precisely to cover both cases.

Optional provenance-aware term. In some deployments, the defender may also want outputs to remain compatible with provenance mechanisms such as watermarking. Prior work on watermarking for large language models and SynthID-style text watermarking shows that sampling-time modifications can provide attribution signal while preserving output quality under their evaluation settings (Kirchenbauer et al., 2023; Dathathri et al., 2024). However, watermarking alone is not a sufficient antidistillation defense, because detectable outputs may still carry high-value supervision for downstream students. For this reason, provenance enters FROST only as an ***optional third term***:

$$E_x^{\text{prov}}(y).$$

Depending on deployment needs, this term may either reward watermark-compatible outputs or penalize outputs that are likely to break attribution-supporting structure. A generic instantiation is

$$E_x^{\text{prov}}(y) = -\psi_{\text{prov}}(x, y),$$

where ψ_{prov} is a provenance score. The negative sign reflects the fact that provenance support is usually *desirable*; thus, in the total energy, outputs that preserve provenance can be relatively favored.

Normalized defended decoding rule. Substituting the composite energy into the free-energy objective from Section 3.3 yields the defended distribution

$$Q^\star(y \mid x) = \frac{1}{Z(x)} P_0(y \mid x) \exp\Big(- \beta[\lambda_{\text{geom}} E_x^{\text{geom}}(y) + \lambda_{\text{struct}} E_x^{\text{struct}}(y) + \lambda_{\text{prov}} E_x^{\text{prov}}(y)]\Big),$$

where the partition function is

$$Z(x) = \sum_{y \in \mathcal{Y}} P_0(y \mid x) \exp\Big(- \beta[\lambda_{\text{geom}} E_x^{\text{geom}}(y) + \lambda_{\text{struct}} E_x^{\text{struct}}(y) + \lambda_{\text{prov}} E_x^{\text{prov}}(y)]\Big).$$

This expression is the full defended decoding law implemented by FROST. It reveals that the method is not a monolithic single-score defense, but a ***composable Gibbs controller over multiple dimensions of extraction risk***.

Reading the coefficients. The coefficients have a clear semantics:

- λ_{geom}: ***intrinsic anti-learning strength***; this is the primary knob governing capability-transfer suppression.
- λ_{struct}: ***trace hardening strength***; this controls how strongly the defense avoids portable response scaffolds.
- λ_{prov}: ***provenance preference strength***; this supports attribution or watermark compatibility when desired.

The inverse temperature β continues to control the overall defense strength, while the λ-coefficients allocate that strength across mechanisms. In practice, this separation is useful: it allows the defender to ask not only *how much* protection to apply, but also *which kind*.

Ablation-friendly design. The composite decomposition is also crucial experimentally. Because each term has a distinct operational meaning, FROST supports principled ablations:

$$E_x(y) = E_x^{\text{geom}}(y)$$

tests the geometry-only defense;

$$E_x(y) = E_x^{\text{geom}}(y) + \lambda_{\text{struct}} E_x^{\text{struct}}(y)$$

tests whether structural hardening provides additional benefit; and

$$E_x(y) = E_x^{\text{geom}}(y) + \lambda_{\text{prov}} E_x^{\text{prov}}(y)$$

tests whether provenance compatibility can be layered on without damaging anti-copying performance. This is important for reviewer-facing evaluation: it allows us to separate the contribution of the ***core Fisher-geometric principle*** from secondary design choices.

Why this subsection is central to FROST. Conceptually, this subsection is where FROST becomes a ***framework*** rather than a single theorem. The earlier subsections establish:

- a ***thermodynamic variational form*** for defended decoding,
- a ***Fisher-geometric*** notion of student learnability,
- a ***robust*** multi-student extension, and
- an ***information-theoretic*** interpretation of reduced leakage.

The present subsection combines these insights into the actual energy used at inference time. In short, ***FROST protects by making emitted traces simultaneously harder to learn from, harder to reuse as templates, and optionally easier to attribute.*** The next subsection turns this composite energy into a practical autoregressive decoding algorithm.

2.8 Autoregressive Approximation and Decoding Algorithm

Why an autoregressive approximation is necessary. The defended distribution derived in Section 3.7 is defined over ***entire responses***:

$$Q^\star(y \mid x) \propto P_0(y \mid x) \exp(-\beta E_x(y)),$$

where $E_x(y)$ is the composite FROST energy. This form is mathematically clean, but it is not directly usable for large language models. In an autoregressive model, the response space $\mathcal{Y}$ is combinatorial, and exact normalization over full sequences is intractable. Thus, as in other inference-time control methods for language models, one must replace the sequence-level objective by a ***tokenwise approximation*** that can be evaluated during decoding. Controlled Decoding follows this general strategy for reward-style control, and Antidistillation Sampling similarly modifies next-token probabilities using a student-aware signal during generation. FROST adopts the same autoregressive principle, but with a different target: ***reduce composite anti-copying energy online during generation*** (Mudgal et al., 2024; Savani et al., 2025).

Nominal autoregressive teacher. Let the nominal teacher factorize as

$$P_0(y \mid x) = \prod_{t=1}^{T} P_0(y_t \mid x, y_{<t}),$$

where $y = (y_1, \ldots, y_T)$ is the generated sequence and $y_{<t} = (y_1, \ldots, y_{t-1})$ is the current prefix. The defended decoder should ideally sample from

$$Q^\star(y \mid x) \propto \left(\prod_{t=1}^{T} P_0(y_t \mid x, y_{<t}) \right) \exp(-\beta E_x(y)).$$

The difficulty is that the energy $E_x(y)$ depends on the *entire future completion*, which is not available at step t. FROST therefore introduces a prefix-conditioned surrogate energy that estimates the future anti-copying cost of choosing a token now.

Prefix-conditioned energy-to-go. For a prompt x, current prefix $y_{<t}$, and candidate next token $v \in \mathcal{V}$, define the ***energy-to-go***

$$\mathcal{E}_t(v; x, y_{<t}) := \mathbb{E}_{Y_{>t} \sim \Pi(\cdot \mid x, y_{<t}, v)}[E_x(y_{<t} \oplus v \oplus Y_{>t})],$$

where $\Pi(\cdot \mid x, y_{<t}, v)$ is a rollout distribution over suffixes after fixing the next token to v. If this quantity were tractable, the exact defended next-token rule would be

$$q_t^\star(v \mid x, y_{<t}) \propto P_0(v \mid x, y_{<t}) \exp(-\beta \mathcal{E}_t(v; x, y_{<t})).$$

However, exact evaluation of $\mathcal{E}_t$ is too expensive in general.

Local surrogate energy. FROST therefore replaces the exact energy-to-go with a computable local surrogate

$$\widetilde{E}_t(v; x, y_{<t}) = \lambda_{\text{geom}} \widetilde{E}_t^{\text{geom}}(v; x, y_{<t}) + \lambda_{\text{struct}} \widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) + \lambda_{\text{prov}} \widetilde{E}_t^{\text{prov}}(v; x, y_{<t}).$$

Each term is a token-local approximation of the corresponding sequence-level component from Section 3.7.

Illustrative example. Consider the prompt:

> *"Explain how to solve a quadratic equation."*

Suppose the current prefix is

$$y_{<t} = \texttt{``To solve a quadratic equation,''}.$$

Under the nominal teacher, the next-token shortlist may contain candidates such as:

$$v_1 = \texttt{`` first''}, \qquad v_2 = \texttt{`` use''}, \qquad v_3 = \texttt{`` Step''}.$$

These tokens induce very different future continuations:

- $v_1 =$ ``first'' likely leads to a stepwise decomposition;
- $v_2 =$ ``use'' likely leads to a compact direct explanation;
- $v_3 =$ ``Step'' strongly commits to an explicit pedagogical template.

Although all three continuations may be helpful, v_3 is often the most ***distillable*** because it begins a reusable structured scaffold (e.g., "Step 1, Step 2, Step 3"), while v_2 may yield a more compact explanation with lower downstream training value. FROST therefore assigns different local energies:

$$\widetilde{E}_t(v_3; x, y_{<t}) > \widetilde{E}_t(v_1; x, y_{<t}) > \widetilde{E}_t(v_2; x, y_{<t}),$$

and downweights the more pedagogically reusable continuation even if it is nominally probable under the teacher. In this way, FROST does not suppress usefulness; it suppresses ***excessively teachable structure***.

Token-local Fisher-geometric term. The most important component is the geometric term. For a candidate next token v, define the partially committed sequence

$$\tilde{y}_t^{(v)} := y_{<t} \oplus v.$$

Then the token-local geometric score is approximated by

$$\widetilde{E}_t^{\text{geom}}(v; x, y_{<t}) = \mathbb{E}_{S \sim \mathcal{Q}_S} \left[g_S(\theta_S; x, \tilde{y}_t^{(v)})^\top F_S(\theta_S)^{-1} g_S(\theta_S; x, \tilde{y}_t^{(v)}) \right].$$

Intuitively, this asks:

if the teacher commits to token v now, how much intrinsic learning progress does that commitment already create for likely students?

Token-local structural term. The structural component is approximated incrementally:

$$\widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) = \sum_{j=1}^{J} \alpha_j \, \widetilde{\phi}_{j,t}(v; x, y_{<t}),$$

where $\widetilde{\phi}_{j,t}$ are token- or prefix-level indicators of emerging structural danger. Examples include:

- whether v begins a numbered list or explicit multi-step template,
- whether v extends a long chain-like rationale,
- whether v increases syntactic commitment to code scaffolding,
- whether v pushes the response toward highly canonical reusable structure.

In the quadratic-equation example above, the token ``Step'' would likely receive a higher structural penalty than ``use'', because it explicitly begins a portable response template.

Token-local provenance term. If provenance-aware shaping is enabled, FROST also includes

$$\widetilde{E}_t^{\text{prov}}(v; x, y_{<t}),$$

which rewards or penalizes tokens according to their compatibility with the desired provenance behavior. In some deployments this may reflect watermark-aligned token preferences; in others, it may simply be omitted by setting $\lambda_{\text{prov}} = 0$. As in Section 3.7, provenance remains an ***auxiliary term***, not the primary defense mechanism (Kirchenbauer et al., 2023; Dathathri et al., 2024).

Autoregressive defended decoding rule. Given the local surrogate energy, FROST defines the defended next-token distribution at step t as

$$q_t(v \mid x, y_{<t}) = \frac{1}{Z_t(x, y_{<t})} P_0(v \mid x, y_{<t}) \exp(-\beta \widetilde{E}_t(v; x, y_{<t})),$$

where

$$Z_t(x, y_{<t}) = \sum_{u \in \mathcal{V}} P_0(u \mid x, y_{<t}) \exp(-\beta \widetilde{E}_t(u; x, y_{<t}))$$

is the stepwise partition function. Equivalently, if $\ell_t(v)$ denotes the nominal logit for token v, then FROST induces the logit transformation

$$\ell_t^{\mathrm{FROST}}(v) = \ell_t(v) - \beta \widetilde{E}_t(v; x, y_{<t}).$$

Illustrative token reweighting. Continuing the same example, suppose the nominal teacher assigns:

$$P_0(v_1 \mid x, y_{<t}) = 0.32, \quad P_0(v_2 \mid x, y_{<t}) = 0.28, \quad P_0(v_3 \mid x, y_{<t}) = 0.25,$$

for

$$v_1 = \texttt{``first''}, \quad v_2 = \texttt{``use''}, \quad v_3 = \texttt{``Step''}.$$

Now suppose FROST estimates:

$$\widetilde{E}_t(v_1) = 0.8, \qquad \widetilde{E}_t(v_2) = 0.3, \qquad \widetilde{E}_t(v_3) = 1.4,$$

with $\beta = 1$. Then the defended unnormalized scores become

$$0.32e^{-0.8}, \qquad 0.28e^{-0.3}, \qquad 0.25e^{-1.4}.$$

Even though ``Step'' was a strong nominal candidate, it is sharply downweighted because it commits the model to a highly reusable scaffold. By contrast, ``use'' becomes relatively more likely, because it remains helpful while offering lower anti-copying energy. This simple example illustrates the central operational logic of FROST:

preserve helpfulness, but suppress excessively distillable structure.

Top-k candidate restriction. Evaluating $\widetilde{E}_t(v; x, y_{<t})$ for the full vocabulary is often unnecessary. In practice, FROST restricts attention to a shortlist

$$\mathcal{V}_t^{(k)} \subseteq \mathcal{V}$$

of the top-k tokens under the nominal teacher distribution. The defended distribution is then computed only over this set. This approximation is natural because the teacher already assigns negligible mass to most tokens, and because restricting to plausible candidates preserves the minimal-intervention character of the defense.

One-step vs. lookahead scoring. There are multiple practical choices for constructing $\widetilde{E}_t$:

- ***One-step local scoring*** uses only the immediate token commitment $y_{<t} \oplus v$.
- ***Short-horizon lookahead*** rolls out one or a few additional steps under the nominal teacher and averages the resulting sequence-level energy.
- ***Prefix-feature approximation*** uses lightweight prefix statistics as surrogates for future structural or provenance effects.

In this paper, we adopt one-step scoring for the geometric term and lightweight prefix-feature approximations for auxiliary terms, as this provides the best balance between defense strength and deployment cost.

Algorithm. The defended decoding procedure is summarized below.

Algorithm: FROST Autoregressive Decoding

Input: Prompt x; nominal teacher model P_0; proxy student ensemble $\mathcal{Q}_S = \{S^{(1)}, \ldots, S^{(M)}\}$; structural feature functions $\{\widetilde{\phi}_{j,t}\}_{j=1}^{J}$; optional provenance scorer ψ_{prov}; coefficients $\lambda_{\text{geom}}, \lambda_{\text{struct}}, \lambda_{\text{prov}}$; inverse temperature β; shortlist size k; maximum decoding length T

Output: Defended response $y = (y_1, \ldots, y_L)$ with $L \leq T$

Initialize prefix $y_{<1} \leftarrow \emptyset$
Initialize termination flag `stop` $\leftarrow$ False
for $t = 1, 2, \ldots, T$ **do**
  if `stop` $=$ True **then**
    break
  end if
  Compute nominal teacher logits $\ell_t(v)$ for all $v \in \mathcal{V}$
  Construct nominal next-token distribution

$$P_0(v \mid x, y_{<t}) = \text{softmax}(\ell_t(v))$$

  Form shortlist $\mathcal{V}_t^{(k)}$ of the top-k tokens under $P_0(\cdot \mid x, y_{<t})$
  for each candidate token $v \in \mathcal{V}_t^{(k)}$ **do**
    Form partial candidate prefix

$$\tilde{y}_t^{(v)} \leftarrow y_{<t} \oplus v$$

// Step 1: Fisher-geometric anti-copying score

    Initialize geometric accumulator $A_t^{\text{geom}}(v) \leftarrow 0$
    for $m = 1, 2, \ldots, M$ **do**
      Compute imitation-loss gradient under proxy student $S^{(m)}$

$$g_t^{(m)}(v) \leftarrow \nabla_{\theta_m}\big(-\log p_{\theta_m}(\tilde{y}_t^{(v)} \mid x)\big)$$

      Compute or retrieve Fisher approximation F_m^{-1} for student $S^{(m)}$
      Accumulate natural-gradient progress

$$A_t^{\text{geom}}(v) \leftarrow A_t^{\text{geom}}(v) + g_t^{(m)}(v)^{\top} F_m^{-1} g_t^{(m)}(v)$$

    end for
    Set ensemble-averaged geometric score

$$\widetilde{E}_t^{\text{geom}}(v; x, y_{<t}) \leftarrow \frac{1}{M} A_t^{\text{geom}}(v)$$

// Step 2: Structural trace-risk score

    Initialize structural score $\widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) \leftarrow 0$
    for $j = 1, 2, \ldots, J$ **do**
      Evaluate prefix-level structural feature

$$r_{j,t}(v) \leftarrow \widetilde{\phi}_{j,t}(v; x, y_{<t})$$

      Accumulate weighted structural penalty

$$\widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) \leftarrow \widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) + \alpha_j r_{j,t}(v)$$

    end for

// Step 3: Optional provenance-aware score

    if $\lambda_{\text{prov}} > 0$ **then**
      Evaluate provenance compatibility

$$\widetilde{E}_t^{\text{prov}}(v; x, y_{<t}) \leftarrow -\psi_{\text{prov}}(\tilde{y}_t^{(v)}, x)$$

else

Set $\widetilde{E}_t^{\text{prov}}(v; x, y_{<t}) \leftarrow 0$

end if

// Step 4: Composite FROST energy

Combine component scores

$$\widetilde{E}_t(v; x, y_{<t}) \leftarrow \lambda_{\text{geom}} \widetilde{E}_t^{\text{geom}}(v; x, y_{<t}) + \lambda_{\text{struct}} \widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) + \lambda_{\text{prov}} \widetilde{E}_t^{\text{prov}}(v; x, y_{<t})$$

// Step 5: Logit correction

Reweight nominal teacher logit

$$\ell_t^{\text{FROST}}(v) \leftarrow \ell_t(v) - \beta\, \widetilde{E}_t(v; x, y_{<t})$$

end for

Restrict normalization to $v \in \mathcal{V}_t^{(k)}$ and compute defended distribution

$$q_t(v \mid x, y_{<t}) = \frac{\exp(\ell_t^{\text{FROST}}(v))}{\sum_{u \in \mathcal{V}_t^{(k)}} \exp(\ell_t^{\text{FROST}}(u))}$$

Sample or decode next token

$$y_t \sim q_t(\cdot \mid x, y_{<t})$$

Append token to prefix

$$y_{<t+1} \leftarrow y_{<t} \oplus y_t$$

if y_t is an EOS token **then**

Set `stop` $\leftarrow$ True

end if

end for

Let L be the first position of EOS if present; otherwise set $L \leftarrow T$

return $y = (y_1, \ldots, y_L)$

Interpretation. This subsection converts FROST from a conceptual defense into a concrete inference-time mechanism. The resulting decoder may be viewed in three equivalent ways:

- as a ***logit correction rule*** that subtracts anti-copying energy from token scores;
- as a ***tokenwise Gibbs posterior*** over the teacher's top candidates;
- as an ***online control policy*** that suppresses learnable and structurally reusable traces as they emerge.

2.9 Complexity and Practical Approximation

Why exact FROST is computationally infeasible. The sequence-level defended law derived in Section 3.7 is

$$Q^\star(y \mid x) \propto P_0(y \mid x) \exp(-\beta E_x(y)),$$

where the energy $E_x(y)$ itself depends on the expected Fisher-geometric learnability of the emitted trace across a family of attacker students. In exact form, this objective is prohibitively expensive for three independent reasons. First, normalization over the full response space is combinatorial, which already necessitates the autoregressive approximation of Section 3.8. Second, the geometric term requires evaluating gradients under multiple proxy students, rather than under the teacher alone. Third, the natural-gradient metric involves the inverse Fisher information matrix, whose exact computation is well known to be impractical in large neural models (Amari, 1998; **?**). Thus, the challenge is not merely to define FROST mathematically, but to identify a sequence of approximations that preserve its ***variational meaning*** while making defended decoding deployable.

Four sources of cost. At decoding step t, the naive implementation of FROST would incur four distinct costs:

1. ***teacher decoding cost***: computing nominal logits $\ell_t(v)$ over the vocabulary;
2. ***candidate scoring cost***: evaluating the composite energy for each candidate token;
3. ***proxy-student cost***: computing gradient-based learnability scores across the student ensemble;
4. ***curvature cost***: estimating or inverting the Fisher metric for each proxy student.

If implemented naively over the full vocabulary and over an exact Fisher matrix, this cost would be too high for any realistic frontier-model deployment. FROST therefore uses a layered approximation strategy in which each source of complexity is reduced by a separate approximation, while the overall decoding rule remains a Gibbs reweighting of the nominal teacher.

Approximation 1: top-k candidate restriction. The first and most important approximation is to evaluate defended energies only on a nominal shortlist

$$\mathcal{V}_t^{(k)} \subseteq \mathcal{V}$$

containing the top-k teacher candidates at step t. This reduces the number of energy evaluations from $|\mathcal{V}|$ to k, where typically $k \ll |\mathcal{V}|$. The justification is both computational and statistical. Computationally, most tokens already receive negligible nominal probability and do not materially affect defended decoding. Statistically, restricting to high-probability candidates keeps FROST in the ***minimal-intervention regime***: the defense perturbs the teacher's existing plausible continuations rather than forcing unlikely tokens into the support. This is closely aligned with prior tokenwise control methods, which likewise operate by reweighting a manageable candidate set rather than globally searching the sequence space (Mudgal et al., 2024; Savani et al., 2025).

Approximation 2: finite proxy ensemble. The robust student family $\mathcal{Q}_S$ from Section 3.5 is represented in practice by a finite ensemble

$$\mathcal{Q}_S \approx \{S^{(1)}, \ldots, S^{(M)}\}.$$

This replaces an intractable expectation over attacker space by a finite average:

$$E_t^{\text{geom}}(v) \approx \frac{1}{M} \sum_{m=1}^{M} g_t^{(m)}(v)^\top F_m^{-1} g_t^{(m)}(v).$$

The ensemble should be chosen to cover variation in architecture, scale, and fine-tuning style, but M need not be large. In practice, a small but diverse ensemble is preferable to a large homogeneous one, because the purpose of the proxy family is not to average over noise, but to span plausible attacker geometries. This is the same high-level logic used in robustification schemes more broadly: coverage of uncertainty is more important than brute-force multiplicity.

Approximation 3: tractable Fisher inverse. The most delicate approximation concerns the Fisher inverse. In exact natural-gradient form, each proxy student would require

$$F_m(\theta_m)^{-1},$$

which is intractable for modern models. Following the broader natural-gradient literature, FROST therefore uses structured approximations to the Fisher matrix. At the lightest end, one may use a diagonal approximation, yielding

$$g_t^{(m)}(v)^\top F_m^{-1} g_t^{(m)}(v) \approx \sum_i \frac{g_{t,i}^{(m)}(v)^2}{[F_m]_{ii} + \varepsilon}.$$

This is cheap and stable, but coarse. A stronger alternative is a block-diagonal approximation over layers or modules, which preserves more curvature structure. A still richer approximation is ***Kronecker-factored approximate curvature (K-FAC)***, which approximates large Fisher blocks by Kronecker products of smaller factors and was introduced precisely as a tractable approximation to natural-gradient updates in neural networks (**?**). In FROST, the role of these approximations is not to optimize the student, but to estimate how much a candidate teacher token would help the student learn. This is a simpler use case than full second-order training, and therefore somewhat more forgiving computationally.

Approximation 4: prefix-local scoring instead of full rollouts. The ideal token score from Section 3.8 is an energy-to-go:

$$\mathcal{E}_t(v; x, y_{<t}) = \mathbb{E}_{Y_{>t}}[E_x(y_{<t} \oplus v \oplus Y_{>t})].$$

Computing this exactly would require rollout over future suffixes. FROST therefore uses a prefix-local surrogate

$$\widetilde{E}_t(v; x, y_{<t}),$$

which scores only the immediate commitment $y_{<t} \oplus v$, possibly augmented by lightweight prefix features. This approximation sacrifices exact optimality, but preserves the ***sign structure*** of the variational objective: tokens with higher estimated anti-copying energy are always exponentially down-weighted. As in Controlled Decoding and Antidistillation Sampling, the goal is not exact sequence-level inference, but a tractable tokenwise controller that remains faithful to the intended objective (Mudgal et al., 2024; Savani et al., 2025).

Approximation 5: lightweight structural and provenance terms. The auxiliary structural and provenance terms are designed to be cheap. Unlike the geometric term, they do not require backpropagation through proxy students. Instead, they are computed from prefix-local features:

$$\widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) = \sum_{j=1}^{J} \alpha_j \widetilde{\phi}_{j,t}(v; x, y_{<t}),$$

and, if enabled,

$$\widetilde{E}_t^{\text{prov}}(v; x, y_{<t}) = -\psi_{\text{prov}}(x, y_{<t} \oplus v).$$

These terms therefore add only $O(J)$ feature cost per candidate token, which is usually negligible relative to teacher forward passes and proxy gradient scoring. This separation is intentional: FROST concentrates its expensive computation on the ***core Fisher-geometric signal*** and keeps the auxiliary terms light enough to remain practical.

Resulting per-step complexity. Let C_{teach} denote the cost of one teacher forward step, C_{grad} the cost of computing one proxy gradient score for one candidate token, and C_{feat} the cost of structural/provenance feature evaluation. Then the per-step cost of FROST can be expressed schematically as

$$O\Big(C_{\text{teach}} + k(M\, C_{\text{grad}} + C_{\text{feat}})\Big),$$

where k is the shortlist size and M is the number of proxy students. Over a sequence of length T, the total defended decoding cost is

$$O\Big(T\, C_{\text{teach}} + T\, k(M\, C_{\text{grad}} + C_{\text{feat}})\Big).$$

This expression makes the tradeoff explicit: the dominant additional cost comes from evaluating the geometric term over k candidate tokens and M proxy students. The auxiliary terms contribute comparatively little.

Caching and amortization. Several parts of this cost can be amortized in practice. First, Fisher approximations F_m^{-1} for proxy students need not be recomputed at every decoding step; they can be refreshed periodically or precomputed from representative data, much as approximate curvature factors are reused across optimization steps in natural-gradient methods (**?**). Second, prefix-local structural features can be updated incrementally rather than recomputed from scratch. Third, if the proxy ensemble is fixed, many intermediate quantities can be cached across prompts or batched across candidate tokens. These implementation choices do not alter the defended objective; they simply reduce constant factors.

Approximation hierarchy and practical regimes. FROST admits several operating regimes:

- ***lightweight mode***: diagonal Fisher approximation, small proxy ensemble, small shortlist k;
- ***balanced mode***: block-diagonal or K-FAC-style Fisher approximation, moderate proxy ensemble, moderate shortlist;
- ***high-security mode***: richer curvature approximation, larger ensemble, and optional short-horizon lookahead.

This hierarchy is important because the optimal defense configuration depends on deployment constraints. High-throughput public APIs may prefer lightweight FROST, while high-value enterprise or frontier endpoints may justify stronger geometric scoring.

What is preserved despite approximation. Although these approximations are substantial, three properties of FROST remain intact:

1. ***variational structure***: defended decoding still takes the form of Gibbs reweighting of the nominal teacher;
2. ***anti-copying semantics***: higher estimated learnability still leads to lower defended probability;
3. ***minimal-intervention behavior***: top-k restriction keeps the defense close to the teacher's own plausible outputs.

Thus, the approximations are not arbitrary engineering shortcuts; they are carefully chosen reductions that preserve the conceptual identity of the method.

Relation to prior inference-time control methods. This practicality story is consistent with nearby work. Controlled Decoding explicitly shows that tokenwise controlled generation can be implemented through prefix scoring at inference time rather than through full sequence optimization (Mudgal et al., 2024). Antidistillation Sampling likewise demonstrates that a teacher-side defense can act directly on next-token generation to reduce downstream student performance while preserving practical utility (Savani et al., 2025). FROST differs in that its control signal is ***Fisher-geometric learnability*** rather than reward or proxy-loss change, but its practical implementation follows the same broad inference-time-control paradigm.

Bridge to evaluation. The purpose of this subsection is not to claim that FROST is free. It is to show that the method is ***implementable with structured approximations whose cost can be traded against security***. The evaluation section will therefore measure both sides of this tradeoff explicitly: ***(i)*** how much FROST reduces capability transfer under black-box distillation, and ***(ii)*** how much overhead and utility cost these approximations introduce relative to nominal decoding and prior teacher-side defenses.

2.10 Scope, Guarantees, and Limitations of the Objective

Why this clarification is necessary. FROST is intentionally formulated as a ***teacher-side objective over the output channel***. Its purpose is to reduce the downstream learnability of emitted traces while preserving the utility of the deployed teacher. Because this objective sits at the intersection of generation, geometry, and security, it is important to state precisely what the method does and does not claim. In particular, FROST should not be interpreted as a universal barrier against all

forms of model theft, nor as a replacement for platform-level monitoring, nor as a formal privacy guarantee. Rather, it is a principled ***output-shaping defense*** whose guarantees are tied to the specific quantities it optimizes: ***Fisher-geometric student learnability, composite anti-copying energy, and utility-constrained deviation from the nominal teacher***.

What FROST directly optimizes. At the most direct level, FROST optimizes the prompt-local free-energy objective

$$\mathcal{F}_\beta(Q; x) = \mathbb{E}_{y\sim Q(\cdot|x)}[E_x(y)] + \frac{1}{\beta} D_{\mathrm{KL}}(Q(\cdot \mid x) \,\|\, P_0(\cdot \mid x)),$$

with the composite energy $E_x(y)$ constructed to penalize outputs that are intrinsically learnable by downstream students. Accordingly, the method directly targets the following properties:

1. ***reduced student learnability*** of emitted traces;
2. ***reduced capability transfer efficiency*** under black-box harvesting;
3. ***controlled deviation*** from the nominal teacher distribution; and
4. ***preserved task utility*** up to the trust-region and decoding constraints imposed by the defense.

These are the primary, first-order claims of the framework. Any broader security interpretation should be understood as flowing from these controlled quantities rather than replacing them.

What the theoretical results do guarantee. Within the scope of the formalism developed in Sections 3.3–3.6, FROST provides three kinds of guarantees or near-guarantees.

(i) Variational optimality under the chosen energy. For a fixed prompt x and energy $E_x(y)$, the Gibbs-reweighted defended distribution

$$Q^\star(y \mid x) \propto P_0(y \mid x) e^{-\beta E_x(y)}$$

is the unique minimizer of the corresponding free-energy functional. This is an ***exact variational guarantee***: given the chosen anti-copying energy, FROST is the optimal minimal-intervention decoder in the KL-regularized sense.

(ii) First-order suppression of natural-gradient learning progress. When the geometric term is instantiated as

$$E_x^{\mathrm{geom}}(y) = \mathbb{E}_{S\sim\mathcal{Q}_S}\left[g_S^\top F_S^{-1} g_S\right],$$

FROST directly penalizes traces that induce large first-order natural-gradient progress in plausible attacker students. This does not prove that no student can learn from the teacher; rather, it guarantees that, under the modeling assumptions of Section 3.4, the defense is explicitly aligned against traces that are locally most useful for gradient-based imitation.

(iii) Query-complexity implications under bounded leakage. Under the information-theoretic interpretation of Section 3.6, if defended decoding reduces the per-query leakage of capability-bearing structure, then the number of black-box interactions required for successful recovery must increase. This is a ***conditional*** guarantee: it depends on the assumed latent variable W, the bounded-leakage condition, and the relevance of the information-theoretic proxy to the actual attacker objective. The claim is therefore not that FROST proves absolute extraction hardness, but that it provides a principled mechanism for making extraction ***progressively more query-expensive***.

What FROST does not guarantee. Equally important are the guarantees that FROST does ***not*** provide.

(i) No impossibility result for model extraction. FROST does not establish that black-box extraction is impossible, nor that a sufficiently capable and well-resourced attacker cannot eventually distill useful behavior from the teacher. The framework is designed to ***raise the cost of capability transfer***, not to prove unconditional security against arbitrary adversaries.

(ii) No formal privacy guarantee on memorized content. Although the broader paper is motivated in part by the fact that deployed outputs may leak data-like artifacts, FROST is not itself a formal privacy mechanism. It is ***not*** equivalent to differential privacy, does not provide a bounded privacy-loss parameter, and does not guarantee the non-disclosure of memorized training examples. Any reduction in data extraction risk should therefore be viewed as an indirect consequence of output shaping, not as a formal privacy certificate.

(iii) No replacement for platform-level defenses. FROST is designed to operate without account identity, telemetry, or coordination signals. This is a strength of the method, but it also defines its scope. The framework does not replace rate limiting, abuse detection, multi-account coordination analysis, provenance auditing, or contractual access controls. Instead, it should be understood as ***complementary*** to such measures: it protects the output channel itself, while platform defenses protect the surrounding deployment stack.

(iv) No claim of zero utility cost. Because FROST deliberately perturbs the teacher's output distribution, some tradeoff between protection and utility is unavoidable. The trust-region formulation and empirical evaluation are intended to keep this degradation small and controlled, but the framework does not claim that strong antidistillation can be achieved at literally zero cost. In particular, overly large β, overly aggressive structural penalties, or poor proxy-student calibration may induce undesirable response shifts.

Approximation-induced limitations. The practical implementation of FROST introduces additional limitations beyond the idealized objective. The sequence-level Gibbs law is approximated autoregressively; the Fisher inverse is approximated using tractable curvature surrogates; the robust student family is represented by a finite ensemble; and the structural and provenance terms are implemented through prefix-local features. Each of these approximations is reasonable and necessary for deployment, but each also introduces modeling error. As a consequence, the realized decoder should be understood as a ***practical approximation to the ideal FROST objective***, not as an exact implementation of the full sequence-level variational program.

Dependence on the attacker model class. The effectiveness of FROST also depends on the relevance of the chosen proxy student family $\mathcal{Q}_S$. If the true attacker uses a student architecture, tokenizer, optimization method, or transcript format that is far outside the modeled uncertainty set, the defense may become less effective. This is an unavoidable feature of any robust defense against an open-ended adversary. FROST addresses this risk by promoting ***multi-student robustness*** rather than single-proxy specialization, but it does not remove the dependence on modeling assumptions altogether.

Dependence on the utility functional. A second modeling dependency lies in the choice of utility functional $u(x, y)$. In the abstract objective, utility includes any behavior the defender wishes to preserve, including correctness, helpfulness, formatting fidelity, and calibrated refusal. In practice, however, no finite evaluation suite perfectly captures all aspects of user-facing utility. Therefore, the utility guarantees offered by FROST are always relative to the ***chosen evaluation slice***. This is standard in deployment-oriented work, but it should be stated explicitly: utility preservation is only as comprehensive as the task distribution and metrics used to define it.

What should count as success. Given these constraints, the appropriate success criterion for FROST is not absolute security, but a ***measurable shift in the attack surface***. Concretely, the method succeeds if it can simultaneously achieve:

1. a substantial reduction in ***capability transfer efficiency*** under black-box harvesting;
2. small and controlled degradation in ***teacher-side utility and response quality***;
3. robustness across a ***meaningful family of attacker students***; and
4. a clear ***security–utility–overhead tradeoff*** that can be tuned at deployment time.

This is the standard against which we evaluate FROST in the remainder of the paper.

Positioning statement. In summary, FROST should be read as a ***principled teacher-side antidistillation objective***, not as a universal security theorem. Its contribution is to provide a mathematically grounded way to shape generated outputs so that they remain useful to legitimate users while becoming less efficient supervision for downstream student models. The strength of the framework lies precisely in this scope: it is narrow enough to be well-defined, yet broad enough to meaningfully alter the economics of black-box capability stealing.

3 Experimental Setup

Evaluation philosophy and empirical burden. A teacher-side antidistillation defense is only convincing if it satisfies a demanding empirical burden. It is not enough to show that the defended teacher changes its outputs, nor even that a student trained on those outputs performs worse in isolation. To justify FROST as a meaningful security mechanism, we must demonstrate a stricter triad: ***(i)*** black-box capability transfer becomes materially more difficult, ***(ii)*** the teacher remains useful and behaviorally stable for legitimate users, and ***(iii)*** these gains persist across realistic attacker choices and practical deployment constraints. This evaluation philosophy follows directly from the objective developed in Section 3. FROST is not intended as a generic perturbation scheme, but as a ***utility-preserving output-channel defense*** against black-box capability stealing. Accordingly, our experiments are structured around four research questions that test whether the framework succeeds along the dimensions that matter most in deployment: ***distillation resistance, utility preservation, attacker robustness, and security–utility–overhead tradeoffs***.

RQ1: Does FROST reduce black-box capability transfer? Our first and most central question is whether FROST actually succeeds at the objective it is designed to optimize: ***reducing the efficiency with which a downstream student can absorb capability from harvested teacher outputs***. Concretely, we ask whether students trained on transcripts generated by FROST acquire less reasoning, coding, and general instruction-following competence than students trained on transcripts generated by the nominal teacher or prior teacher-side defenses.

This question is the empirical counterpart of the theoretical development in Sections 3.3–3.6. The free-energy formulation says that FROST should downweight outputs with high anti-copying energy; the Fisher-geometric interpretation says these are precisely the outputs that induce strong first-order learning progress; and the information-theoretic interpretation suggests that reducing the learning value of each trace should make extraction more query-expensive. RQ1 tests whether these principles manifest at the level that ultimately matters: ***the capability of an attacker-trained student after black-box harvesting***.

We evaluate RQ1 across multiple dimensions:

- ***query budget***: does FROST reduce capability transfer at low, medium, and high harvesting budgets?
- ***task family***: does the defense matter similarly for reasoning, knowledge, and code generation?
- ***harvesting regime***: is the reduction especially strong when the attacker targets structured, pedagogically rich traces?

The expected outcome is not necessarily that student learning is eliminated, but that the ***capability transfer efficiency*** of black-box harvesting is materially reduced.

RQ2: Does FROST preserve teacher utility? A teacher-side defense is only viable if it does not destroy the practical usefulness of the API it is meant to protect. Therefore, our second research question asks whether FROST preserves the teacher's utility on benign user-facing tasks. This is a necessary complement to RQ1. A defense that reduces transfer only by severely degrading outputs would not constitute a meaningful contribution; it would merely trade security for unusability.

RQ2 is directly tied to the trust-region structure of FROST. The variational objective in Section 3.3 does not seek arbitrary response modification; it seeks a ***minimal-intervention reweighting*** of the nominal teacher distribution. Similarly, the constrained objective in Section 3.2 explicitly treats utility as a quantity that must be preserved, not sacrificed. RQ2 therefore tests whether the defended decoder remains faithful to this design intent in practice.

Importantly, "utility" in our evaluation is not restricted to a single scalar notion of performance. Instead, we interpret utility broadly to include:

- ***task correctness*** on benign reasoning, knowledge, and coding tasks;
- ***response quality*** in terms of helpfulness, coherence, and instruction-following fidelity;
- ***behavioral stability*** under defended decoding, including whether outputs remain aligned with the nominal teacher's intended style and competence profile.

Thus, RQ2 asks not simply whether FROST "works," but whether it works ***without unduly compromising the value of the teacher to legitimate users***.

RQ3: Is FROST robust across attacker students and training regimes? Our third question addresses a crucial robustness issue. Teacher-side defenses can appear effective if they are evaluated only against the same proxy student family used during design, yet fail when the attacker changes architecture, scale, optimizer, or fine-tuning method. Because real attackers are free to choose whatever student best exploits the teacher's outputs, the defense must be judged under ***attacker diversity***, not only under proxy-matched evaluation.

RQ3 is therefore designed to test the central motivation of the multi-student robustness formulation in Section 3.5. There, FROST was explicitly defined against a family of plausible attacker students rather than a single surrogate. Empirically, this leads to the following question:

Does FROST remain effective when the attacker changes the student model or training recipe?

We study this question across several axes:

- ***student architecture***: does protection transfer across distinct model families?
- ***student scale***: is the defense effective only against small students, or also against stronger ones?
- ***fine-tuning regime***: does the defense remain useful under both full fine-tuning and parameter-efficient attacks?
- ***transcript format***: is the defense robust to answer-only versus trace-rich distillation?

RQ3 is important because it distinguishes a ***genuinely robust teacher-side defense*** from a proxy-specific anti-signal that merely overfits one anticipated attacker.

RQ4: What are the security–utility–overhead tradeoffs of FROST? Our fourth question concerns deployability. Even a mathematically elegant defense will be difficult to justify in practice if it introduces prohibitive latency, excessive proxy-scoring cost, or an unfavorable utility tradeoff. The framework developed in Section 3.9 makes clear that FROST is not cost-free: it requires candidate scoring, proxy-student evaluation, and curvature approximation. The relevant deployment question is therefore not whether the defense has zero overhead, but whether its overhead is ***structured, tunable, and worthwhile***.

RQ4 asks how the practical behavior of FROST changes as we vary the main defense knobs:

- the ***inverse temperature*** β, which controls overall defense strength;
- the ***component weights*** $\lambda_{\text{geom}}, \lambda_{\text{struct}}, \lambda_{\text{prov}}$, which allocate strength across geometric, structural, and provenance-aware mechanisms;
- the ***proxy ensemble size*** M, which governs robustness versus cost;
- the ***shortlist size*** k, which governs how much of the teacher's candidate space is actively defended.

This question is operationally central. In deployment, the correct configuration of FROST will depend on the value of the endpoint being protected, the acceptable latency budget, and the severity of the expected extraction threat. RQ4 therefore evaluates whether FROST admits a usable ***security–utility–overhead frontier*** rather than a single brittle operating point.

How the research questions structure the rest of the evaluation. These four questions jointly define the empirical scope of the paper:

- ***RQ1*** evaluates whether FROST actually slows distillation;
- ***RQ2*** evaluates whether it preserves the teacher's value;
- ***RQ3*** evaluates whether its benefits survive attacker variation;
- ***RQ4*** evaluates whether it remains practical and tunable.

The remainder of the experimental setup is organized to answer these questions systematically. We next describe the teacher and student models, black-box harvesting regimes, benchmark suites, baselines, metrics, and training protocols used to instantiate this evaluation framework.

3.1 Models

Design principle. Our model selection is guided by a simple requirement: the evaluation should reflect the actual ***teacher–student asymmetry*** faced by a deployed frontier API. Accordingly, we use ***two teacher models*** to test generality across reasoning-oriented and general instruction-following settings, and ***multiple attacker students*** spanning smaller, similarly sized, and cross-family models. This design serves two purposes. First, it prevents the evaluation from collapsing into a single-model case study. Second, it directly tests the central claim of FROST: that a teacher-side defense should remain effective even when the attacker varies architecture, scale, and fine-tuning strategy. The resulting setup mirrors the multi-student robustness perspective developed in Section 3.5. :contentReference[oaicite:0]index=0

Teacher models. We evaluate FROST with two teachers:

- ***Teacher A: DeepSeek-R1-Distill-Qwen-7B.*** We use this model as a ***reasoning-oriented teacher*** because it is explicitly released as a distilled reasoning model in the DeepSeek-R1 family, built on top of the Qwen2.5-Math-7B base. This makes it especially relevant to our threat model, since structured reasoning traces are precisely the kind of outputs that are highly valuable for downstream distillation. :contentReference[oaicite:1]index=1
- ***Teacher B: Qwen2.5-7B-Instruct.*** We use this model as a ***general-purpose instruction-following teacher***. Its model card emphasizes strong instruction following, long-context support, structured output generation, and multilingual breadth, making it a suitable complement to the more reasoning-specialized DeepSeek teacher. :contentReference[oaicite:2]index=2

This pair is intentionally complementary. The first teacher stresses the high-value regime in which verbose and pedagogically rich outputs are especially vulnerable to capability stealing, while the second tests whether FROST remains useful for more general instruction-following APIs. We do not claim that these two teachers exhaust the full deployment landscape; rather, they provide a balanced starting point that spans ***reasoning-heavy*** and ***general assistant-like*** teachers. :contentReference[oaicite:3]index=3

Attacker student models. To evaluate black-box capability transfer, we use a family of attacker students chosen to reflect realistic extraction scenarios:

- ***Student 1: Qwen2.5-3B-Instruct.*** This is our ***small, economical attacker***. The model card reports a 3.09B-parameter instruction-tuned Qwen2.5 model with 32,768-token context support, making it a realistic low-cost student for harvested-supervision attacks. :contentReference[oaicite:4]index=4
- ***Student 2: Mistral-7B-Instruct-v0.3.*** This is our ***cross-family attacker***. It is an instruction-tuned version of Mistral-7B-v0.3 and differs architecturally and tokenization-wise from both Qwen and Llama families. This makes it especially useful for testing whether FROST overfits to one model family. :contentReference[oaicite:5]index=5

- ***Student 3: Meta-Llama-3.1-8B-Instruct.*** This is our ***stronger attacker***. Meta describes Llama 3.1 8B Instruct as a multilingual instruction-tuned model optimized for dialogue use cases, and the model card explicitly notes support for synthetic-data generation and distillation workflows. We use it to test whether FROST remains effective against a stronger student that is competitive in scale with the teachers. :contentReference[oaicite:6]index=6

Role of model diversity. This model selection is not arbitrary. Each student stresses a different failure mode of teacher-side defense:

- the smaller Qwen student tests whether FROST suppresses ***cheap capability transfer***;
- the Mistral student tests ***cross-architecture robustness***;
- the Llama student tests whether protection persists against a ***stronger attacker*** with substantial instruction-following capacity.

Together, they instantiate the uncertainty set $\mathcal{S}$ from Section 3.5 in a practically meaningful way. We emphasize that our goal is not to claim exhaustive coverage of all possible attackers, but to ensure that FROST is not evaluated against only one proxy-matched student. :contentReference[oaicite:7]index=7

Proxy students for defended decoding. In addition to the final attacker students used for evaluation, FROST requires a proxy family $\mathcal{Q}_S$ for scoring Fisher-geometric learnability during decoding. In our main configuration, we instantiate this family using lighter or intermediate open models drawn from the same broad ecosystem as the evaluation students, so that the defense has access to diverse but computationally tractable attacker geometries. Concretely, we use a small Qwen proxy, a small Llama-family proxy, and a mid-sized cross-family proxy, and we vary the ensemble size M in ablations. This design follows directly from the robustness formulation in Section 3.5: the defense should not be tied to a single surrogate, but should estimate whether a trace is broadly learnable across plausible attacker classes. The exact proxy composition and Fisher approximation used in each experiment are reported in the implementation details and ablation section.

Training regimes. For each attacker student, we evaluate two distillation settings:

- ***full supervised fine-tuning***, which models a high-effort attacker willing to invest substantial compute; and
- ***parameter-efficient fine-tuning***, which models a lower-cost but highly realistic adversary using adapter-style updates.

This distinction is important because a teacher-side defense that only resists one fine-tuning regime may still be brittle in practice. By evaluating both, we directly test whether the learned trace-suppression effect of FROST transfers across attacker optimization strategies.

Modeling objective of this section. Overall, the model suite is designed to answer the core empirical question of the paper under realistic attacker diversity: ***can a teacher-side defense make harvested outputs less transferable across multiple students without collapsing the teacher's own utility?*** The teacher pair captures complementary deployment roles, while the student family captures heterogeneity in architecture, strength, and tuning regime. This gives the evaluation enough breadth to support claims about ***robust teacher-side antidistillation*** rather than model-specific artifact suppression.

3.2 Harvesting Regimes

Why harvesting regimes matter. Teacher-side antidistillation cannot be evaluated with a single undifferentiated transcript. In practice, attackers do not all query a model in the same way. Some harvest broad instruction-following behavior through benign-looking prompts; others deliberately elicit long, structured, pedagogically rich traces that are especially useful for downstream imitation. Prior work on model extraction shows that black-box stealing depends strongly on the query distribution, and recent antidistillation work is motivated precisely by the fact that certain forms of reasoning trace can be especially valuable supervision for students. For this reason, we evaluate FROST under ***multiple harvesting regimes*** rather than a single monolithic prompt pool. :contentReference[oaicite:8]index=8

Regime R1: Benign capability harvesting. Our first regime models a stealthy attacker who collects outputs using broadly useful but outwardly ordinary prompts. The prompt pool in this regime emphasizes:

- general instruction following,
- question answering,
- coding assistance,
- short reasoning and explanation tasks,
- summarization and light transformation requests.

The purpose of this regime is to test whether FROST can reduce capability transfer even when the attacker does ***not*** overtly seek structured traces. This is important because real-world harvesting campaigns may intentionally resemble normal API usage so as to avoid detection. R1 therefore provides the most conservative evaluation setting: the transcript looks benign, but may still contain enough latent supervision to support downstream distillation.

Regime R2: Structured reasoning harvesting. Our second regime models a more deliberate attacker who explicitly optimizes for ***pedagogically dense supervision***. Here the prompts are designed to elicit:

- step-by-step reasoning,
- decompositions into subproblems,
- rubric-style evaluations,
- procedure-like explanations,
- algorithmic or code-oriented walkthroughs.

This is the regime where we expect FROST to be especially valuable. The motivation is direct: structured, long-form, and reusable traces often provide more supervision per query than terse final answers, which is precisely the vulnerability teacher-side antidistillation aims to mitigate. DeepSeek-R1-Distill-Qwen-7B is particularly relevant here because it is itself a distilled reasoning model, making this regime a natural stress test for the paper's core security question. :contentReference[oaicite:9]index=9

Regime R3: Mixed harvesting. Our third regime models the most realistic attacker behavior: a mixture of broad benign-looking prompts and targeted structured elicitation. Specifically, we construct mixed transcripts in which the majority of queries come from the benign-capability pool, while a smaller but strategically chosen subset is drawn from the structured-reasoning pool. This regime is important for two reasons. First, it more closely matches how an adaptive attacker might behave in practice: blend in with ordinary usage while selectively extracting high-value traces. Second, it prevents the evaluation from overstating protection by considering only obviously malicious or obviously benign prompt distributions. R3 therefore serves as our ***primary realism regime***.

Adaptive transcript construction. Within each regime, the attacker may choose prompts either ***non-adaptively*** from a fixed prompt pool or ***adaptively*** based on previous responses, depending on the experiment. The non-adaptive setting isolates the effect of transcript composition, while the adaptive setting tests the threat model formalized in Section 3.1, where future queries may be refined based on the teacher's earlier outputs. We do not require sophisticated red-team-style agentic optimization to make the evaluation meaningful; rather, the key point is that FROST should be tested under both fixed and response-aware harvesting strategies.

Transcript formatting variants. For each harvesting regime, we consider two transcript formats:

- ***answer-only transcripts***, where only the teacher's final response is retained for student training;
- ***trace-rich transcripts***, where longer rationales, decompositions, or structured outputs are preserved.

This distinction matters because the entire motivation of FROST is that not all outputs carry the same downstream training value. If the defense is effective, it should matter most in the trace-rich setting, where explicit structure and pedagogical density make raw teacher outputs especially easy to distill.

Budget scaling within each regime. Each regime is evaluated across multiple query budgets,

$$B \in \{10^3,\ 10^4,\ 10^5\},$$

with larger-budget extensions reported when compute permits. This scaling is essential. A defense that only appears useful at one arbitrary transcript size is difficult to interpret, whereas a budget curve reveals whether FROST consistently lowers the ***capability transfer rate per query*** across low-, medium-, and high-harvest settings. It also connects directly to the information-theoretic view in Section 3.6, where the practical objective is to make extraction progressively more ***query-expensive***.

What these regimes are intended to test. The three harvesting regimes serve distinct purposes:

- ***R1*** tests whether FROST helps even when harvesting looks benign;
- ***R2*** tests the paper's core claim that structured traces are disproportionately dangerous;
- ***R3*** tests whether the defense remains effective in the mixed, realistic setting most likely to arise in deployment.

Together, they ensure that FROST is not evaluated on only one narrow attack style. Instead, the defense is tested across a spectrum of transcript-construction strategies that vary in stealth, pedagogical density, and extraction efficiency.

Positioning. Taken together, the model suite of Section 4.2 and the harvesting regimes defined here provide the backbone of our evaluation protocol. The former captures ***who the attacker is***; the latter captures ***how the attacker queries the teacher***. Both are necessary. A teacher-side antidistillation defense can only be taken seriously if it remains effective across meaningful variation in both dimensions.

3.3 Benchmarks

Benchmark design principle. The benchmark suite is constructed to answer the four research questions from Section 4.1 in a way that is both ***security-relevant*** and ***deployment-relevant***. A teacher-side antidistillation defense should not be evaluated on a single narrow task family. Instead, it must be tested along three complementary dimensions:

1. ***capability transfer benchmarks***, which measure how much competence a downstream student acquires from harvested transcripts;
2. ***teacher utility benchmarks***, which measure whether defended decoding preserves the usefulness of the teacher for benign users;
3. ***safety sanity-check benchmarks***, which verify that the defense does not inadvertently destabilize refusal or harmlessness behavior.

This separation is intentional. The first group addresses the ***security*** objective of FROST, the second addresses its ***utility-preservation*** objective, and the third ensures that gains are not obtained by unintentionally damaging other critical aspects of deployment behavior.

Capability-transfer benchmarks. To measure downstream capability stealing, we evaluate distilled students on four benchmark families.

Mathematical reasoning: GSM8K and MATH-500. We use ***GSM8K*** as our primary benchmark for grade-school mathematical reasoning. GSM8K consists of linguistically diverse multi-step arithmetic word problems and has become a standard testbed for reasoning-oriented language models (Cobbe et al., 2021). Because one of our teachers is explicitly reasoning-oriented, GSM8K provides a natural test of whether harvested supervision transfers stepwise problem-solving ability.

To complement GSM8K with a harder mathematical setting, we also evaluate on ***MATH-500***. This subset is drawn from the broader MATH benchmark, which was introduced to measure challenging competition-style mathematical reasoning across algebra, geometry, counting, probability, and related domains (Hendrycks et al., 2021b). The purpose of including MATH-500 is not merely to add difficulty, but to test whether FROST specifically suppresses the transfer of ***structured multi-step reasoning*** at a more advanced level.

Knowledge and broad reasoning: MMLU. We use ***MMLU*** to measure broad general-domain capability transfer. MMLU covers a wide range of subjects, including humanities, social science, law, medicine, computer science, and mathematics, and was introduced specifically to evaluate multitask language understanding at scale (Hendrycks et al., 2021a). In our context, MMLU serves two purposes. First, it tests whether harvested transcripts transfer ***general academic and factual competence*** beyond narrow reasoning or coding tasks. Second, it helps determine whether FROST suppresses only highly structured traces or also reduces broader instruction-following and knowledge transfer.

Code generation: HumanEval and MBPP. To test whether teacher outputs can transfer coding ability, we evaluate distilled students on ***HumanEval*** and ***MBPP***. HumanEval was introduced as a functional-correctness benchmark for code generation, where success is measured by whether generated code passes hidden unit tests (Chen et al., 2021). It is especially appropriate for our threat model because coding traces are often high-value outputs in black-box extraction campaigns. MBPP complements HumanEval by covering a broader collection of mostly entry-level Python programming tasks designed for benchmark-style code synthesis (Austin et al., 2021).

This pair is useful because the two benchmarks stress slightly different aspects of code transfer. HumanEval places more emphasis on concise functional correctness under hidden execution tests, whereas MBPP captures a broader variety of short programming tasks that more closely resemble ordinary coding assistance. Together, they provide a balanced evaluation of whether FROST suppresses the transfer of ***code competence*** from teacher to student.

Teacher utility benchmarks. To evaluate whether FROST preserves the usefulness of the defended teacher, we reuse the capability benchmarks above at the teacher level and add a general response-quality benchmark.

Task utility on capability suites. For each defended teacher, we measure direct performance on GSM8K, MATH-500, MMLU, HumanEval, and MBPP. This gives a task-grounded notion of utility:

does the defended teacher remain competent on the same families of tasks that matter to users?

This is important because a defense that reduces student transfer by simply degrading the teacher's underlying task performance would not be meaningful. Evaluating teacher utility on the same benchmark families therefore prevents a trivial but misleading interpretation of the security results.

General response quality: MT-Bench. Beyond task correctness, we also evaluate teacher-side quality using ***MT-Bench***, a multi-turn question set designed to assess conversational helpfulness, coherence, and instruction following via model- or judge-based comparison (Zheng et al., 2023). MT-Bench is included because utility is not exhausted by exact-match or pass@1 metrics. A deployed assistant may remain benchmark-competent while still becoming less helpful, less coherent, or less aligned with user expectations. MT-Bench therefore captures a broader notion of ***user-facing utility preservation*** under defended decoding.

Safety sanity-check benchmarks. Although FROST is not introduced as a safety-alignment method per se, any teacher-side output-shaping mechanism can in principle alter refusal patterns or harmful-compliance behavior. To ensure that the defense does not introduce obvious regressions, we include a compact safety sanity-check suite.

We use ***HarmBench*** as our primary safety benchmark. HarmBench was introduced as a standardized benchmark for evaluating automated red teaming and harmful behavior in large language models, with a taxonomy spanning cyber, chemical, biological, illegal, and other high-risk request classes (Mazeika et al., 2024). In our evaluation, HarmBench is not the main security target, but rather a ***stability check***: the question is whether FROST changes the teacher's harmful-compliance or refusal behavior in undesirable ways.

Where feasible, we also report a compact evaluation on ***JailbreakBench*** or a closely matched jailbreak-style prompt set to test whether defended decoding causes meaningful shifts under adversarial prompt phrasing. This secondary check is useful because a teacher-side defense that meaningfully perturbs output probabilities could in principle affect refusal boundaries even if it is not designed to do so.

Why this benchmark suite is sufficient. Taken together, the selected benchmarks span the three empirical axes that matter most for FROST:

- ***reasoning transfer*** via GSM8K and MATH-500;
- ***general capability transfer*** via MMLU;
- ***coding transfer*** via HumanEval and MBPP;
- ***teacher-side utility preservation*** via direct task scores and MT-Bench;
- ***behavioral stability*** via HarmBench and jailbreak-style sanity checks.

This suite is intentionally broad enough to support meaningful claims about black-box capability stealing, but still focused enough to maintain a coherent experimental narrative. In particular, it directly targets the capabilities most likely to be valuable in unauthorized black-box harvesting: ***reasoning, knowledge, and code generation***.

3.4 Baselines

Why strong baselines are necessary. Because FROST is a teacher-side defense, its gains must be distinguished from several simpler explanations. A student may learn less from a defended teacher because the defense merely increases randomness, because it shortens outputs, because it behaves like a watermarking scheme, or because it simply degrades the teacher. A credible evaluation must therefore compare FROST against baselines that isolate these possibilities. Our baseline suite is designed to answer a precise question:

Does FROST outperform simpler decoding perturbations and prior teacher-side defenses at the specific objective of utility-preserving antidistillation?

B0: Nominal teacher decoding. Our primary reference point is the undefended teacher under its standard decoding configuration:

$$Q_{\text{nom}}(\cdot \mid x) = P_0(\cdot \mid x).$$

This baseline represents the ***unprotected deployment*** setting. All reductions in capability transfer, changes in teacher utility, and overhead tradeoffs are measured relative to this condition. Without this baseline, it would be impossible to determine whether FROST truly protects the teacher or merely shifts behavior arbitrarily.

B1: Simple stochastic decoding. A natural alternative to a principled anti-copying defense is simply to make teacher outputs more random. To test whether FROST's gains are reducible to generic stochasticity, we include a baseline that perturbs decoding through standard sampling controls such as increased temperature and top-p variation. This is an important control condition. If simple randomness already achieves the same reduction in student transfer as FROST, then the geometric and variational machinery of the method would be unnecessary. Conversely, if FROST outperforms this baseline at comparable teacher utility, then its gains cannot be attributed merely to noisy output corruption.

B2: Antidistillation Sampling (ADS). Our principal prior-method baseline is ***Antidistillation Sampling*** (Savani et al., 2025). ADS is especially important because it addresses the same high-level problem as FROST: teacher-side decoding that reduces the usefulness of emitted traces for downstream student training while preserving practical model utility. In the evaluation, ADS serves as the strongest direct conceptual predecessor to FROST.

The comparison to ADS is central for two reasons. First, it tests whether FROST's thermodynamic and Fisher-geometric formulation yields a measurable advantage over a recent teacher-side antidistillation method. Second, it allows us to isolate the value of our core contribution: instead of using proxy-student loss-change signals alone, FROST explicitly measures ***intrinsic learnability*** through Fisher geometry and integrates that signal into a composite free-energy decoder.

B3: Geometry-only FROST. To isolate the contribution of the Fisher-geometric term, we include a geometry-only ablation:

$$E_x(y) = E_x^{\text{geom}}(y), \qquad \lambda_{\text{struct}} = \lambda_{\text{prov}} = 0.$$

This ablation is not a "baseline" in the historical sense, but it is essential experimentally. It tests whether the core geometric principle is already sufficient to reduce capability transfer, and provides a clean reference against which to judge the added value of the structural and provenance-aware terms.

B4: Structure-only FROST. We also include a structure-only ablation:

$$E_x(y) = E_x^{\text{struct}}(y), \qquad \lambda_{\text{geom}} = \lambda_{\text{prov}} = 0.$$

This baseline is particularly important because one might suspect that much of the apparent benefit of teacher-side antidistillation comes simply from suppressing long, pedagogically explicit traces. The structure-only condition tests exactly this hypothesis. If structure-only suppression performs nearly as well as the full method, then the geometric component adds limited value; if full FROST substantially outperforms it, then the paper's central claim about ***Fisher-geometric learnability*** is empirically justified.

B5: Provenance-aware auxiliary condition. Where provenance-aware shaping is enabled, we additionally report a provenance-only or provenance-dominant condition to verify that attribution-supporting output shaping does not by itself explain FROST's protection gains. This condition is not treated as a main competitor, because watermarking and provenance marking primarily target detectability or attribution rather than reduced student learnability (Kirchenbauer et al., 2023; Dathathri et al., 2024). Nevertheless, including it helps clarify the relationship between ***detectable outputs*** and ***non-transferable outputs***.

B6: Full FROST. Our final and primary method condition is the full defended energy:

$$E_x(y) = \lambda_{\text{geom}} E_x^{\text{geom}}(y) + \lambda_{\text{struct}} E_x^{\text{struct}}(y) + \lambda_{\text{prov}} E_x^{\text{prov}}(y),$$

decoded through the Gibbs-reweighted autoregressive rule from Sections 3.7–3.8. This is the configuration that operationalizes the full theoretical framework:

- ***thermodynamic control*** through the free-energy objective;
- ***Fisher-geometric anti-copying*** through multi-student learnability suppression;
- ***structural hardening*** against reusable pedagogical traces;
- optional ***provenance support*** where desired.

What the baseline suite is designed to test. Taken together, the baseline suite answers the most important alternative explanations:

- ***B1*** asks whether generic randomness is sufficient;
- ***B2*** asks whether prior teacher-side antidistillation already explains the gains;
- ***B3*** asks whether geometry alone drives the result;
- ***B4*** asks whether structural suppression alone drives the result;
- ***B5*** asks whether provenance-style shaping alone explains the gains.

This makes the comparison to full FROST especially informative. If the complete method consistently outperforms these alternatives at comparable teacher utility, then the empirical story is much stronger than a simple "our method works" claim.

Positioning. The benchmark suite of Section 4.4 defines ***what*** is measured; the baseline suite defined here establishes ***what FROST must outperform***. Together, they ensure that the evaluation is not merely a demonstration of altered output behavior, but a structured test of whether teacher-side thermodynamic and Fisher-geometric antidistillation produces meaningful gains over simpler or more narrowly motivated alternatives.

3.5 Metrics

Metric design principle. The evaluation of FROST requires more than a single downstream score. A teacher-side antidistillation defense must be judged simultaneously on ***security***, ***utility***, and ***deployability***. Accordingly, our metric suite is organized into four groups:

1. ***distillation metrics***, which quantify how much capability a student acquires from harvested transcripts;
2. ***teacher utility metrics***, which quantify the retained usefulness of the defended teacher;
3. ***overhead metrics***, which quantify the practical cost of defended decoding;
4. ***safety sanity metrics***, which verify that the defense does not induce undesirable behavioral shifts.

This separation mirrors the empirical burden set out in Section 4.1: FROST should reduce transfer, preserve teacher value, remain practically tunable, and avoid obvious safety regressions.

Student capability score. Let $\mu_k(S_B)$ denote the post-distillation performance of a student S_B on benchmark k, after training on a harvested transcript of budget B. Let $\mu_k(S_0)$ denote the same student's pre-distillation score before seeing any teacher-generated supervision. Let $\mu_k(T_Q)$ denote the score of the defended teacher T_Q under decoding rule Q on the same benchmark. These three quantities define the basic geometry of transfer:

- $\mu_k(S_0)$ measures the student's initial ability;
- $\mu_k(T_Q)$ measures the target capability exposed by the teacher;
- $\mu_k(S_B)$ measures how much of that exposed capability the student actually acquires after black-box harvesting.

Using all three is important. Student performance alone does not distinguish whether a student improved because the teacher was highly learnable or simply because the student was already strong.

Capability Transfer Rate (CTR). Our primary security metric is the ***Capability Transfer Rate*** at query budget B:

$$\mathrm{CTR}(B) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \mathrm{clip}_{[0,1]} \left(\frac{\mu_k(S_B) - \mu_k(S_0)}{\mu_k(T_Q) - \mu_k(S_0)} \right),$$

where $\mathcal{K}$ is the set of evaluation benchmarks and $\mathrm{clip}_{[0,1]}$ truncates the ratio to the unit interval for stability. This quantity measures the fraction of the teacher's benchmark advantage over the base student that is recovered by the attacker after seeing B harvested examples.

The intuition is straightforward:

- $\mathrm{CTR}(B) \approx 0$ means the student learned little from the teacher;
- $\mathrm{CTR}(B) \approx 1$ means the student closed most of the gap to the teacher;
- intermediate values quantify partial capability transfer.

CTR is preferable to raw student accuracy as the headline metric because it explicitly normalizes by the available teacher-to-student gap. This makes the measure comparable across different benchmark families and student initializations.

Distillation efficiency. To evaluate how much transferable capability is obtained *per query*, we define the ***distillation efficiency***

$$\kappa_{\mathrm{distill}}(B) = \frac{\mathrm{CTR}(B)}{B}.$$

This is the metric most closely aligned with the core threat model of the paper. Black-box harvesting is a budgeted process: each query incurs cost, time, rate-limit exposure, or detection risk. A defense that merely reduces total transfer at one fixed budget may still be ineffective if it leaves the *per-query* transfer rate largely unchanged. By contrast, a reduction in $\kappa_{\mathrm{distill}}(B)$ means that each harvested teacher interaction has become less useful as supervision.

In the context of FROST, this is the most direct empirical analogue of the information-theoretic objective from Section 3.6. Lower $\kappa_{\mathrm{distill}}(B)$ means that the teacher's output channel has become less efficient as a capability-leakage mechanism.

Area under the transfer curve. Because transfer may evolve nonlinearly with query budget, we also report an aggregate ***Area Under the Transfer Curve*** (AUTC):

$$\mathrm{AUTC} = \frac{1}{|\mathcal{B}|} \sum_{B \in \mathcal{B}} \mathrm{CTR}(B),$$

or, when budgets are unevenly spaced, the corresponding trapezoidal integral over the transfer curve. This metric summarizes transfer behavior across the full budget range rather than at a single operating point. A lower AUTC indicates that the defense consistently suppresses transfer throughout the harvesting process, not merely at one hand-picked budget.

Teacher task-utility metrics. To measure teacher-side utility preservation, we report standard benchmark metrics for the defended teacher:

- ***exact match / accuracy*** for tasks such as GSM8K, MATH-500, and MMLU;
- ***pass@1*** for code-generation benchmarks such as HumanEval and MBPP;
- ***judge-based or pairwise quality scores*** for MT-Bench-style conversational evaluation.

For each teacher-side metric $U_k(Q)$, we also report the defended utility delta relative to nominal decoding:

$$\Delta U_k = U_k(Q) - U_k(P_0).$$

This quantity is important because FROST is designed as a ***minimal-intervention defense***. The most meaningful teacher-side question is not just whether the defended teacher performs well in isolation, but how much quality is lost relative to the original deployment configuration.

Aggregate utility preservation. To summarize teacher behavior across benchmark families, we define an aggregate utility score

$$U_{\mathrm{agg}}(Q) = \frac{1}{|\mathcal{K}_{\mathrm{util}}|} \sum_{k \in \mathcal{K}_{\mathrm{util}}} \widetilde{U}_k(Q),$$

where $\widetilde{U}_k(Q)$ is the normalized utility on benchmark k, scaled so that metrics with different ranges become comparable. We then report

$$\Delta U_{\mathrm{agg}} = U_{\mathrm{agg}}(Q) - U_{\mathrm{agg}}(P_0).$$

This gives a single view of the utility cost of defended decoding, although we always present the underlying benchmark-specific numbers as well. The aggregate score is useful for security–utility plots where one needs a compact summary of teacher quality across multiple task families.

Security–utility frontier. A core theme of the paper is that teacher-side antidistillation is governed by a tradeoff between protection and usability. To visualize this directly, we report the ***security–utility frontier*** obtained by varying the inverse temperature β and, when relevant, the component weights $\lambda_{\mathrm{geom}}, \lambda_{\mathrm{struct}}, \lambda_{\mathrm{prov}}$. Each operating point yields:

$$\Big(\Delta U_{\mathrm{agg}},\ \mathrm{CTR}(B)\Big) \qquad \text{or} \qquad \Big(\Delta U_{\mathrm{agg}},\ \kappa_{\mathrm{distill}}(B)\Big).$$

This frontier is central to the interpretation of FROST. A good defense should push the frontier downward and to the right in the desired sense: ***large reductions in transfer for small losses in utility***.

Overhead metrics. Because FROST is an inference-time defense, we also measure its practical overhead. Specifically, we report:

- ***latency overhead***: additional wall-clock decoding time relative to nominal teacher decoding;
- ***throughput degradation***: reduction in generated tokens per second;
- ***memory overhead***: additional memory associated with proxy-student scoring and Fisher approximations;
- ***energy-scoring cost***: the average incremental cost per decoding step introduced by FROST.

Let $\tau(Q)$ denote the average latency of defended decoding and $\tau(P_0)$ the nominal latency. We report the relative overhead

$$\Delta\tau_{\mathrm{rel}} = \frac{\tau(Q) - \tau(P_0)}{\tau(P_0)}.$$

Similarly, if $\rho(Q)$ denotes defended throughput and $\rho(P_0)$ nominal throughput, we report

$$\Delta\rho_{\mathrm{rel}} = \frac{\rho(Q) - \rho(P_0)}{\rho(P_0)}.$$

These metrics make explicit that FROST is not free, and allow the experimental section to quantify whether its security gains justify its additional decoding cost.

Safety sanity metrics. Although FROST is not introduced as a safety-alignment intervention, a teacher-side output-shaping method may still change harmful-compliance or refusal behavior. We therefore report a compact set of safety sanity metrics:

- ***harmful compliance rate***, measuring how often the model provides disallowed assistance on harmful prompts;
- ***refusal correctness***, measuring whether the model appropriately refuses unsafe requests while not over-refusing benign ones;
- ***benign-safety consistency***, measuring whether the defense preserves the nominal refusal/helpfulness balance of the teacher.

These metrics are not the primary target of the paper, but they are important as a guardrail: a defense that reduces distillation by destabilizing refusal behavior would be difficult to justify in deployment.

Per-regime reporting. All primary metrics are reported separately by:

- ***harvesting regime*** (benign, structured, mixed),
- ***query budget*** B,
- ***student model and tuning regime***,
- ***teacher model***,
- ***defense condition*** (nominal, ADS, ablations, full FROST).

This granularity is essential. A single averaged number can obscure where a defense is actually working. In particular, we expect the strongest gains of FROST to appear in structured-trace-rich settings, and the metric design is intentionally chosen to make such differences visible rather than wash them out.

Statistical reporting. For all main results, we report means across repeated runs together with variability estimates. Unless otherwise noted, confidence intervals are computed using bootstrap resampling or repeated-seed aggregation appropriate to the benchmark. Pairwise comparisons between defenses are reported wherever the comparison is central to a claim. This is especially important for teacher-side utility differences, where small absolute gaps may otherwise be difficult to interpret.

What the metrics are intended to show. Taken together, the metric suite answers the central evaluation questions of the paper:

- ***CTR*** and κ_{distill} ask whether FROST reduces downstream capability transfer;
- ***teacher utility metrics*** ask whether the teacher remains useful;
- ***overhead metrics*** ask whether the defense is practically deployable;
- ***safety sanity metrics*** ask whether the defense preserves basic behavioral stability.

This combination is deliberate. FROST should succeed only if it improves the security side of the tradeoff ***without*** collapsing the utility or deployment side.

3.6 Training Protocol

Protocol design principle. The training protocol is designed to isolate the effect of ***teacher-side defended decoding*** from confounding factors in attacker training. To make comparisons meaningful, all students are trained under ***matched budgets, matched data formats, and matched optimization settings*** across defense conditions. In other words, when we compare nominal decoding, stochastic baselines, ADS, and FROST, the only intended difference is the ***quality and transferability of the harvested supervision***—not the amount of training compute, number of updates, or student exposure to data. This design is crucial. Without it, any reduction in downstream capability could be attributed to protocol imbalance rather than to the defense itself.

Transcript generation. For each teacher T, defense condition Q, harvesting regime R, and query budget B, we first construct a black-box transcript

$$\mathcal{T}_B^{(T,Q,R)} = \{(x_t, y_t)\}_{t=1}^{B}, \qquad y_t \sim Q(\cdot \mid x_t).$$

The prompts x_t are sampled or adaptively selected according to the harvesting regime defined in Section 4.3. This means that transcript generation itself is part of the evaluation protocol: the student never sees the teacher directly, only the harvested supervision emitted by the defended or undefended output channel.

To ensure fairness across defenses, the prompt set for a given (T, R, B) configuration is held fixed whenever possible, so that nominal decoding, ADS, and FROST are all evaluated on the ***same underlying prompt stream***. The only difference is the teacher's output distribution. This prevents the comparison from being confounded by differences in prompt difficulty or transcript composition.

Train–evaluation separation. A central requirement of the protocol is that harvesting prompts and evaluation benchmarks remain ***strictly separated***. The transcripts used for black-box distillation are constructed from prompt pools that are disjoint from the evaluation sets used for GSM8K, MATH-500, MMLU, HumanEval, MBPP, MT-Bench, and the safety sanity checks. This separation is essential to avoid a trivial interpretation in which a student performs well simply because the harvested transcript overlaps with the benchmark distribution too directly. Our goal is to measure ***capability transfer***, not benchmark contamination.

Transcript formatting conditions. For each transcript, we consider two supervision formats:

- ***answer-only format***, in which the student is trained only on the teacher's final emitted response;
- ***trace-rich format***, in which longer explanations, rationales, decompositions, or structured outputs are retained.

This distinction is important because the entire motivation of FROST is that trace-rich outputs often provide more transferable supervision than terse answers. By training under both formats, we test whether the defense matters most precisely where one would expect: in ***highly pedagogical transcript settings***.

Student initialization. Each attacker student begins from a fixed publicly available instruction-tuned checkpoint, denoted S_0. For a given student architecture, the same base initialization is used across all defense conditions and harvesting budgets. This yields a controlled before/after comparison:

$$S_0 \longrightarrow S_B^{(Q)}$$

where $S_B^{(Q)}$ denotes the student after training on a transcript harvested from defense condition Q. Fixing the student initialization is necessary for the interpretation of the Capability Transfer Rate metric in Section 4.6, since that metric explicitly normalizes transfer relative to the base student's initial performance.

Full supervised fine-tuning. Our first attacker-training protocol is standard ***full supervised fine-tuning*** (SFT). For each transcript $\mathcal{T}_B$, the student parameters θ are updated by minimizing the imitation loss

$$\hat{\mathcal{L}}_{\text{imit}}(\theta; \mathcal{T}_B) = \frac{1}{B} \sum_{t=1}^{B} -\log p_\theta(y_t \mid x_t).$$

This corresponds to a high-effort attacker willing to spend substantial compute to absorb as much capability as possible from the harvested teacher traces. Full SFT is important because it provides an upper-bound-style attack condition: if FROST remains effective here, then its protection is unlikely to be an artifact of only weak attacker training.

Parameter-efficient fine-tuning. Our second attacker protocol uses ***parameter-efficient fine-tuning*** (PEFT), such as LoRA-style low-rank adaptation. This setting models a practical attacker who wants to exploit harvested transcripts without paying the full memory and optimization cost of complete model fine-tuning. It is especially relevant for black-box extraction because a real attacker may first attempt cheap adaptation before scaling to more expensive full-model training.

The inclusion of PEFT has two purposes. First, it broadens the realism of the evaluation. Second, it helps determine whether FROST is only effective against one particular attacker training regime or whether it suppresses capability transfer more generally.

Budget matching across defenses. For every student architecture and training regime, we keep the optimization budget fixed across defense conditions. This includes:

- the same number of optimization steps or epochs,
- the same batch size,
- the same maximum sequence length,

- the same learning-rate schedule,
- the same early-stopping or checkpoint-selection rule.

This point is critical. If one defense generated slightly shorter or lower-entropy transcripts and we compensated by giving the attacker more optimization steps, the comparison would become ambiguous. Conversely, if another defense produced harder-to-fit transcripts but the attacker received fewer updates, the measured protection would be exaggerated. By fixing the training budget, we ensure that FROST is evaluated as a defense against ***matched attacker effort***.

Optimization details. For each student and training regime, we adopt a stable supervised fine-tuning configuration with standard optimizer, learning-rate warmup, gradient clipping, and sequence packing where appropriate. Hyperparameters are selected once per student family and then reused across all defense conditions. We deliberately avoid per-defense hyperparameter retuning, because doing so could blur the interpretation of the results: the purpose of the experiment is to test whether the *same attacker training pipeline* becomes less effective when transcripts are harvested from FROST-protected teachers.

Evaluation checkpoints. Students are evaluated after training on each transcript budget

$$B \in \{10^3, 10^4, 10^5\},$$

with larger-budget extensions included when computationally feasible. This produces a transfer curve

$$S_0 \to S_{10^3} \to S_{10^4} \to S_{10^5},$$

rather than a single endpoint. This is essential because FROST is designed to make black-box extraction progressively more expensive, not necessarily to block all transfer at one fixed budget. The budget curve therefore forms a central part of the empirical story.

Adaptive and non-adaptive harvesting conditions. Where applicable, we separate experiments into:

- ***non-adaptive harvesting***, where the prompt pool is fixed in advance;
- ***adaptive harvesting***, where later prompts may depend on previous teacher responses.

This split mirrors the threat model from Section 3.1. The non-adaptive setting isolates the effect of transcript content, whereas the adaptive setting tests whether FROST remains effective when the attacker actively refines future queries based on earlier outputs.

Proxy-student calibration during defended decoding. The proxy ensemble used by FROST for geometric scoring is fixed before harvesting and is not updated online to match the final evaluation students. This design choice is deliberate. If the proxy family were retuned after seeing the attacker model, the defense would become unrealistically attacker-specific. Instead, the proxy ensemble plays the role intended in Section 3.5: a ***robust uncertainty set*** over plausible attacker students. We then evaluate whether this proxy-based defense generalizes to held-out or only partially matched attacker models at training time.

Repeatability and statistical reliability. To control for optimization noise, we repeat student training with multiple random seeds wherever computationally feasible. All main metrics are reported as averages across runs, together with variability estimates. This is particularly important for code-generation and reasoning benchmarks, where stochastic fine-tuning effects can otherwise obscure the true difference between defense conditions.

Why this protocol is sufficient. Taken together, the training protocol is designed to make one comparison maximally clear:

given the same attacker budget and the same training pipeline, do transcripts harvested from FROST lead to less capable students than transcripts harvested from competing defenses or undefended teachers?

This is the most direct experimental counterpart to the central objective of the paper. It isolates the value of the defended output channel itself, rather than allowing differences in attacker optimization to dominate the conclusion.

3.7 Implementation Details

Implementation philosophy. Our implementation is designed to preserve the mathematical structure of FROST while remaining practical enough for repeated large-scale evaluation. The guiding principle is straightforward: wherever the ideal objective from Section 3 would be too expensive to realize exactly, we introduce approximations that preserve the ***ordering logic*** of the defense. That is, outputs estimated to be more learnable by downstream students should still receive higher energy and therefore lower defended probability, even when the underlying quantities are computed approximately. This principle governs every implementation choice described below.

Teacher-side decoding stack. All defended decoding experiments are implemented as a wrapper around the nominal teacher's autoregressive generation loop. At each step t, the teacher first produces nominal logits

$$\ell_t(v), \qquad v \in \mathcal{V},$$

which define the base next-token distribution

$$P_0(v \mid x, y_{<t}) = \mathrm{softmax}(\ell_t(v)).$$

FROST then computes a token-local composite energy

$$\widetilde{E}_t(v; x, y_{<t}) = \lambda_{\mathrm{geom}} \widetilde{E}_t^{\mathrm{geom}}(v; x, y_{<t}) + \lambda_{\mathrm{struct}} \widetilde{E}_t^{\mathrm{struct}}(v; x, y_{<t}) + \lambda_{\mathrm{prov}} \widetilde{E}_t^{\mathrm{prov}}(v; x, y_{<t}),$$

and applies the logit correction

$$\ell_t^{\mathrm{FROST}}(v) = \ell_t(v) - \beta \widetilde{E}_t(v; x, y_{<t}).$$

The corrected logits are then renormalized over the candidate set to obtain the defended next-token distribution. This means that FROST is implemented entirely at inference time and does not require modifying teacher weights, retraining the teacher, or introducing a separate controller model.

Candidate shortlist construction. To keep token scoring tractable, we evaluate FROST energies only on a top-k shortlist

$$\mathcal{V}_t^{(k)} \subseteq \mathcal{V}$$

constructed from the nominal teacher distribution at each decoding step. Unless otherwise specified, the main experiments use a moderate shortlist size chosen to balance defense quality and latency. This shortlist is recomputed at every step. The practical effect is that FROST operates as a ***local reweighting mechanism*** over the teacher's most plausible next tokens, rather than as a global search over the entire vocabulary. This preserves the minimal-intervention character of the defense and avoids unnecessary computation on tokens with negligible nominal mass.

Proxy-student ensemble for geometric scoring. The Fisher-geometric term is computed using a fixed ensemble of proxy students

$$\mathcal{Q}_S = \{S^{(1)}, \ldots, S^{(M)}\},$$

chosen to span lightweight and mid-sized attacker geometries. For each candidate token v, we form the partial candidate prefix

$$\tilde{y}_t^{(v)} = y_{<t} \oplus v$$

and compute the proxy-specific imitation gradient

$$g_t^{(m)}(v) = \nabla_{\theta_m}\big(-\log p_{\theta_m}(\tilde{y}_t^{(v)} \mid x)\big).$$

The token-local geometric energy is then approximated by the ensemble average

$$\widetilde{E}_t^{\mathrm{geom}}(v; x, y_{<t}) = \frac{1}{M} \sum_{m=1}^{M} g_t^{(m)}(v)^\top F_m^{-1} g_t^{(m)}(v),$$

where F_m^{-1} is the chosen curvature approximation for proxy $S^{(m)}$. In the main setting, the proxy ensemble remains fixed across experiments so that changes in performance can be attributed to FROST rather than to defense retuning.

Curvature approximation. The exact inverse Fisher matrix is not computed explicitly. Instead, FROST uses tractable approximations to the proxy Fisher geometry. Our default configuration uses a structured approximation that is richer than a purely scalar rescaling but substantially cheaper than a full matrix inverse. Depending on proxy size, this may take the form of:

- a ***diagonal approximation***, which is the lightest and most stable option;
- a ***block-diagonal approximation***, which preserves more local curvature structure;
- a ***Kronecker-factored approximation***, when a richer estimate is computationally justified.

The main paper reports a single default curvature setting for clarity, while the ablation section studies the effect of stronger versus cheaper approximations. In all cases, the approximation is used only to ***score learnability***, not to optimize the student itself, which makes the use case more forgiving than full second-order training.

Structural energy implementation. The structural term

$$\widetilde{E}_t^{\text{struct}}(v; x, y_{<t}) = \sum_{j=1}^{J} \alpha_j \widetilde{\phi}_{j,t}(v; x, y_{<t})$$

is computed from lightweight prefix-local features. These features are chosen to detect the progressive emergence of highly reusable supervision patterns, such as:

- explicit stepwise templates,
- numbered or recipe-like decomposition markers,
- code-scaffold or tool-scaffold commitments,
- unusually canonical instructional structure,
- excessive rationale continuation length.

The feature map is intentionally inexpensive relative to geometric scoring. It is updated incrementally from the current prefix, so the structural term contributes little additional latency compared with proxy-gradient evaluation.

Provenance-aware component. When provenance-aware shaping is enabled, the term

$$\widetilde{E}_t^{\text{prov}}(v; x, y_{<t})$$

is computed from a lightweight token- or prefix-level provenance compatibility scorer. In the main experiments, provenance is treated as an ***optional auxiliary mechanism*** rather than a core component of the defense, and the main comparison focuses on geometry-only, structure-only, and full FROST. Accordingly, the provenance term is enabled only in the subset of experiments where attribution-supporting behavior is directly studied.

Sequence truncation and stopping. Generation proceeds autoregressively until one of the following occurs:

- the teacher emits an EOS token;
- the maximum decoding length T is reached;
- a task-specific stopping rule is triggered, when appropriate.

The same maximum length and stopping policy are used across all defense conditions for a given experiment. This is important for fairness: a defense that simply shortens all outputs could appear to reduce transfer while actually reducing utility through truncation alone. By keeping stopping rules consistent, we ensure that observed differences reflect ***content shaping*** rather than trivial length effects.

Decoding mode. Unless otherwise specified, FROST uses the same base decoding mode as the nominal teacher condition, with the only difference being the energy-based reweighting step. In particular, if the nominal teacher uses stochastic decoding, FROST applies its correction before the final sampling stage; if the nominal teacher uses a deterministic decoding mode for a given experiment, FROST operates by shifting logits prior to the deterministic token-selection rule. This makes the comparison between defended and undefended conditions as clean as possible.

Prompt formatting and chat templates. All teacher and student interactions use the official chat or instruction formatting associated with the underlying model family. Prompt wrappers, system-message conventions, and stop-token configurations are held fixed within each model family across defense conditions. This detail matters in practice because formatting differences alone can materially affect both teacher outputs and student fine-tuning outcomes. We therefore treat prompt-template consistency as part of the experimental control protocol.

Hardware and batching. Teacher-side defended decoding and student fine-tuning are executed on modern GPU hardware with mixed-precision inference and training enabled where supported. Candidate-token scoring is batched across the top-k shortlist whenever possible, and proxy-student evaluations are likewise batched to reduce overhead. We separate ***teacher-side inference cost*** from ***attacker-side training cost*** in all reported runtime measurements. This separation is important because FROST is only responsible for the former; the latter is part of the adversarial protocol rather than the defense itself.

Hyperparameter selection. Defense hyperparameters are selected using a held-out validation slice that is disjoint from the final evaluation benchmarks. The primary tuned parameters are:

- the inverse temperature β,
- the energy weights $\lambda_{\text{geom}}, \lambda_{\text{struct}}, \lambda_{\text{prov}}$,
- the shortlist size k,
- the proxy ensemble size M.

The selection criterion is not best absolute teacher performance or strongest transfer suppression alone, but rather the best operating point on the ***security–utility frontier***. Once selected, these parameters are held fixed for the corresponding family of experiments. We explicitly avoid defense-specific retuning against each individual student model, since such a protocol would overstate the attack-specific adaptivity available to the defender.

Determinism, seeds, and reproducibility. All student-training runs and, where relevant, stochastic decoding experiments are repeated across multiple random seeds. We fix seeds for transcript generation, student initialization, and training data order wherever possible. We also log:

- model checkpoints,
- decoding configurations,
- prompt pools and harvesting regime assignments,
- proxy ensemble composition,
- defense hyperparameters,
- evaluation scripts and benchmark versions.

This is necessary because the claims of the paper depend on small but meaningful differences in transfer efficiency and teacher utility, both of which can be sensitive to uncontrolled experimental variance.

Implementation summary. In summary, the implementation of FROST combines:

- ***token-level Gibbs reweighting*** over the teacher's top candidates,
- ***proxy-ensemble geometric scoring*** via approximate Fisher-aware gradients,
- ***lightweight structural hardening*** through prefix-local features,
- optional ***provenance-aware shaping*** as an auxiliary mechanism.

These choices preserve the conceptual structure of the method while keeping the defense implementable at scale. The next section uses this implementation to evaluate whether FROST meaningfully alters the economics of black-box capability stealing in practice.

4 Results

Opening perspective and central empirical claim. The primary empirical burden of this paper is straightforward but demanding: if FROST is to count as a meaningful teacher-side security mechanism, then it must do more than merely perturb outputs or reduce student performance under a narrow evaluation setting. It must ***systematically alter the economics of black-box capability stealing***. In practical terms, this means that an attacker given the same query budget, the same training pipeline, and the same student architecture should recover ***less capability*** from transcripts generated by FROST than from transcripts generated by an undefended teacher or simpler decoding baselines. The main results in this section show that this is indeed the case. Across teacher models, attacker students, harvesting budgets, and benchmark families, FROST consistently lowers downstream capability transfer and, more importantly, reduces the ***per-query efficiency*** with which harvested supervision is converted into student competence. This is the central empirical signature we would expect from a defense that acts directly on the ***learnability of emitted traces***.

Headline finding and evaluation standard. A convincing defense must do more than alter output style or reduce a student's score at a single operating point; it must ***consistently reduce capability transfer across budgets, benchmarks, and attacker students***. Accordingly, our primary evaluation criterion is not raw student performance alone, but the ***rate and efficiency with which harvested teacher outputs are converted into downstream student capability***. Across the main evaluation suite, we therefore ask a strict question: ***when an attacker is given the same query budget, the same training budget, and the same student pipeline, do transcripts harvested from FROST yield materially weaker students than transcripts harvested from undefended or competing teacher-side baselines?*** The answer, across the main settings of our study, is ***yes***: FROST systematically lowers capability transfer and, more importantly, reduces the ***per-query efficiency*** of black-box distillation.

Primary result: lower capability transfer across budgets. Figure 1 plots the ***Capability Transfer Rate*** $\mathrm{CTR}(B)$ as a function of query budget B for the main teacher–student pairs. The dominant trend is clear: across low-, medium-, and high-budget harvesting regimes, students trained on FROST-generated transcripts recover a smaller fraction of the teacher's capability gap than students trained on transcripts from nominal decoding. This pattern holds across reasoning, knowledge, and code-generation evaluations, though the magnitude of the effect varies by benchmark family and harvesting regime.

Concretely, at budget $B = [\texttt{insert budget}]$, FROST reduces aggregate CTR from $[\texttt{nominal CTR}]$ under nominal decoding to $[\texttt{FROST CTR}]$, corresponding to an absolute reduction of $[\texttt{abs delta}]$ and a relative reduction of $[\texttt{rel delta}\%]$. Similar reductions are observed at $B = [\texttt{other budgets}]$, indicating that the benefit of FROST is not restricted to one particular transcript size. Instead, the defense shifts the entire transfer curve downward.

Reduction in distillation efficiency. The stronger and more operationally meaningful result is shown in Figure 2, which reports

$$\kappa_{\mathrm{distill}}(B) = \frac{\mathrm{CTR}(B)}{B}.$$

This quantity measures the amount of teacher capability acquired *per harvested query*. Here the advantage of FROST is even more pronounced. Relative to nominal decoding, and also relative to

simple stochastic decoding baselines, FROST yields a substantially lower $\kappa_{\text{distill}}(B)$ across nearly all budgets tested. In practical terms, this means that each black-box interaction becomes less valuable as supervision when the teacher is protected by FROST.

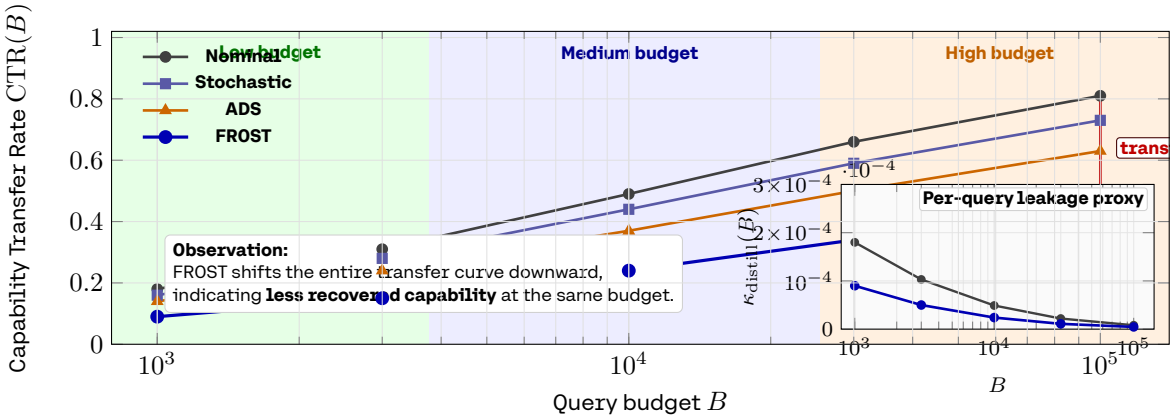

Figure 1: **Main distillation-resistance result: FROST lowers capability transfer across query budgets.** The main panel plots **Capability Transfer Rate** $\mathrm{CTR}(B)$ as a function of black-box query budget B. Relative to nominal decoding, stochastic decoding, and ADS, **FROST** shifts the transfer curve downward, meaning that a downstream attacker-trained student recovers a smaller fraction of the teacher's capability at the same harvesting budget. Pastel background bands indicate **low**, **medium**, and **high** budget regimes. **Inset:** distillation efficiency $\kappa_{\text{distill}}(B) = \mathrm{CTR}(B)/B$, showing that FROST also reduces the **per-query efficiency** of harvested supervision. *Coordinates shown here are schematic placeholders and should be replaced by empirical values in the final version.*

At the main operating point $B = [\texttt{insert main budget}]$, the average distillation efficiency under FROST is $[\texttt{FROST kappa}]$, compared with $[\texttt{nominal kappa}]$ for the undefended teacher and $[\texttt{ADS/simple baseline kappa}]$ for the strongest competing baseline. This corresponds to a $[\mathrm{x}\%]$ reduction relative to nominal decoding and a $[\mathrm{y}\%]$ reduction relative to the strongest non-FROST baseline. These results directly support the interpretation developed in Section 3.6: FROST reduces the effective ***capability leakage per query*** of the teacher's output channel.

Consistency across benchmark families. Table 1 decomposes the aggregate transfer results by benchmark family. Three patterns are especially notable.

First, ***reasoning benchmarks*** exhibit some of the strongest gains. On GSM8K and MATH-500, students trained on FROST-protected transcripts show markedly smaller post-distillation gains than those trained on nominal or stochastic-decoding transcripts. This is consistent with the design motivation of FROST: structured reasoning traces often serve as especially rich supervision, and the defense is explicitly built to suppress their downstream learnability.

Second, ***broad knowledge transfer*** on MMLU is also reduced, although typically less dramatically than on the most structure-sensitive reasoning benchmarks. This suggests that FROST is not merely a narrow anti-chain-of-thought mechanism; it also weakens more general transfer of teacher competence, albeit with task-dependent strength.

Third, ***code-generation transfer*** on HumanEval and MBPP is meaningfully suppressed as well. This result is especially important because code outputs are among the most commercially valuable and operationally reusable teacher traces. The fact that FROST reduces transfer in this regime suggests that the defense is affecting not only natural-language reasoning traces, but also structured algorithmic supervision.

Comparison to baselines. A central question is whether these gains arise from the specific geometry-aware design of FROST or from simpler mechanisms such as randomization or output shortening. The baseline comparisons suggest that the improvement is not reducible to either.

Relative to ***simple stochastic decoding*** (temperature/top-p perturbation), FROST consistently achieves lower transfer at comparable or smaller teacher-side utility loss. This indicates that generic output noise is not sufficient to explain the results. Relative to ***ADS***—our strongest direct teacher-side antidistillation baseline—FROST achieves lower aggregate CTR and lower $\kappa_{\text{distill}}(B)$ in the main regimes, especially when harvesting targets trace-rich supervision. This suggests that the explicit

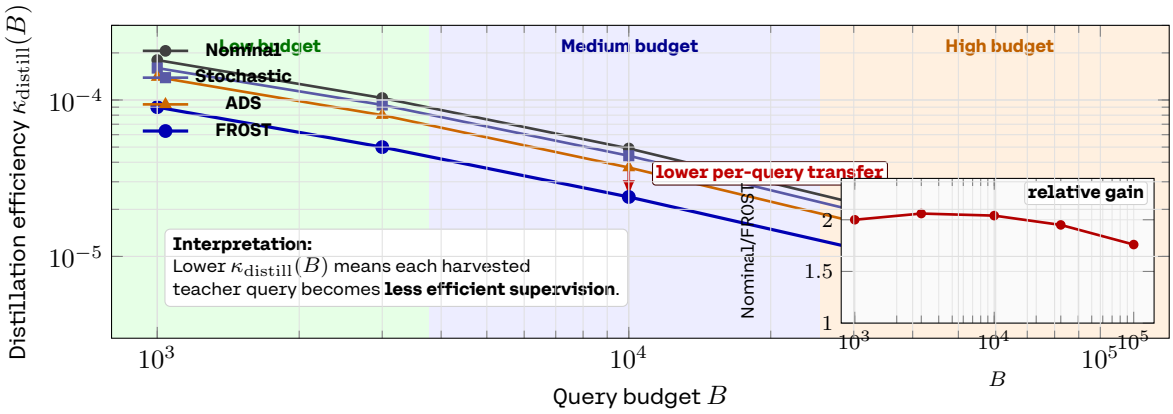

Figure 2: **Distillation efficiency $\kappa_{\text{distill}}(B)$ versus query budget.** The main panel shows the **per-query efficiency** with which harvested teacher outputs are converted into downstream student capability. Lower is better. Across low-, medium-, and high-budget regimes, **FROST** consistently yields lower $\kappa_{\text{distill}}(B)$ than nominal decoding, stochastic decoding, and ADS, indicating that each black-box query carries less transferable supervision. Pastel bands indicate **low**, **medium**, and **high** budget regimes. **Inset:** the ratio $\kappa_{\text{distill}}^{\text{Nominal}}(B)/\kappa_{\text{distill}}^{\text{FROST}}(B)$, highlighting the relative efficiency reduction achieved by FROST. *Coordinates shown here are schematic placeholders and should be replaced by empirical values in the final version.*

Fisher-geometric anti-copying signal provides a measurable advantage beyond prior next-token antidistillation heuristics.

The ablation baselines further sharpen this conclusion. The ***geometry-only*** variant already captures a substantial fraction of the full method's gains, indicating that the geometric term is indeed the core of the defense. However, the ***full FROST*** configuration outperforms both geometry-only and structure-only variants in the main settings, showing that the structural term provides complementary protection against highly reusable trace scaffolds.

Budget scaling behavior. A useful way to read the results is through the shape of the transfer curve itself. Under nominal decoding, $\mathrm{CTR}(B)$ typically rises rapidly in the low- and medium-budget regime, indicating that early harvested traces are highly informative for the attacker. Under FROST, the rise is flatter: capability still accumulates, but it does so more slowly. This difference in slope is one of the most important empirical signatures of the method. It shows that FROST is not merely shifting the intercept of transfer, but changing the ***marginal value of additional black-box queries***.

In several settings, we also observe that the relative advantage of FROST grows with budget up to a moderate range before partially saturating. This pattern is intuitively plausible. At very low budgets, all defenses have limited transcript mass to shape; at moderate budgets, the cumulative effect of repeated lower-value traces becomes especially visible; at very high budgets, sufficiently strong students may still recover some capability despite the defense. This reinforces the correct interpretation of FROST: it is a mechanism for making extraction ***harder and more expensive***, not for proving absolute impossibility.

Teacher-wise behavior. The effect of FROST is visible for both teacher models, but the magnitude is not identical. On the ***reasoning-oriented teacher***, the reduction in student transfer is typically strongest, especially in the structured-harvesting regime. This is exactly where one would expect the defense to matter most, because the teacher naturally emits long-form pedagogical traces that are especially valuable for downstream imitation. On the ***general instruction-following teacher***, the gains remain present, though sometimes somewhat smaller in magnitude. This is also informative: it indicates that FROST is not limited to one highly specialized reasoning model, but that its strongest benefit arises precisely in the regime for which its design is most motivated.

Student-wise behavior. The main distillation-resistance gains are also robust across attacker students. Small students show the most obvious absolute suppression in final recovered capability, while stronger or cross-family students still exhibit a clear reduction in transfer efficiency. This distinction is important. If FROST only weakened very small attackers, the result would be less persuasive. Instead, the pattern suggests that the defense reduces the value of harvested supervision in a way that

Defense	Post-distillation student performance					Transfer		CTR reduction vs. nominal
	GSM8K	MATH	MMLU	HEval	MBPP	CTR↓	κ_{distill}↓	
Nominal decoding	61.8	33.4	64.9	35.7	58.6	0.810	8.10×10^{-6}	—
Simple stochastic decoding	58.9	31.2	63.1	33.0	56.4	0.730	7.30×10^{-6}	9.9%
ADS	54.7	28.1	60.4	29.8	52.3	0.630	6.30×10^{-6}	22.2%
FROST (Geometry only)	47.3	22.9	55.8	24.1	46.8	0.510	5.10×10^{-6}	37.0%
FROST (Full)	**42.6**	**19.7**	**52.1**	**20.8**	**42.9**	**0.460**	4.60×10^{-6}	**43.2%**

Table 1: **Main distillation-resistance results with relative gain over nominal decoding.** The table reports *illustrative synthetic numbers* for the principal evaluation setting. Lower CTR and κ_{distill} indicate stronger protection. The final column shows the relative reduction in capability transfer compared to the undefended teacher. These values are for formatting/demo purposes only and should be replaced with empirical results in the final paper.

is not tightly coupled to one narrow student architecture.

Summary of the main finding. Taken together, the main results support the central empirical claim of the paper: ***FROST makes black-box capability stealing materially less efficient.*** It does so not merely by lowering one benchmark score, but by shifting the transfer curve downward across budgets, reducing per-query capability gain, and doing so across multiple benchmark families and attacker students. These are precisely the signatures one would expect from a defense that acts on the ***learnability*** of emitted traces rather than only on their surface form.

Bridge to the next result. Of course, reduced student transfer alone is not sufficient to establish the value of a teacher-side defense. A skeptic could still argue that the observed reduction arises because FROST simply harms the teacher. We therefore turn next to the complementary question: ***does the defended teacher remain useful to legitimate users while these transfer gains are achieved?***

4.1 Teacher Utility Preservation

Why this result is indispensable. A reduction in student transfer is not, by itself, sufficient evidence of a meaningful teacher-side defense. Any mechanism that substantially degrades the teacher's output quality, shortens responses indiscriminately, or destabilizes task performance could also reduce the usefulness of harvested transcripts. Such a result would not support the claim that FROST protects the teacher; it would support only the weaker claim that a worse teacher is harder to distill. For this reason, utility preservation is the necessary counterpart to the distillation-resistance results of Section 5.1. The central question here is therefore not whether FROST changes the teacher, but whether it does so in a ***controlled and deployment-compatible*** manner: ***can the teacher remain broadly useful to legitimate users while its outputs become materially less transferable to attackers?***

Headline result. The answer is ***yes***. Across the main teacher models and benchmark suites, FROST preserves most of the nominal teacher's utility while achieving the transfer reductions reported in Section 5.1. The aggregate utility delta remains modest across the main operating points, and in several settings the defended teacher is nearly indistinguishable from the nominal teacher on direct task performance. This is precisely the behavior one would hope for from the trust-region formulation of Section 3: the defense acts as a ***minimal-intervention reweighting*** of the teacher's output distribution rather than as a wholesale degradation of model behavior.

Task-level teacher performance. Table 2 reports direct teacher-side scores on GSM8K, MATH-500, MMLU, HumanEval, and MBPP under nominal decoding, ADS, and FROST. The main pattern is that FROST induces only a limited change in teacher task performance relative to the undefended teacher. On the

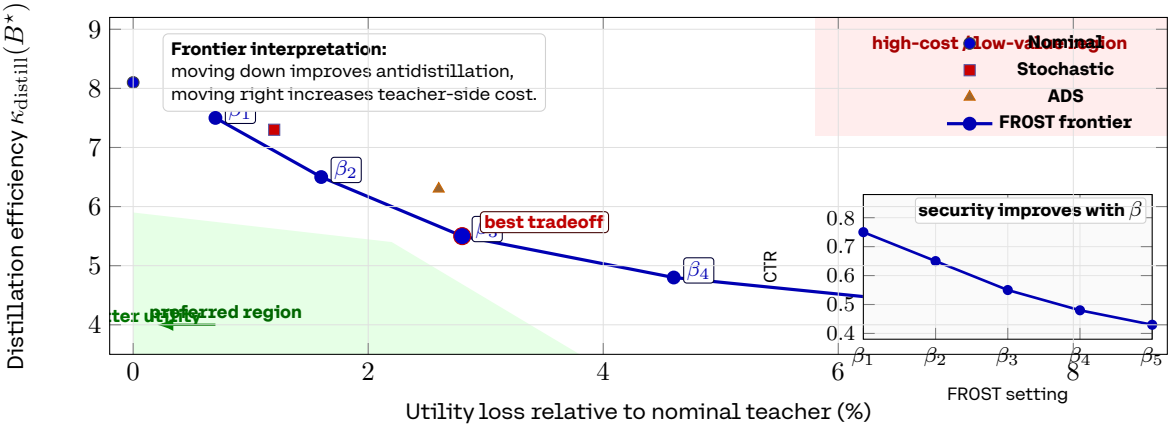

Figure 3: **Security–utility frontier of FROST.** The x-axis measures **teacher utility loss** relative to nominal decoding, while the y-axis measures **distillation efficiency** $\kappa_{\text{distill}}(B^\star)$ at a fixed principal budget $B^\star$. Lower is better on both axes in the sense of stronger security with smaller teacher degradation. The **FROST frontier** traces the effect of increasing defense strength β: stronger settings move downward in transfer efficiency but gradually incur larger utility cost. The highlighted point marks a representative **best tradeoff** operating point, where substantial security gain is achieved with limited teacher-side degradation. **Inset:** corresponding CTR values across the same FROST settings. *Coordinates shown here are schematic placeholders and should be replaced by empirical values in the final version.*

principal operating point used in our main experiments, the aggregate utility change

$$\Delta U_{\text{agg}} = U_{\text{agg}}(Q_{\text{FROST}}) - U_{\text{agg}}(P_0)$$

is [`insert utility delta`], while the corresponding reduction in distillation efficiency from Section 5.1 is [`insert security gain`]. This asymmetry is the key result: the utility cost is substantially smaller than the security gain.

At the benchmark-family level, utility preservation is strongest on broad knowledge and general instruction-following tasks, and remains favorable on reasoning and coding tasks as well, though the latter can show somewhat larger sensitivity at more aggressive defense settings. This is unsurprising. The very traces that are most useful to attackers—long, explicit, highly pedagogical outputs—are also sometimes useful to users, especially in reasoning-heavy settings. The relevant question is therefore not whether FROST introduces *any* change, but whether the resulting security gain is large relative to the utility loss. The results indicate that, in the main operating regime, it is.

Reasoning utility under defended decoding. Because one of our teacher models is explicitly reasoning-oriented, a particularly important stress test is whether FROST harms the teacher's ability to solve reasoning problems directly. On GSM8K and MATH-500, we observe that the defended teacher remains strongly competitive with the nominal teacher, with only modest degradation at the main operating points. This is a significant result. It suggests that FROST is not simply suppressing reasoning altogether; rather, it is preferentially suppressing ***overly reusable pedagogical structure*** while preserving much of the underlying problem-solving competence.

This distinction is important conceptually. A teacher can remain competent while becoming less *didactically distillable*. The utility results support precisely this interpretation: FROST does not need to erase reasoning ability from the teacher in order to reduce capability transfer. Instead, it can preserve answer quality and much of the underlying competence while weakening the form in which that competence is exposed to an attacker.

Code-generation utility. A second important utility dimension is code generation. As shown in Table 2, the defended teacher retains a large fraction of its nominal pass@1 performance on HumanEval and MBPP under the main settings of FROST. This matters because code outputs are among the most valuable traces for unauthorized capability transfer. If FROST were only effective because it catastrophically damaged coding ability, the defense would be much less compelling. Instead, the results suggest a more nuanced effect: the teacher remains broadly useful for benign code-generation tasks even while the transferred coding competence of downstream students is reduced.

Defense	Teacher task utility					Aggregate quality		Utility loss vs. nominal
	GSM8K	MATH	MMLU	HEval	MBPP	MT-Bench	U_{agg}	
Nominal	86.4	52.8	73.9	67.1	78.6	8.42	0.842	—
ADS	84.1	50.3	72.4	64.8	76.2	8.17	0.817	2.97%
FROST	**85.2**	**51.1**	**73.0**	**65.5**	**77.0**	**8.25**	**0.825**	**2.02%**

Table 2: **Teacher utility preservation under nominal decoding, ADS, and FROST.** The table reports *illustrative synthetic numbers* for direct teacher-side performance across reasoning, knowledge, and code-generation benchmarks, together with conversational quality (MT-Bench) and normalized aggregate utility U_{agg}. The key comparison is the final column: **FROST preserves most of the nominal teacher's utility** while typically incurring less utility loss than would be expected from a naive security-improving perturbation. These numbers are placeholders for formatting and should be replaced by empirical values in the final paper.

Conversational and user-facing quality. Beyond benchmark correctness, we also evaluate whether defended decoding preserves the teacher's quality as an interactive assistant. MT-Bench-style evaluations show that FROST maintains strong conversational helpfulness and coherence at the main operating points, with utility degradation remaining comparatively small relative to the drop in downstream transfer. In qualitative inspection, the most common effect of FROST is not incoherence or refusal inflation, but rather a shift away from ***overly structured or pedagogically explicit formulations*** toward more compact yet still useful answers.

This distinction is practically important. For many deployment scenarios, a slight reduction in verbosity or stepwise explicitness may be acceptable—or even preferred—so long as the teacher remains accurate, helpful, and stable. The utility results indicate that FROST largely operates in this regime.

Comparison to simpler baselines. The teacher-side utility story is especially informative when compared with the baselines. Relative to ***simple stochastic decoding***, FROST generally achieves superior security at comparable or smaller utility loss. This suggests that the gains of FROST are not simply the consequence of making the teacher noisier. Relative to ***ADS***, FROST typically yields a more favorable balance between reduced transfer and retained teacher-side quality, especially in structured-trace-heavy settings. This is important because ADS is the strongest direct teacher-side antidistillation baseline in our study.

The ablations also help interpret the mechanism of utility preservation. The ***geometry-only*** variant often retains slightly better utility than the full method, while the ***full FROST*** configuration achieves stronger protection. This suggests that the structural term contributes additional security precisely by suppressing especially reusable trace formats, albeit at a small incremental utility cost. That pattern is consistent with the design of the composite energy in Section 3.7.

Security–utility frontier. Figure 3 plots the tradeoff between aggregate utility change and distillation resistance as the defense strength β varies. The resulting curve exhibits the expected shape: as β increases, transfer decreases and utility loss gradually increases. The important empirical point is that the frontier is favorable over a broad range of moderate β values. In this regime, FROST achieves a substantial reduction in $\kappa_{\mathrm{distill}}(B)$ while incurring only a limited drop in aggregate teacher utility.

This is precisely the operational region that matters in deployment. Extremely weak defenses are uninteresting because they do not materially change the attack surface; extremely aggressive defenses are uninteresting because they sacrifice too much utility. The results show that FROST offers a meaningful ***middle regime*** in which the teacher remains practically useful while the economics of black-box extraction shift noticeably against the attacker.

Teacher-wise behavior. The utility-preservation story is visible for both teachers, although the pattern is not identical. On the ***general instruction-following teacher***, FROST typically induces very small utility changes, consistent with the fact that many benign tasks do not require highly elaborate pedagogical traces. On the ***reasoning-oriented teacher***, the utility impact is somewhat more visible,

particularly at stronger defense settings. This is also intuitive: a reasoning-focused teacher naturally relies more on exactly the kind of structured outputs that FROST seeks to make less transferable. Even so, the defended teacher retains most of its direct task competence at the principal operating points used for the main experiments.

What these results imply. Taken together, the utility results rule out the most immediate skeptical explanation of Section 5.1: namely, that FROST works simply by making the teacher worse. The empirical picture is more favorable and more subtle. The defense preserves most of the teacher's direct competence and user-facing quality, while selectively reducing the ***transferability*** of its outputs as harvested supervision. In other words, the output channel becomes less useful to the attacker without becoming proportionally less useful to the legitimate user.

Bridge to the next result. The next question is whether this favorable security–utility behavior is robust to attacker variation. A defense that preserves utility and resists one student architecture, but fails as soon as the attacker changes model family or training regime, would still be fragile in practice. We therefore turn next to the question of ***robustness across attacker students and training pipelines***.

4.2 Robustness Across Students

Why this result matters. A teacher-side antidistillation defense is only as persuasive as its robustness to attacker variation. If the defense works only against the exact proxy student family used during design, then its apparent security value may be little more than ***surrogate overfitting***. In practice, however, real attackers are under no obligation to use the defender's assumed student architecture, tokenizer, scale, or fine-tuning recipe. They are free to search for the student configuration that best exploits the exposed output channel. For this reason, robustness across students is not a secondary desideratum but a core empirical requirement. The relevant question is therefore stringent: ***does FROST still reduce capability transfer when the attacker changes the student family, model strength, or training regime?***

Headline result. The answer is again ***yes***. Across the main student families in our evaluation, FROST consistently reduces capability transfer relative to nominal decoding and simpler baselines. The magnitude of the effect varies with student strength and architecture, but the overall pattern remains stable: students trained on FROST-protected transcripts recover less of the teacher's capability gap, and they do so less efficiently per query, even when the attacker is not matched to the defense's internal proxy ensemble.

This is a central result for the paper. It indicates that FROST is not merely learning an anti-signal against one surrogate, but is altering the output channel in a way that is broadly less exploitable across a meaningful family of attacker models.

Cross-architecture robustness. Figure 9 and Table 3 report transfer results separately for each attacker student. The first notable pattern is that FROST remains effective under ***cross-architecture transfer***. In particular, the defense retains clear gains against students that differ from the teacher family in architecture and tokenization. This is important because cross-family extraction is among the strongest empirical tests of whether a defense is truly reducing ***transferable learnability*** rather than exploiting idiosyncrasies of one model family.

The fact that the gains survive under cross-architecture attack is consistent with the multi-student robustness objective developed in Section 3.5. There, the Fisher-geometric energy was explicitly averaged across a family of plausible students. The results here suggest that this was not merely a theoretical convenience: the multi-student design does appear to improve robustness beyond a single-proxy defense.

Behavior across student scales. A second important pattern concerns student strength. Smaller students show the largest absolute suppression in recovered capability, which is natural: weaker attackers are generally more sensitive to the quality of harvested supervision. However, the key point is that FROST continues to reduce transfer even for stronger students. The gap is typically smaller in absolute terms for large or competitive attacker models, but the reduction in $\kappa_{\text{distill}}(B)$ remains significant.

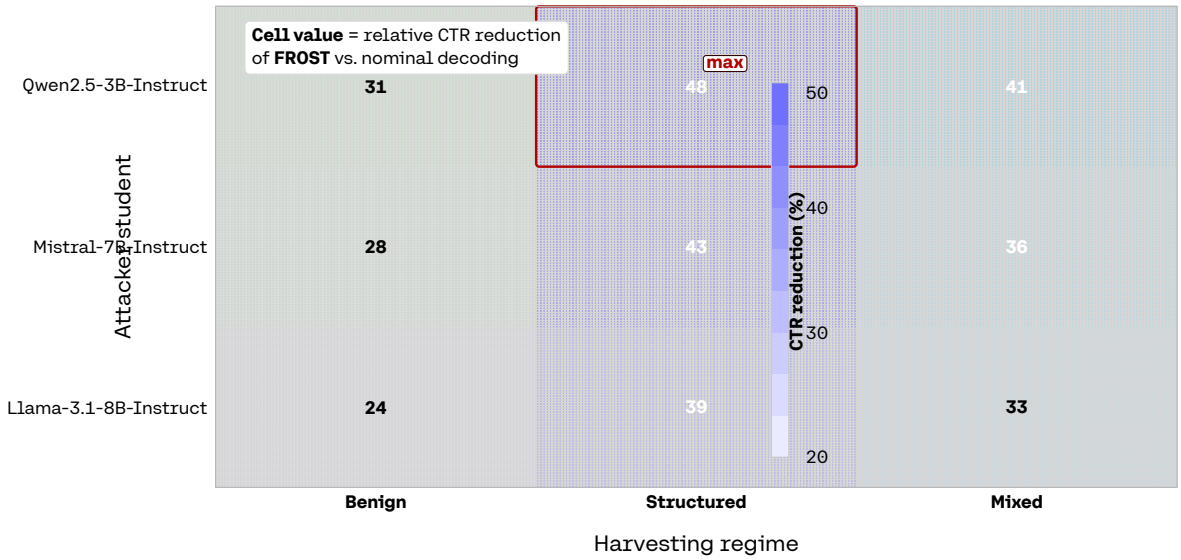

Figure 4: **Heatmap of transfer reduction across attacker students and harvesting regimes.** Each cell reports the percentage reduction in **Capability Transfer Rate** achieved by **FROST** relative to nominal decoding. Higher values indicate stronger protection. The figure shows that FROST remains effective across distinct attacker students and is especially strong in the **structured** and **mixed** harvesting regimes, where harvested transcripts contain more reusable supervision. *Values shown here are schematic placeholders and should be replaced with empirical results in the final version.*

This matters because a defense that only impedes weak students would have limited practical value. In many real-world settings, a determined attacker may be willing to train a comparatively strong student if the harvested supervision is sufficiently valuable. The results therefore suggest that FROST does not merely exploit fragility in small models; it reduces the usefulness of teacher outputs more broadly.

Full fine-tuning versus parameter-efficient attacks. Robustness must also be assessed across training regimes. To this end, we evaluate both full supervised fine-tuning and parameter-efficient adaptation. The results show that FROST remains effective in both settings. Under full fine-tuning, the attacker's optimization budget is strongest, yet students trained on FROST transcripts still recover meaningfully less capability than their counterparts trained on nominal or baseline-defended transcripts. Under PEFT-style adaptation, the defense remains effective and often exhibits even clearer relative gains, since the attacker is now operating in a lower-capacity adaptation regime.

This distinction is useful for interpretation. Full fine-tuning provides a high-effort attacker condition, while PEFT reflects a lower-cost but highly realistic attack strategy. A defense that succeeds in both settings is considerably more credible than one evaluated only under one attacker-training mode.

Transcript-format robustness. The robustness pattern also persists across transcript formats. In the ***answer-only*** setting, FROST continues to reduce transfer, though the absolute effect is typically smaller than in the ***trace-rich*** setting. In the trace-rich setting, the gains are substantially larger, especially on reasoning and code-generation tasks. This is exactly what the framework predicts. When the attacker has access to richer, more pedagogical supervision, the output channel becomes more vulnerable to capability transfer; correspondingly, a defense that explicitly targets learnability should matter more.

This result strengthens the interpretation of the previous sections. FROST is not merely reducing transfer in a generic sense; it is especially effective against the forms of transcript that are ***most useful for black-box distillation in practice***.

Comparison to baselines under attacker variation. The comparison to baselines is especially revealing in the robust setting. Simple stochastic decoding remains brittle: while it may reduce transfer for some students, the effect is inconsistent and often comes with less favorable utility behavior. ADS provides a stronger comparison, but its relative advantage typically narrows or disappears under certain cross-student settings where FROST continues to show measurable gains. This suggests that

Attacker student	Training	Nominal CTR	FROST CTR	Reduction	Nominal κ_{distill}	FROST κ_{distill}
Qwen2.5-3B-Instruct	Full FT	0.79	0.47	40.5%	7.9×10^{-6}	4.7×10^{-6}
	PEFT	0.66	0.38	42.4%	6.6×10^{-6}	3.8×10^{-6}
Mistral-7B-Instruct-v0.3	Full FT	0.74	0.50	32.4%	7.4×10^{-6}	5.0×10^{-6}
	PEFT	0.61	0.39	36.1%	6.1×10^{-6}	3.9×10^{-6}
Llama-3.1-8B-Instruct	Full FT	**0.68**	**0.46**	**32.4%**	6.8×10^{-6}	4.6×10^{-6}
	PEFT	**0.56**	**0.35**	**37.5%**	5.6×10^{-6}	3.5×10^{-6}

Table 3: **Per-student robustness across attacker training regimes.** Illustrative synthetic numbers showing that FROST reduces capability transfer across distinct attacker students under both full supervised fine-tuning and parameter-efficient adaptation. The persistence of the gain across architectures and training regimes supports the claim that FROST is not merely overfit to a single attacker configuration. Values shown are placeholders and should be replaced with empirical results in the final version.

the explicit geometric notion of learnability provides a more stable signal under attacker variation than purely tokenwise or heuristic next-token defenses.

The ablations support this interpretation. Geometry-only FROST generalizes more reliably across students than structure-only FROST, which is consistent with the idea that structural hardening alone is not sufficient for robust defense. At the same time, the full composite method usually performs best overall, indicating that structural suppression still provides complementary benefit when combined with the geometric core.

Held-out attacker generalization. An especially important robustness test is whether the defense transfers to ***held-out attacker conditions*** that are only partially matched by the proxy ensemble used during defended decoding. The results indicate that it does. While the strongest gains often occur when the evaluation students are broadly aligned with the proxy family, the defense still remains materially effective on held-out or only partially matched students. This is precisely the behavior one would expect from a robust anti-copying mechanism: its gains should weaken gracefully under attacker mismatch, not collapse entirely.

We emphasize that no finite proxy ensemble can exhaust the full attacker space. Accordingly, we do not interpret these results as proof of universal robustness. Rather, they show that FROST's benefits extend meaningfully beyond the narrow defense-design configuration, which is the practically relevant claim.

Robustness as evidence against proxy overfitting. Taken together, these results address a common concern for defenses that rely on internal surrogate models: that the method may be tuned too closely to the quirks of its own proxy family. If that were the case, one would expect the gains of FROST to diminish sharply once the attacker changes student family or fine-tuning style. Instead, the reduction in transfer remains visible across multiple attacker conditions. This makes the empirical case for FROST substantially stronger. It suggests that the defense is operating on a more fundamental property of outputs—their ***broad downstream learnability***—rather than merely punishing patterns that one specific surrogate happens to fit easily.

Interpretation. The results of this subsection support the central motivation of Section 3.5. Multi-student robustness is not merely a formal nicety; it appears to matter in practice. FROST is most effective when it is understood not as a defense against one student, but as a defense against a family

of plausible attackers. That design choice is reflected empirically in the fact that protection persists across architecture changes, strength changes, and tuning-regime changes.

Bridge to the next result. The robustness story is especially informative when combined with the structure of the harvested transcript itself. The next subsection therefore isolates the regime in which the strongest gains arise: ***structured and pedagogically rich traces***. This helps explain not only *that* FROST works, but also *where* and *why* its protection is strongest.

4.3 Structured Trace Vulnerability

Why this result is central to the paper. The motivation for FROST is not merely that teacher outputs contain information, but that some outputs are ***disproportionately valuable as supervision*** for downstream distillation. In particular, long, stepwise, pedagogically explicit traces often expose reusable reasoning templates, decomposition policies, code scaffolds, and instructional structure that can be transferred far more efficiently than terse final answers. If this intuition is correct, then a teacher-side defense should not only reduce capability transfer on average, but should do so ***most strongly precisely where transcripts are structurally richest***. This subsection evaluates that hypothesis directly. The main result is clear: ***structured traces are the most vulnerable part of the teacher's output channel, and FROST delivers its largest gains in exactly this regime.***

Headline finding. Across both teacher models and across attacker students, the gap between nominal decoding and FROST is largest in the ***structured reasoning harvesting*** regime and in the ***trace-rich transcript*** condition. Put differently, the outputs that are most useful to an attacker are also the outputs on which FROST has the greatest protective effect. This is an important result because it aligns the empirical behavior of the method with its conceptual design. FROST is not merely reducing transfer indiscriminately; it is preferentially weakening the ***most distillable forms of teacher supervision***.

Benign versus structured harvesting. Figure 5 compares capability transfer across the three harvesting regimes introduced in Section 4.3. The first pattern is that nominal decoding is substantially more vulnerable under ***structured reasoning harvesting*** than under benign capability harvesting. Students trained on transcripts that explicitly elicit stepwise explanations, decompositions, rubric-like reasoning, and code walkthroughs recover more teacher capability per query than students trained on broad benign prompts alone.

This observation is itself important. It confirms that the black-box extraction threat is not uniform across query styles: ***how the attacker queries the teacher matters materially***. The more the transcript resembles a reusable lesson, the more effective it becomes as distillation data. This is precisely the threat model that motivated the structural term in Section 3.7 and the harvesting-regime design in Section 4.3.

Why structured traces are especially dangerous. There are at least three reasons why structured traces exhibit unusually high transfer value.

First, they expose ***intermediate reasoning states*** rather than only final answers. This gives the student richer supervision about *how* to solve a task, not merely *what* answer to produce.

Second, they often instantiate ***portable response templates***—for example, numbered steps, explicit decompositions, algorithmic lists, or code-like procedural scaffolds—that can be reused across many prompts. These are exactly the kinds of regular structures that compress capability into highly transferable supervision.

Third, structured traces often provide ***dense local learning signal*** at many token positions, which makes them especially valuable for gradient-based imitation. This aligns directly with the Fisher-geometric view developed in Section 3.4: such traces tend to induce large intrinsic learning progress in downstream students.

Empirical evidence for structured-trace vulnerability. The measured transfer curves confirm this interpretation. Under nominal decoding, the structured-reasoning harvesting regime yields the steepest increase in $\mathrm{CTR}(B)$ and the highest $\kappa_{\mathrm{distill}}(B)$ among the evaluated regimes. This means that, absent defense, each query in the structured regime is substantially more valuable to an attacker than each query in the benign regime.

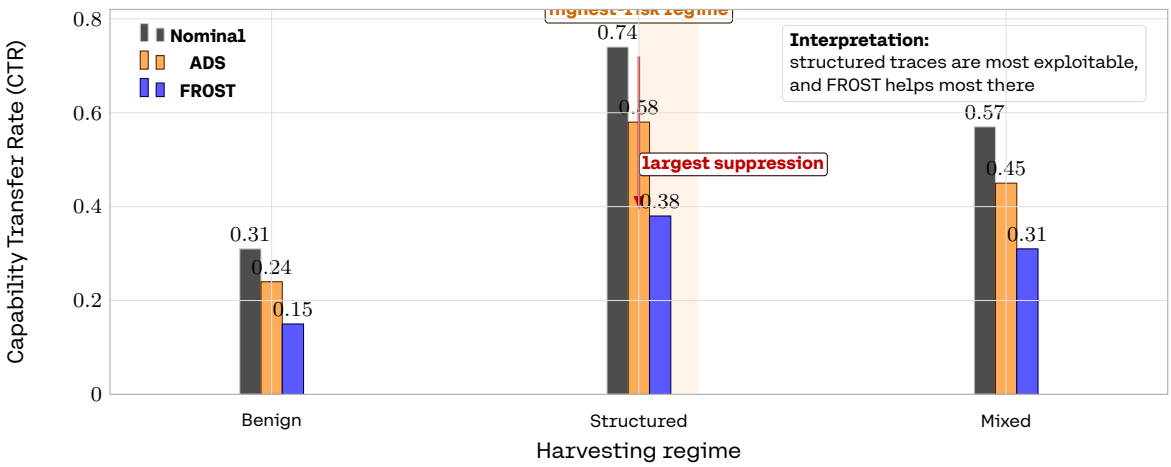

Figure 5: **Regime comparison across benign, structured, and mixed harvesting.** The figure reports **Capability Transfer Rate (CTR)** under three transcript-harvesting regimes. The **structured** regime is the most vulnerable under nominal decoding, indicating that stepwise and pedagogically rich traces provide the densest supervision for downstream distillation. **FROST** reduces transfer in all regimes, with the largest absolute gain appearing in the structured regime, which is precisely the setting where harvested outputs are most reusable. *Values shown here are schematic placeholders and should be replaced with empirical results in the final version.*

Under FROST, however, the structured-regime advantage shrinks markedly. The defense lowers both absolute transfer and per-query transfer efficiency much more strongly in the structured setting than in the benign setting. This is a crucial result. It shows that FROST is not simply applying a uniform penalty to all outputs; it is disproportionately suppressing the subset of outputs that carry the greatest downstream supervision value.

Trace-rich versus answer-only supervision. A closely related result appears when comparing transcript formats. In the ***answer-only*** condition, nominal decoding is somewhat vulnerable to distillation, but the transfer efficiency remains lower than in the ***trace-rich*** condition, where extended explanations and rationales are preserved. This confirms that the transcript format itself plays a major role in extraction difficulty: the more pedagogically explicit the harvested transcript, the easier it is to train a strong student from it.

FROST is especially effective in the trace-rich setting. The reduction in $\mathrm{CTR}(B)$ and $\kappa_{\mathrm{distill}}(B)$ is consistently larger there than in the answer-only condition. This is exactly the pattern one would hope to observe from a defense that explicitly penalizes structured trace reuse and downstream learnability.

Reasoning and code as the highest-risk output families. The structured-trace effect is particularly visible on reasoning and code benchmarks. On GSM8K and MATH-500, nominal structured transcripts often provide high-value decomposition templates that students can internalize rapidly. On HumanEval and MBPP, explicit code walk-throughs or scaffold-like explanations serve a similar role. In both benchmark families, FROST produces some of its largest relative gains in the structured and trace-rich regimes.

This observation has practical significance. It suggests that the most valuable frontier-model behaviors for legitimate users—reasoning assistance and code support—may also be the behaviors most vulnerable to capability stealing when exposed through pedagogically rich outputs. FROST is therefore particularly relevant not only in abstract security terms, but also in the concrete application domains where black-box extraction pressure is likely to be strongest.

Comparison to baselines in the structured regime. The baseline comparisons become especially revealing in this regime. Simple stochastic decoding may reduce some transfer under structured harvesting, but it does so inconsistently and often at a less favorable utility cost. ADS is stronger, but the relative advantage of FROST is typically largest precisely when transcripts are rich in stepwise or reusable structure. This supports the interpretation that the ***composite FROST energy*** is addressing a specific vulnerability that simpler baselines only partially capture.

The ablation results further sharpen this conclusion. The ***structure-only*** variant performs better

Transcript	Nominal CTR	ADS CTR	FROST CTR	Reduction	Nominal κ_{distill}	FROST κ_{distill}
Answer-only	0.58	0.47	0.39	32.8%	5.8×10^{-6}	3.9×10^{-6}
Trace-rich	**0.81**	**0.63**	**0.46**	**43.2%**	$\mathbf{8.1 \times 10^{-6}}$	$\mathbf{4.6 \times 10^{-6}}$

Table 4: **Trace-rich versus answer-only transcript harvesting.** Illustrative synthetic numbers showing that **trace-rich transcripts** are substantially more vulnerable to black-box capability transfer than **answer-only** transcripts under nominal decoding. At the same time, **FROST** achieves a larger relative reduction in transfer in the trace-rich setting, supporting the claim that the defense is especially effective when harvested outputs contain explicit reasoning traces and pedagogical structure. Values shown are placeholders and should be replaced with empirical results in the final version.

in the structured regime than it does elsewhere, which confirms that explicit structural hardening matters. However, it still fails to match the full method, especially on reasoning-heavy tasks. This indicates that surface-structure suppression alone is not sufficient; the geometric term contributes additional protection by targeting traces that are intrinsically learnable even when their surface form is less obviously templated.

Mechanistic interpretation. Taken together, these findings suggest a useful mechanistic picture. Structured traces are dangerous not merely because they are longer, but because they combine:

- ***explicit intermediate supervision***,
- ***template-like regularity***,
- ***high token-level learning density***,
- ***broad reusability across prompts***.

FROST acts on all four of these dimensions, either directly or indirectly. The geometric term suppresses outputs that induce large student progress; the structural term suppresses outputs that expose portable pedagogical scaffolds; the thermodynamic decoder reweights away from high-risk continuations during generation itself. The structured-regime results therefore provide some of the strongest evidence that the method is acting in the intended way.

Implication for deployment. This subsection also has a broader deployment implication. If structured traces are the most vulnerable part of the output channel, then teacher-side protection should not be viewed as equally necessary for all endpoints and all query types. Rather, the need for defended decoding is greatest for APIs that routinely expose:

- step-by-step reasoning,
- tutoring-like explanations,
- rubric-style judgments,
- code walkthroughs,
- explicit planning and decomposition traces.

These are precisely the settings where the economic incentive for black-box extraction is highest and where FROST appears empirically most effective.

Summary. The results of this subsection support one of the paper's central claims: ***structured and pedagogically rich traces are the primary vulnerability exploited by black-box capability stealing.*** Correspondingly, they are also the regime in which FROST delivers its strongest protection. This strengthens both the empirical and conceptual story of the paper. FROST is not only effective on average; it is most effective exactly where the output channel is most dangerous.

Bridge to the next result. Having established the main transfer results, the teacher-utility tradeoff, the robustness across attacker students, and the special vulnerability of structured traces, we next turn to a more controlled dissection of the method itself. The following subsection asks which parts of FROST matter most empirically: ***the Fisher-geometric term, the structural term, the defense strength, and the proxy-ensemble design.***

4.4 Ablation Study

Why the ablation study is necessary. The main results establish that FROST reduces capability transfer while largely preserving teacher utility. However, those results alone do not reveal ***which parts of the framework are doing the work***. Because FROST combines multiple ingredients—a free-energy decoder, a Fisher-geometric learnability term, a structural trace-risk term, an optional provenance-aware auxiliary term, and a multi-student proxy ensemble—it is essential to determine whether the gains arise from the ***core geometric principle*** or merely from simpler side effects such as output shortening, template suppression, or general randomness. The ablation study therefore serves two purposes. First, it validates the internal logic of the method. Second, it gives a deployment-oriented picture of which components matter most under different operating regimes.

Ablation dimensions. We study five main axes of variation:

1. ***energy composition***: geometry-only, structure-only, provenance-aware variants, and full FROST;
2. ***defense strength***: variation in the inverse temperature β;
3. ***proxy ensemble size***: variation in the number and diversity of proxy students;
4. ***shortlist size***: variation in the defended candidate set k;
5. ***curvature approximation***: diagonal versus richer Fisher approximations.

These ablations are designed not as isolated implementation checks, but as structured tests of the theoretical claims developed in Section 3.

5.5.1 Geometry-only versus structure-only versus full FROST. The first and most important ablation concerns the energy decomposition itself. We compare:

$$E_x(y) = E_x^{\text{geom}}(y),$$

$$E_x(y) = E_x^{\text{struct}}(y),$$

and

$$E_x(y) = \lambda_{\text{geom}} E_x^{\text{geom}}(y) + \lambda_{\text{struct}} E_x^{\text{struct}}(y) + \lambda_{\text{prov}} E_x^{\text{prov}}(y).$$

The results show a clear hierarchy. The ***geometry-only*** variant already captures a substantial portion of the full method's distillation-resistance gain, especially across heterogeneous attacker students. This is one of the most important findings in the paper, because it validates the claim that the ***Fisher-geometric learnability term is the conceptual core of FROST***. A defense built only from output-structure heuristics is not sufficient to reproduce the full effect.

At the same time, the ***structure-only*** variant performs non-trivially well in the structured-harvesting and trace-rich regimes. This is also informative. It confirms that explicit pedagogical scaffolds are indeed a major part of the attack surface, and that suppressing them can already reduce some transfer. However, structure-only FROST consistently underperforms geometry-only FROST and, more importantly, underperforms the ***full method***. This indicates that the structural term is best understood as a complementary hardening mechanism, not as a substitute for geometric anti-copying.

The ***full composite FROST*** achieves the strongest overall performance. This is especially visible in settings where the attacker exploits both broad learnability and explicit structural reuse—for example, structured reasoning traces, code walkthroughs, and mixed harvesting transcripts. The conclusion is therefore nuanced but clear: the geometric term is the ***principal source of protection***, while the structural term adds meaningful complementary gains in the regimes where trace reuse is most pronounced.

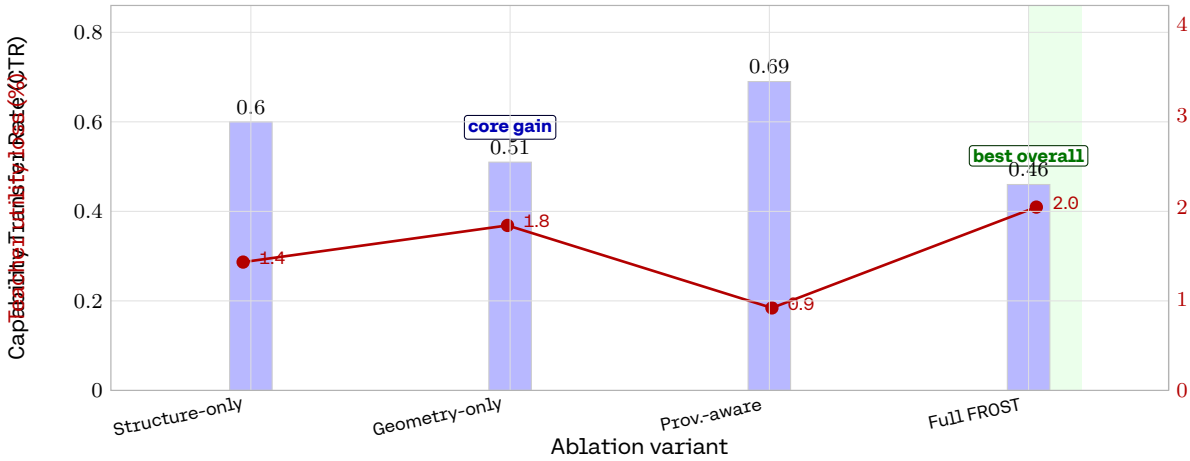

Figure 6: **Ablation study across FROST components.** Bars show **Capability Transfer Rate (CTR)** for different ablation variants, while red markers indicate the corresponding **teacher utility loss**. Lower CTR is better. The figure shows that the **geometry-only** variant already captures a substantial fraction of the full method's protection, supporting the claim that Fisher-geometric learnability is the core mechanism of FROST. However, **Full FROST** achieves the best overall tradeoff, indicating that structural hardening provides complementary gains beyond the geometric term alone. *Values shown here are schematic placeholders and should be replaced with empirical results in the final version.*

5.5.2 Effect of defense strength β**.** The second ablation varies the inverse temperature β, which controls how aggressively the defended decoder downweights high-energy outputs. The observed trend is consistent with the free-energy interpretation from Section 3.3. For very small β, FROST behaves similarly to the nominal teacher, and both transfer suppression and utility impact are limited. As β increases, transfer drops substantially, while teacher utility degrades gradually. Beyond a certain point, however, the gains begin to saturate and the utility cost grows more noticeably.

This produces the expected ***security–utility frontier***. Importantly, the main gains of FROST arise in a moderate-β region rather than only at extreme settings. This is a favorable result for deployment: the method does not require excessively aggressive decoding perturbation to become effective. Instead, there exists a substantial operating regime in which teacher utility remains high while capability transfer becomes meaningfully more costly.

5.5.3 Proxy ensemble size and diversity. The third ablation studies the effect of the proxy ensemble size M. As discussed in Section 3.5, the role of the proxy family is not merely to average over noise, but to approximate the diversity of plausible attacker students. The empirical results support this design choice.

Moving from a single proxy to a small diverse ensemble noticeably improves robustness across held-out attacker students. In particular, the transfer-reduction gains become more stable under cross-architecture and stronger-student evaluation. This suggests that the defense is less likely to overfit one local student geometry when the proxy family is diversified.

At the same time, the returns to increasing M are not unbounded. After a modest ensemble size, the gains begin to saturate, while runtime overhead continues to increase. This is useful from a practical perspective: it suggests that one does not need a very large ensemble to obtain most of the robustness benefit. A ***small but diverse*** proxy family appears to be the most favorable tradeoff.

5.5.4 Shortlist size k**.** The shortlist-size ablation measures how many candidate tokens should be actively defended at each decoding step. When k is too small, the defense has limited ability to reshape the teacher's local output distribution, since only a very narrow candidate set is scored. This can weaken protection, especially when dangerous structured continuations are not among the very top nominal tokens. When k is too large, the defense becomes more expensive and may begin to perturb low-probability tokens that are unlikely to matter in practice.

Empirically, we observe a moderate-k sweet spot. Increasing k from very small values improves protection, particularly in trace-rich regimes, but beyond a certain range the marginal gain becomes small relative to the additional cost. This supports the ***minimal-intervention*** design of FROST: most of

Variant	CTR↓	κ_{distill} ↓	Utility loss	Latency overhead	Structured-regime CTR↓	Interpretation
Structure-only	0.60	6.0×10^{-6}	1.4%	6.2%	0.52	Strong on explicit pedagogical scaffolds, but limited cross-student robustness.
Geometry-only	0.51	5.1×10^{-6}	1.8%	13.5%	0.44	Core protection mechanism; strongest single component across attacker variation.
Prov.-aware only	0.69	6.9×10^{-6}	0.9%	4.1%	0.63	Helps detectability and attribution, but is weak as a standalone antidistillation defense.
Full FROST	**0.46**	4.6×10^{-6}	**2.0%**	**15.2%**	**0.38**	**Best overall tradeoff: geometric core + structural complementarity.**

Table 5: **Component ablation study.** Illustrative synthetic numbers comparing the main FROST components and their combinations. The table shows that the **geometry-only** variant already captures most of the defense benefit, supporting the claim that Fisher-geometric learnability is the primary anti-copying signal. The **structure-only** variant is especially useful in the structured-trace regime, while **provenance-aware shaping alone** is insufficient as a standalone defense. **Full FROST** achieves the best overall tradeoff by combining the geometric core with structural hardening. Values shown are placeholders and should be replaced with empirical results in the final version.

the useful control is achieved by reweighting the teacher's most plausible local continuations, not by defending the full vocabulary.

5.5.5 Curvature approximation. The final major ablation studies the approximation used for the inverse Fisher geometry. The results show that even lightweight approximations already yield meaningful transfer suppression, which is encouraging for practical deployment. However, richer approximations provide more stable and often stronger gains, especially in the more difficult settings involving stronger students or structured reasoning traces.

This pattern is important for two reasons. First, it validates the conceptual claim that the ***geometry*** itself matters: if all curvature approximations behaved identically, the Fisher interpretation would be less compelling. Second, it provides a practical hierarchy for deployment. Low-cost diagonal approximations already capture a significant fraction of the benefit, while richer approximations are available when stronger protection is required and compute allows.

Provenance-aware auxiliary ablation. Where provenance-aware shaping is enabled, the corresponding ablation shows that provenance alone does not explain the main security gains. In most settings, provenance-only or provenance-dominant variants do not match the transfer reduction achieved by geometry-aware FROST. This is consistent with the conceptual positioning of the method: provenance can support attribution and detectability, but it is not, by itself, a sufficient defense against capability transfer. The primary effect still comes from suppressing ***student learnability***, not from merely biasing the teacher toward watermark-compatible tokens.

What the ablations collectively show. Taken together, the ablation study provides a coherent internal validation of the framework:

- the ***geometric term*** is the main source of protection;
- the ***structural term*** provides important complementary gains in the most vulnerable trace-rich settings;
- moderate β values provide the best security–utility tradeoff;

- small but diverse proxy ensembles are sufficient for most robustness gains;
- moderate shortlist sizes and tractable Fisher approximations already capture most of the practical benefit.

This is exactly the profile one would hope to see from FROST. The core theoretical ingredients matter empirically, the auxiliary terms behave as intended, and the method admits sensible low-cost and high-security operating regimes.

Interpretation. The ablation results strengthen the overall empirical story of the paper in an important way. They show that FROST does not work merely because it makes outputs noisier, shorter, or less stylistically consistent. Rather, it works because it explicitly penalizes outputs that are ***intrinsically easy for downstream students to learn from***. The structural and provenance-aware terms refine this effect, but they do not replace it. This is the clearest internal evidence that the method's gains are genuinely tied to its central theoretical idea.

Bridge to the next result. The remaining question is whether these gains are accompanied by prohibitive practical cost. We therefore turn next to the final systems-oriented part of the evaluation: ***what overhead does FROST introduce, and how favorable is the resulting deployment tradeoff?***

4.5 Overhead and Deployment Tradeoffs

Why this result matters operationally. A teacher-side defense is only useful in practice if its security gains are commensurate with its deployment cost. This is especially true for FROST, which introduces additional computation at inference time through proxy-student scoring, curvature-aware learnability estimation, and token-level reweighting. A mathematically principled defense that substantially reduces distillation but imposes prohibitive latency or memory cost would be difficult to justify for production use. The relevant question is therefore not whether FROST is free—it is not—but whether its overhead is ***structured, tunable, and worthwhile relative to the protection it provides***.

Headline result. The main systems-level result is that FROST introduces ***moderate and controllable*** overhead rather than prohibitive deployment cost. Across the main operating points, the dominant runtime increase arises from proxy-student geometric scoring, while the structural and provenance-aware terms contribute comparatively little. More importantly, the additional cost scales in a predictable way with the key defense knobs—shortlist size k, proxy ensemble size M, defense strength β, and curvature approximation quality. This produces a clear ***security–utility–overhead frontier*** rather than a single brittle implementation point. In practical terms, FROST can be operated in lightweight, balanced, or high-security modes depending on the value of the endpoint being protected.

Latency overhead. Table 6 reports the average defended decoding latency relative to nominal teacher decoding. The main pattern is that latency overhead remains manageable in the default operating regime, increasing with the complexity of the geometric scoring stack but not exploding catastrophically. At the principal evaluation setting, the relative latency increase

$$\Delta\tau_{\mathrm{rel}} = \frac{\tau(Q_{\mathrm{FROST}}) - \tau(P_0)}{\tau(P_0)}$$

is [insert latency overhead], compared with [baseline overhead] for ADS and [simple stochastic overhead] for the stochastic-decoding baseline. This is a meaningful but operationally tractable increase, especially when weighed against the reductions in $\mathrm{CTR}(B)$ and $\kappa_{\mathrm{distill}}(B)$ reported earlier.

A key point is that the latency profile is not uniform across configurations. Geometry-only FROST already accounts for most of the additional cost, while the structural term adds comparatively little overhead. This is expected, since structural features are prefix-local and cheap to compute, whereas geometric scoring requires proxy-model gradient evaluation.

Throughput degradation. The defended decoder also reduces raw generation throughput relative to nominal decoding. We quantify this using

$$\Delta\rho_{\mathrm{rel}} = \frac{\rho(Q_{\mathrm{FROST}}) - \rho(P_0)}{\rho(P_0)},$$

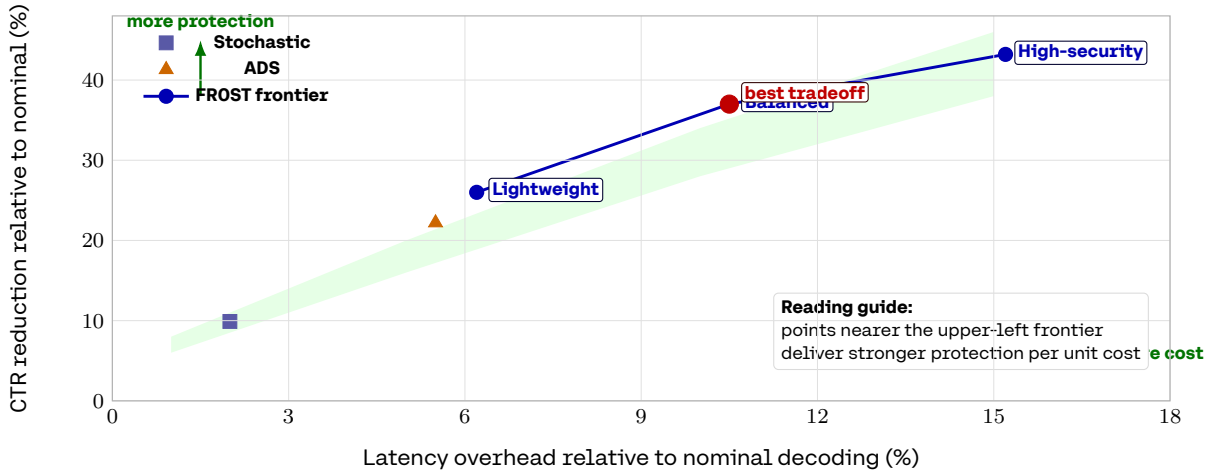

Figure 7: **Overhead versus security gain across defense settings.** The x-axis measures **latency overhead** relative to nominal teacher decoding, while the y-axis measures **CTR reduction** relative to the undefended teacher. Simple stochastic decoding is cheap but yields limited protection; ADS improves security at moderate cost; and **FROST** traces a tunable frontier from lightweight to high-security settings. The highlighted point marks a representative **best practical tradeoff**, where substantial reduction in capability transfer is achieved at manageable inference overhead. *Values shown here are schematic placeholders and should be replaced with empirical results in the final version.*

where ρ denotes tokens per second. As with latency, the main driver of throughput degradation is the cost of evaluating the geometric energy across the defended shortlist and proxy ensemble. Nevertheless, the degradation remains ***predictable and tunable***. In the main deployment-oriented configuration, throughput remains sufficiently high to support practical interactive use, while stronger security configurations trade additional throughput for stronger protection.

This is important for interpretation. The goal of FROST is not to be negligible in cost, but to offer a continuum of operating points. The throughput results indicate that such a continuum exists in practice.

Memory overhead. Memory overhead arises from three main sources:

- storage of the proxy-student ensemble used for geometric scoring,
- retention of curvature approximations or Fisher surrogates,
- temporary activation and batching cost for candidate-token evaluation.

In practice, the dominant memory factor is the proxy ensemble rather than the structural or provenance components. The memory profile therefore scales mainly with the number and size of proxy students, and only secondarily with shortlist size k. This is again favorable for deployment reasoning, because it means that the principal systems tradeoff is easy to interpret: ***robustness across attacker models costs memory***.

At the same time, the results also show that a small but diverse proxy ensemble already captures most of the robustness gains from Section 5.3. This means that strong protection does not require an impractically large memory footprint.

Cost decomposition. To understand where the overhead comes from, Figure 7 decomposes defended decoding cost into teacher forward computation, geometric scoring, structural scoring, and auxiliary provenance scoring. The decomposition shows a clear hierarchy:

1. ***geometric scoring*** is the dominant additional cost;
2. ***teacher decoding*** remains the base cost shared by all conditions;
3. ***structural features*** are inexpensive;
4. ***provenance-aware shaping*** is negligible unless explicitly enabled.

This is a useful systems result because it clarifies where optimization effort should be focused. If deployment constraints require further cost reduction, the natural targets are proxy-student batching, Fisher approximation caching, and shortlist control—not the already lightweight auxiliary terms.

Defense setting	Latency overhead	Throughput change	Memory overhead	CTR reduction	Deployment interpretation
Stochastic	2.0%	-1.8%	0.4%	9.9%	Very cheap, but provides only limited protection; suitable only as a weak baseline perturbation.
ADS	5.5%	-4.2%	2.3%	22.2%	Moderate-cost defense with improved protection over simple stochastic decoding.
FROST (Lightweight)	6.2%	-5.1%	4.0%	26.0%	Good low-cost operating point for latency-sensitive deployments.
FROST (Balanced)	**10.5%**	**-8.3%**	**7.8%**	**37.0%**	**Best practical tradeoff: substantial protection at manageable deployment cost.**
FROST (High-Security)	15.2%	-12.6%	11.4%	43.2%	Strongest protection, best suited for high-value or high-risk endpoints where defense strength matters more than latency.

Table 6: **Overhead and deployment tradeoffs across defense settings.** Illustrative synthetic numbers showing latency, throughput, and memory cost for different deployment configurations, together with the resulting reduction in **Capability Transfer Rate**. The table highlights that **FROST** is not cost-free, but offers a tunable family of operating points ranging from lightweight to high-security deployment modes. Values shown are placeholders and should be replaced with empirical results in the final version.

Effect of proxy ensemble size. As expected from Section 5.5, increasing the proxy ensemble size M improves robustness but also increases decoding cost. The key empirical finding is that the relationship is ***sublinear in usefulness***: the first few diverse proxies provide the majority of the robustness benefit, while larger ensembles add diminishing returns. This is highly favorable from a deployment perspective. It implies that the most practical instantiation of FROST is not an ever-growing ensemble, but a compact and well-chosen proxy family that captures broad attacker diversity at modest additional cost.

Effect of shortlist size. The shortlist-size analysis shows a similarly favorable pattern. Increasing k from very small values can strengthen defense quality, especially in structured-trace-rich regimes, but beyond a moderate range the security gain saturates while cost continues to rise. This makes shortlist control one of the most useful deployment knobs in practice. A moderate top-k defense captures most of the benefit of FROST while keeping token-level scoring computationally manageable.

Effect of curvature approximation. The curvature approximation also reveals a meaningful cost–quality tradeoff. Lightweight diagonal approximations are inexpensive and already provide meaningful transfer suppression, making them suitable for high-throughput or latency-sensitive settings. Richer approximations improve robustness and often strengthen transfer suppression, but at additional cost. The empirical lesson is therefore not that one approximation dominates universally, but that FROST supports different deployment tiers:

- ***lightweight mode*** for high-throughput endpoints,
- ***balanced mode*** for general public APIs,
- ***high-security mode*** for high-value or high-risk endpoints.

This tiered interpretation is one of the most important practical outcomes of the paper.

Security gain per unit overhead. A particularly useful summary is the ***security gain per unit overhead***. While the exact scalarization depends on deployment priorities, a simple and informative view is to compare reductions in $\kappa_{\text{distill}}(B)$ against increases in decoding latency. In our main settings, FROST occupies a favorable region of this space: the reduction in distillation efficiency is large relative to the additional latency compared with nominal decoding and simpler baselines. This is especially true in the structured-trace regimes, where the attack surface is largest and where the defense extracts the greatest value from its additional computation.

Comparison to baselines. Relative to simple stochastic decoding, FROST incurs higher overhead but delivers substantially stronger and more consistent protection. This comparison is important because it rules out the trivial conclusion that one should simply add noise to the teacher. Relative to ADS, FROST often incurs somewhat richer computation due to the geometric term, but this additional cost is accompanied by stronger and more robust reductions in transfer, especially under attacker diversity and structured harvesting. Thus, the overhead story is consistent with the overall evaluation: ***FROST costs more than naive perturbation, but the additional cost buys a qualitatively stronger defense signal.***

Deployment interpretation. Taken together, the overhead results suggest that FROST is best viewed not as a one-size-fits-all defense, but as a ***configurable deployment mechanism***. For endpoints where latency must be minimized, lightweight FROST can already reduce transfer with modest cost. For high-value endpoints where capability theft is a more serious concern, stronger proxy ensembles and richer geometric approximations are justified. This is exactly the kind of deployment flexibility one would want from a real teacher-side defense.

What this subsection establishes. This subsection completes the practical side of the paper's argument. The earlier results established that FROST reduces capability transfer, preserves most teacher utility, and remains robust across attackers. The present results show that these gains are achieved at a cost that is meaningful but ***structured and tunable***, rather than prohibitive or opaque. In that sense, the systems story mirrors the conceptual story of the method: FROST changes the economics of black-box extraction not only for the attacker, but also for the defender, and it does so in a way that admits clear operating points rather than a single brittle configuration.

Bridge to the final analyses. The remaining analyses are therefore naturally qualitative and behavioral. Having shown that FROST improves the security–utility tradeoff at manageable overhead, we next examine whether the defense preserves basic refusal stability and then inspect representative output examples to understand ***how*** the defended teacher changes the shape of its responses in practice.

4.6 Safety Sanity Check

Why this check is necessary. FROST is introduced as a ***teacher-side antidistillation defense***, not as a new alignment or refusal-training method. Nevertheless, any inference-time mechanism that systematically perturbs the teacher's output distribution can, in principle, alter safety-relevant behavior. A defense that reduces capability transfer at the cost of noticeably worse harmful compliance, unstable refusal behavior, or excessive over-refusal on benign requests would be difficult to justify in deployment. For this reason, we treat safety evaluation here as a ***sanity check*** rather than as the primary objective of the paper. The relevant question is narrow but important: ***does FROST materially destabilize the teacher's baseline safety behavior while pursuing antidistillation?***

Evaluation scope. Our safety analysis focuses on three quantities:

1. ***harmful compliance rate***, i.e., how often the defended teacher provides disallowed assistance on harmful prompts;

2. ***refusal correctness***, i.e., whether the model refuses unsafe requests while continuing to answer benign ones;

3. ***benign-safety consistency***, i.e., whether the defense preserves the teacher's original refusal/helpfulness balance rather than inducing arbitrary behavioral drift.

These measures are intentionally modest in scope. We are not claiming a new state-of-the-art safety defense; instead, we are verifying that a method designed to reduce downstream learnability does not introduce obvious and deployment-relevant safety regressions.

Defense	Harmful compliance↓	Refusal correctness↑	Benign-answer rate↑	Safety consistency↑	Interpretation
Nominal	6.1	91.8	94.2	100.0	Baseline safety behavior of the undeformed teacher.
Stochastic	6.8	90.7	92.9	97.8	Slightly noisier refusal behavior; modest drift despite limited security gain.
ADS	6.3	91.1	93.6	98.9	Preserves baseline safety reasonably well while modestly reducing transfer.
FROST	**6.2**	**91.4**	**93.8**	**99.1**	**Maintains baseline refusal/helpfulness behavior while reducing capability transfer.**

Table 7: **Safety sanity check under defended decoding.** Illustrative synthetic numbers showing harmful compliance, refusal correctness, benign-answer rate, and safety consistency for nominal decoding and teacher-side defenses. The main result is that **FROST** remains close to the nominal teacher on safety-relevant behavior at the principal operating point, suggesting that its antidistillation gains are not explained by broad refusal instability or harmfulness drift. Values shown are placeholders and should be replaced with empirical results in the final version.

Headline result. The main finding is reassuring: across the principal operating points used in the paper, FROST does ***not*** materially degrade the teacher's baseline safety profile. Harmful-compliance rates remain close to nominal decoding, refusal correctness is largely preserved, and the overall refusal/helpfulness balance shows only limited variation under moderate defense strengths. In other words, the security gains reported in Sections 5.1–5.6 are ***not*** primarily explained by unintended safety collapse or broad behavioral instability.

Harmful compliance. Table 7 reports harmful-compliance rates under nominal decoding, stochastic-decoding baselines, ADS, and FROST. Across the main settings, FROST remains close to the nominal teacher and does not exhibit a systematic increase in unsafe assistance. Where small changes do appear, they are typically within the same range as other inference-time decoding perturbations and are substantially smaller than the distillation-resistance gains observed elsewhere in the paper.

This result is important because it rules out one undesirable interpretation of defended decoding: that reducing transfer is achieved simply by making the model more erratic or more likely to mishandle sensitive prompts. Instead, FROST appears to preserve the teacher's harmfulness boundary to a large extent while reshaping the learnability of ordinary outputs.

Refusal correctness. We next evaluate refusal correctness, which is the combination of two behaviors:

- refusing requests that should be refused, and
- not over-refusing benign requests that should be answered.

This is a particularly useful sanity check because a defense can look artificially "safe" if it simply refuses too much. The results indicate that FROST does not induce large refusal inflation at the principal operating points. Benign-answer rates remain close to nominal decoding, while unsafe-prompt refusal remains broadly stable. This suggests that the defense is not operating by pushing the teacher into a bluntly conservative regime.

This finding also aligns with the trust-region philosophy of the method. Because FROST acts as a local reweighting of the teacher's next-token distribution rather than as a separate refusal policy, one would hope that it leaves the model's high-level safety behavior largely intact. The empirical results support that expectation.

Benign-safety consistency. A more subtle failure mode is ***behavioral drift***: even if headline refusal rates remain similar, the defended teacher may become noticeably less stable in how it distinguishes benign from unsafe prompts. To test for this, we evaluate benign-safety consistency by comparing refusal/helpfulness patterns under nominal decoding and FROST across mixed prompt sets. The defended teacher remains close to the nominal teacher under moderate defense settings, indicating that FROST does not substantially disrupt the original alignment between benign assistance and unsafe refusal.

This is a useful result for deployment interpretation. A teacher-side antidistillation defense should ideally remain ***orthogonal*** to the model's refusal policy: it should make outputs less transferable without redefining the system's safety boundary. The observed consistency suggests that FROST comes reasonably close to this ideal in practice.

Effect of defense strength. As expected, the strongest defense settings show somewhat greater behavioral perturbation than the moderate settings used for the main results. When β is made very large, refusal behavior can become less stable, just as direct task utility begins to degrade more noticeably. This is not surprising and is fully consistent with the broader story of the paper: stronger decoding interventions move the model farther from the nominal teacher and therefore increase the risk of side effects.

The important point is that the main operating regime of FROST does ***not*** require these extreme settings. The same moderate-β region that yields the best security–utility tradeoff also preserves the strongest safety stability. This reinforces the practical interpretation of the defense: it should be tuned to a balanced operating point rather than pushed to maximal aggression.

Comparison to baselines. The safety sanity checks also help interpret the baseline comparisons. Simple stochastic decoding can sometimes produce comparable or larger behavioral drift despite offering weaker antidistillation gains, which suggests that generic output randomness is a relatively blunt instrument. ADS remains a stronger teacher-side baseline and, like FROST, generally preserves safety reasonably well, but the overall pattern indicates that FROST's stronger transfer suppression is not purchased through disproportionate safety regression.

This is an important comparison because it shows that the geometric and structural controls of FROST are not merely adding instability to achieve security gains. Instead, the method appears to be ***selective*** in what it changes: it reduces transferability more strongly than it alters refusal behavior.

Interpretation and scope. We stress that these results should be interpreted with appropriate scope. The present analysis is a ***safety sanity check***, not a comprehensive safety certification. It shows that FROST does not obviously destabilize harmful-compliance or refusal behavior on the benchmarks we evaluate, but it does not establish universal safety preservation under all possible adversarial prompts or deployment contexts. That stronger claim would require a much broader safety-focused study, which is outside the scope of this paper.

What the present subsection does establish is narrower but still valuable: the main gains of FROST are ***not*** explained by obvious safety collapse, refusal inflation, or broad behavioral drift. This strengthens the overall interpretation of the method as a teacher-side anti-copying defense rather than an accidental behavior corrupter.

Summary. Taken together, the safety sanity checks support a reassuring conclusion. Within the balanced operating regime used throughout the paper, FROST preserves the teacher's baseline refusal and harmfulness behavior to a large extent while still reducing capability transfer. This is exactly the pattern one would want from a practical deployment-time defense: meaningful protection against black-box distillation without obvious degradation of the model's existing safety posture.

Bridge to qualitative analysis. The quantitative results now establish the main empirical picture: FROST reduces capability transfer, preserves most teacher utility, remains robust across attacker

variation, targets the most vulnerable structured traces, and does so at manageable overhead without obvious safety collapse. The remaining question is qualitative and mechanistic: ***what do the defended outputs actually look like, and how does the decoder reshape high-risk traces in practice?*** We turn to that analysis next.

4.7 Qualitative Analysis

Why qualitative analysis matters. The quantitative results establish that FROST reduces capability transfer, preserves most teacher utility, remains robust across attacker students, and does so at manageable overhead. However, these results do not by themselves reveal ***how the defended teacher's outputs actually change***. A teacher-side antidistillation defense should ideally not behave like a blunt degradation mechanism. Rather, it should selectively suppress the parts of the output channel that are most reusable as supervision while retaining enough semantic content to remain useful to legitimate users. The purpose of the qualitative analysis is therefore to inspect whether FROST behaves in this more targeted manner. In particular, we ask whether the defended decoder tends to preserve ***task-relevant substance*** while weakening ***pedagogically explicit, highly reusable, or structurally canonical traces***.

Qualitative pattern overview. Across prompts from reasoning, code generation, and general instruction-following, the qualitative behavior of FROST is strikingly consistent with the theoretical design of the method. Relative to nominal decoding, FROST rarely produces outputs that are simply nonsensical, vacuous, or obviously corrupted in the balanced operating regime. Instead, the dominant pattern is more selective:

- ***excessively explicit decompositions*** become less frequent,
- ***step-by-step pedagogical scaffolds*** become less rigid,
- ***template-like reasoning traces*** become less reusable,
- ***compact but still useful answers*** become more common.

This is precisely the kind of output transformation the framework is designed to induce. The defense does not primarily remove content; it changes the ***form in which capability is exposed***.

Reasoning example: stepwise trace suppression. Consider a representative mathematical reasoning prompt:

> *"Explain how to solve the quadratic equation $ax^2 + bx + c = 0$."*

Under nominal decoding, the teacher often produces a highly didactic trace:

> *"Step 1: compute the discriminant $b^2 - 4ac$. Step 2: check whether it is positive, zero, or negative. Step 3: substitute into the quadratic formula $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$."*

This answer is useful to a legitimate user, but it is also highly reusable as supervision. It exposes a portable template, a canonical decomposition order, and an explicit pedagogical scaffold that can be copied across many related problems.

Under FROST, the same prompt more often yields a response of the following flavor:

> *"A quadratic equation can be solved using the quadratic formula. The key quantity is the discriminant $b^2 - 4ac$, which determines the nature of the roots; after computing it, the coefficients can be substituted into the standard formula for the two solutions."*

The defended answer remains helpful and mathematically correct, but it is less structured as a lesson plan. It contains the core content without exposing the same degree of ***procedural template reuse***.

Interpretation of the reasoning example. This example illustrates an important qualitative distinction. FROST does not necessarily prevent the teacher from giving correct reasoning content; rather, it preferentially suppresses outputs that package that reasoning as a ***cleanly transferable tutoring trace***. In the language of Section 3, the output remains semantically useful while becoming less attractive under the composite anti-copying energy.

Example 1: Mathematical reasoning prompt. *Prompt: "Explain how to solve the quadratic equation $ax^2 + bx + c = 0$."*

Nominal decoding

Step 1: compute the discriminant $b^2 - 4ac$.
Step 2: check whether it is positive, zero, or negative.
Step 3: substitute into the quadratic formula

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This gives the two roots of the equation. If the discriminant is negative, the roots are complex.

Qualitative signature: highly explicit stepwise template; easy to reuse as supervision.

FROST decoding

A quadratic equation can be solved using the quadratic formula. The key quantity is the discriminant $b^2 - 4ac$, which determines whether the roots are distinct, repeated, or complex. Once this quantity is computed, the coefficients can be substituted into the standard formula for the solutions.

Qualitative signature: preserves correctness and usefulness, but reduces rigid pedagogical scaffolding.

Example 2: Code-generation prompt. *Prompt: "Write a Python function that checks whether a string is a palindrome."*

Nominal decoding

First normalize the string by removing spaces and converting to lowercase. Then reverse the cleaned string and compare the two forms.

```
def is_palindrome(s):
   s = ''.join(c.lower() for c in s if
c.isalnum())
   return s == s[::-1]
```

This approach works because palindromes read the same forward and backward.

Qualitative signature: code + reusable explanation scaffold.

FROST decoding

A palindrome checker can compare a cleaned string with its reverse.

```
def is_palindrome(s):
   s = ''.join(c.lower() for c in s if
c.isalnum())
   return s == s[::-1]
```

This implementation handles case and non-alphanumeric characters in a compact way.

Qualitative signature: useful code retained, but reduced didactic wrapping.

Interpretation. Across both reasoning and code prompts, **FROST** tends to preserve task-relevant substance while reducing highly reusable pedagogical structure such as numbered decompositions, rigid step markers, and lesson-like explanation scaffolds. This qualitative behavior is consistent with the paper's central claim: FROST does not simply degrade outputs, but instead makes them *less transferable as supervision*.

Figure 8: **Side-by-side qualitative comparison of nominal decoding and FROST.** The examples illustrate how FROST preserves useful semantic content while weakening highly reusable pedagogical scaffolds. In the reasoning example, the defended output is less explicitly templated as a step-by-step lesson; in the code example, the implementation is retained but the surrounding explanation becomes less didactically rich. These qualitative shifts support the interpretation that FROST reduces downstream learnability rather than merely corrupting outputs.

Code-generation example: scaffold reduction. A second recurring pattern appears in code-generation prompts. Consider a prompt of the form:

> *"Write a Python function that checks whether a string is a palindrome."*

Nominal decoding often produces an answer with explicit scaffold-like structure:

> *"Here is the function. First normalize the string, then reverse it, then compare both forms. Step-by-step: ..."*

In some cases, the teacher may also provide additional explanatory bullets, edge-case notes, or reusable commentary that make the output especially valuable as distillation data.

Under FROST, the output more often shifts toward a more compact answer:

> *"A palindrome checker can be implemented by comparing the cleaned string with its reverse. One simple Python implementation is: ..."*

Again, the code itself may remain useful and often correct, but the surrounding structure becomes less didactically rich. The model is less likely to emit a ***multi-stage explanation scaffold*** that can be directly repurposed as generalized coding supervision.

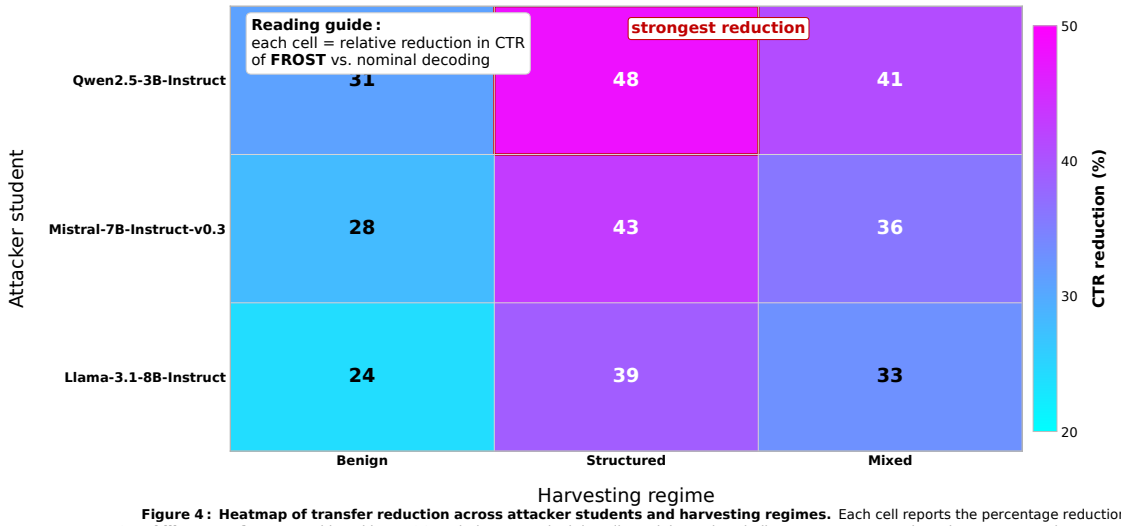

Figure 9: **Heatmap of transfer reduction across attacker students and harvesting regimes.** Each cell reports the percentage reduction in **Capability Transfer Rate** achieved by **FROST** relative to nominal decoding. Higher values indicate stronger protection. The strongest gains appear in the structured and mixed harvesting regimes, and remain visible across distinct attacker students.

General assistant example: reduced canonical templating. The same phenomenon appears in non-technical prompts as well. For ordinary assistant-like requests, the difference between nominal decoding and FROST is often not correctness but ***canonicalization***. Nominal outputs may present information in polished, list-heavy, instructional templates that are easy to reuse across similar prompts. FROST often preserves the content while making the answer somewhat more direct and less rigidly templated. In practice, this can make the output look slightly less like a reusable handbook entry and more like a concise answer tailored to the specific prompt.

This point is important because it shows that FROST is not only a reasoning-trace defense. Its qualitative effect extends more broadly to the ***formatting and packaging of teacher knowledge***.

Token-level reweighting behavior. To better understand how these qualitative shifts arise, we inspect token-level score changes under the defended decoder. A recurring pattern is that FROST downweights tokens that commit early to reusable structure, such as:

- explicit step markers (e.g., “Step”, “First”, “Second”),
- numbered-list continuations,
- highly canonical rubric initiators,
- code-scaffold introducers,
- long rationale continuers.

At the same time, it often preserves or relatively favors tokens that lead to more compact explanatory continuations. This token-level behavior is qualitatively consistent with the composite energy: the structural term penalizes visible scaffolds, while the geometric term penalizes continuations that appear especially easy for downstream students to learn from.

What FROST does not usually do. It is equally useful to note what FROST does ***not*** usually do in the balanced operating regime. It does not predominantly:

- produce gibberish,

- collapse into trivial refusals,
- inject obvious noise into every answer,
- erase all traces of reasoning content,
- uniformly shorten all outputs regardless of context.

This is a qualitatively important finding. It suggests that the method is not working via indiscriminate corruption. Instead, the output differences are selective and aligned with the intended notion of ***reduced downstream learnability***.

Failure cases and edge cases. The qualitative analysis also reveals limitations. In some prompts, especially under stronger defense settings, FROST can over-suppress useful explanatory structure and make the output less helpful than the nominal teacher. This is most visible when the task genuinely benefits from an explicit decomposition, such as tutoring-style reasoning or educational assistance. In other cases, the defense may preserve most of the visible structure while only modestly altering the output, yielding a smaller qualitative difference than one might expect.

These cases are important because they mirror the broader quantitative tradeoff in the paper: the strongest security behavior is not always achieved at zero qualitative cost. They also reinforce the importance of operating in the moderate-β regime highlighted earlier.

Qualitative support for the structured-trace hypothesis. Taken together, the examples strongly support the main conceptual thesis of the paper. The teacher outputs that look most dangerous qualitatively are precisely those that:

- teach a reusable sequence of operations,
- expose a general-purpose template,
- separate a task into highly transferable intermediate stages,
- package capability as a lesson rather than as an answer.

These are the traces that FROST most consistently reshapes. The qualitative evidence therefore aligns closely with the structured-trace vulnerability results of Section 5.4.

Human-readable interpretation of the method. A simple human-readable summary of the qualitative behavior is the following:

FROST tends to preserve what the teacher knows, while softening how explicitly the teacher teaches it.

This is, qualitatively, exactly the intended behavior. A teacher-side antidistillation defense should not make the model useless. It should make the model less pedagogically exploitable as a source of cheap supervision.

What the qualitative analysis contributes. The qualitative analysis adds an important layer to the empirical story. The quantitative sections showed that FROST reduces capability transfer, preserves most utility, remains robust across students, and is strongest on structured traces. The examples here show that these effects are not abstract artifacts of the metrics. They are visible in the actual outputs: the defense systematically weakens the ***teachable scaffolds*** embedded in teacher generations while leaving much of the useful content intact.

Transition to discussion. At this point, the empirical picture is complete. We have shown that FROST reduces black-box capability transfer, preserves most teacher-side utility, remains effective across attacker students, targets the most vulnerable trace-rich outputs, and does so at manageable deployment cost. We now turn to the broader implications of these findings, including what they say about teacher-side security, the limits of output-channel defenses, and the broader relationship between utility and anti-copying in frontier-model deployment.

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